40 Cancer of the Colon

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INTRODUCTION

A more thorough understanding of the molecular basis for this disease, coupled with the development of new therapeutic approaches, has dramatically altered the way in which patients with colorectal cancer (CRC) are managed. This chapter and the one (see Chapter 41) that follows will provide an up-to-date description of the current state of the science and outline a multidisciplinary approach to the patient with colon or rectal cancer.

MOLECULAR BIOLOGY

CRC is a solid tumor that arises and progresses as the result of defects in DNA repair or chromosome stability and through accumulation of somatic mutations and epigenetic changes. It is one of the few cancers known to develop in animals from a specific native population of intestinal stem cells (ISCs). This dual appreciation of molecular defects and the cell of origin is profoundly informed by studies of common somatic mutations, well-predisposition syndromes, and the signals that sustain normal ISCs.

MULTISTEP MODELS OF COLORECTAL CANCER AND GENETIC INSTABILITY

Nearly all CRCs develop within benign precursor polyps, where gatekeeper mutations initiate epithelial overgrowth by constitutive activation of the Wnt signaling pathway and additional mutations combine to promote invasion and metastasis (Fig. 40.1A). Pedunculated polyps larger than 1 cm confer the highest risk, with approximately 15% progressing to invasive cancer over 10 years. The prevalence of polyps in the U.S. population, estimated at 50% by age 70 years, dwarfs the 5% lifetime risk of developing CRC because few polyps become invasive and aberrations that promote malignancy accumulate over decades.[1] Nevertheless, endoscopic resection of adenomas reduces CRC incidence and mortality. Hyperplastic polyps confer low cancer risk, but approximately 8% of sporadic CRCs originate in sessile adenomas with serrated histology, characteristic nuclear morphology, and little to no dysplasia.[2] These CRCs may represent a distinct subtype, with high microsatellite instability (MSI-hi), frequent *BRAF* mutation, the CpG island methylator phenotype (CIMP), and a poor prognosis (Fig. 40.1B).

Inherited CRC syndromes and sporadic tumors together highlight the importance of proper DNA repair in preventing CRC. Scores to thousands of somatic DNA aberrations accumulate as a result of hallmark genetic instability, which is acquired in a limited number of stereotypic ways (see Fig. 40.1). About 80% of CRCs show widespread chromosomal instability (CIN): gains, losses, and translocations that produce various gene amplifications, deletions, and rearrangements. Chromosomal segregation defects, mediated by segregation factors such as BUB1, may underlie CIN, but few genes are implicated directly. CRCs with CIN show a genome-wide bias toward C:G to T:A transitions at 5′-CpG-3′, and fewer mutations in 5′-TpC-3′, dinucleotides than breast cancer, for example.[3] On average, 17 genes are deleted or amplified to 12 or more copies per CRC,[4] with deletions resulting in loss of heterozygosity (LOH). In aggregate, the oncogenes *ERBB2*, *MYC*, *KRAS*, *MYB*, *IGF2*, *CCND1*, and *CDK8* are amplified or overexpressed in most cases, usually along with neighboring genes, but nearly half the copy number alterations also occur in other cancers. CRCs thus reflect perturbation of selected pathways of replicative and tissue homeostasis, some common to many cancers and others restricted to CRC. Specific cytogenetic anomalies correlate poorly with clinical outcomes or disease features. Recent evidence suggests that beyond driving widespread gene alterations, CIN causes excessive spillage of DNA into the cytosol, hence activating non-canonical nuclear factor kappa B (NF-κB) signaling, which promotes metastasis.[5]

Although the remaining approximately 15% of CRCs are euploid, they carry thousands of small insertions and deletions (indels) or point mutations near nucleotide repeat tracts—collectively designated as MSI-hi—as a result of defective DNA mismatch repair (MMR; see Fig. 40.1B).[6] Up to one-third of these cases occur in the setting of the familial Lynch syndrome, discussed at length in the following text. MSI-hi tumors usually arise in the ascending colon, have a high frequency of *BRAF*V600E mutation, resist adjuvant 5-fluorouracil treatment, and respond better to immunotherapy with programmed cell death protein 1 (PD-1) blockade than other cancers.[7] Stage for stage, aneuploidy in CIN confers a worse prognosis than MSI-hi disease.[8] Some authors consider CIMP-positive serrated CRCs separately, but because their molecular, clinical, and pathologic features overlap with MSI-hi tumors, The Cancer Genome Atlas (TCGA) and other groups pool the two CRC types into a single "hypermutated" category.[9]

Among the approximately 3% of sporadic CRCs cases that are hypermutated but lack MSI, the majority carry somatic mutations in the exonuclease domain of *POLE*.[9] Moreover, rare kindreds with a dominantly inherited high risk of CRC carry specific germline defects in *POLE* or *POLD1*, the proofreading exonuclease genes for the leading- and lagging-strand DNA polymerases ε and δ. Although the tumors carry thousands of mutations, microsatellite tracts are stable; polymerase proofreading-associated polyposis (PPAP) thus represents a third "mutator" mechanism in CRC.

MUTATIONAL AND EPIGENETIC LANDSCAPES IN COLORECTAL CANCER

After accounting for the different paths to genome instability, mutational profiles are similar in colon and rectal cancers. Constitutive activation of the Wnt pathway—triggered by gatekeeper mutations in *APC*, *CTNNB1*, *RNF43*, or *RSPO* genes—initiates adenoma formation, as discussed in the following text. Although the classic progression sequence (see Fig. 40.1) has its origins in the frequencies of various mutations at different stages of disease,

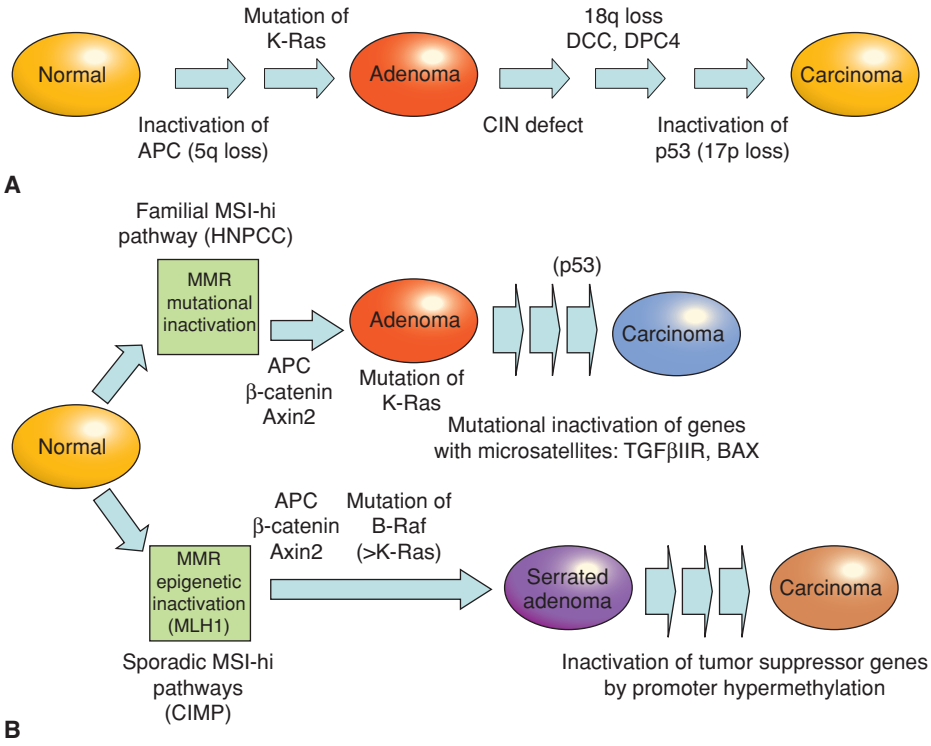


Figure 40.1 Genetic pathways to colorectal carcinoma. All colorectal cancers (CRCs) arise within benign adenomatous precursors, fueled by mutations that serially enhance malignant behavior. Mutations that activate the Wnt signaling pathway seem to be necessary initiating events, after which two possible courses contribute to the accumulation of additional mutations. **A:** Chromosomal instability is a feature of up to 80% of CRCs and is commonly associated with activating KRAS point mutations and loss of regions that encompass *P53* and other tumor suppressors on 18q and 17p, often but not necessarily in that order. **B:** About 20% of CRCs are euploid but defective in DNA mismatch repair (MMR), resulting in high microsatellite instability (MSI-hi). MMR defects may develop sporadically, associated with CpG island methylation (CIMP), or as a result of familial predisposition in hereditary nonpolyposis colorectal cancer (HNPCC). Mutations accumulate in the *KRAS* or *BRAF* oncogenes, in *p53* tumor suppressor, and in microsatellite-containing genes vulnerable to MMR defects, such as *TGFβIIR*. Epigenetic inactivation of the MMR gene *MLH1* and activating *BRAF* point mutations are especially common in serrated adenomas, which progress, in part, through the silencing of tumor suppressor genes by promoter hypermethylation. Progression from adenoma to CRC takes years to decades, a process that accelerates in the presence of MMR defects. CIN, chromosomal instability.

CRCs vary substantially in which genes are mutated and in which order. Early studies estimated an average of 81 mutations per CRC, with a few mutations (*APC*, *KRAS*, *TP53*, *PIK3CA*) occurring frequently, others of known function (e.g., *BRAF*V600E) less commonly, and most mutations detected only in a handful of cases.[3] Genome-scale studies of hundreds of CRCs and matched normal colonic mucosa corroborate these findings and reliably catalog the genetic landscape of CRC (Table 40.1).[9,10] Few novel mutations uncovered in the latter studies lie in genes that constitute "druggable" targets, such as kinases, and some may have pleiotropic effects on cell survival, growth, and metastasis. In aggregate, genes that modify the epigenome represent the largest new class of frequently mutated genes, and MSI-hi tumors in particular show recurrent mutations in *ARID1A*, a chromatin remodeling factor. CRCs arising in African Americans have a partially distinct profile, including mutations in *EPHA6* and *FLCN*,[11] and approximately 10% of CRCs carry monoallelic missense mutations in *SOX9*, a transcription factor that is highly expressed and active in intestinal crypt stem and progenitor cells.[12]

CRCs progress by activating or disrupting pathways that involve many gene products; some genes in vital pathways are more prone to mutation than others, and mutation of a crucial gene may obviate the need for additional mutations in the same pathway. Thus, *KRAS* and *BRAF* mutations, which together occur in about half of all CRCs but are mutually exclusive, represent alternative means to activate intracellular epidermal growth factor receptor (EGFR) signaling. Although these examples conform to the paradigm that frequent occurrence of a genetic aberration reflects the selective advantage it confers on tumor cells, the mere presence of other mutations does not necessarily signify a pathogenic role. "Passenger" alterations, whether associated with CIN or MSI, confer no advantage and may even be detrimental to tumors. Two features are therefore used to impute "driver" status: recurrent alteration, as detected in large cohorts, and experimental demonstration of a contribution to malignancy, which is laborious and imperfect. Infrequent events that may contribute to neoplasia tend to concentrate in genes that control cell adhesion, signaling, DNA topology, and the cell cycle. Common mutations in sporadic CRCs rarely correlate with specific histologic or clinical features, but certain genotypes do delineate disease subtypes and sensitivity to certain therapies. For example, *KRAS* and *BRAF* mutations preclude clinical response to EGFR antibodies[13,14] and MSI-hi CRCs are uniquely sensitive to treatment with immune checkpoint inhibitors.[15]

Prominent modes of epigenetic control in mammalian cells include DNA methylation at CpG dinucleotides and covalent modification of certain histone residues. Compared to normal colonic epithelium, CRCs and even benign adenomas show 8% to 15% lower *total* DNA methylation.[16,17] Global DNA hypomethylation may reduce the fidelity of chromosome segregation, owing to reduced pericentromeric methylation and reversal of

TABLE 40.1

Recurrent Somatic Gene Mutations in Human Colorectal Cancers

Gene	Frequency	CRC Class	Known Cellular Function
Oncogenes			
KRAS	35%–40%	CIN	RTK signaling
PIK3CA	18%–20%	Mostly CIN	RTK signaling
BRAF	7%–15%	MSI-hi, CIMP	RTK signaling
NRAS	9%	CIN	RTK signaling
ERBB3	~8%	CIN	RTK signaling
CTNNB1	~5%	All	Wnt pathway activation
Tumor Suppressor Genes			
APC	85%	All	Wnt pathway regulation
RNF43	18%	Mainly MSI-hi	Wnt pathway regulation
TP53	50%	Mainly CIN	Stress, hypoxia response
SMAD4	10%	CIN	TGF-β signaling
FBXW7	10%	CIN	Ubiquitin ligase
SOX9	5%–10%	CIN	Wnt-dependent ISC function
TGFBR2		MSI-hi	TGF-β signaling
MSH2, etc.		MSI-hi	DNA mismatch repair
POLE		Hypermutated (non-MSI)	DNA polymerase ε
Epigenetic Modifier Genes (Roles Are Emerging)			
ARID1A		MSI-hi	Chromatin remodeling
SIN3A			Transcriptional repression
SMARCA5			Chromatin remodeling
NCOR1			Transcriptional repression
JARID2			Histone modification
TET1, 2, 3			DNA demethylation

CRC, colorectal cancer; CIN, chromosomal instability; RTK, receptor tyrosine kinase; MSI-hi, high microsatellite instability; CIMP, CpG island methylator phenotype; TGF-β, transforming growth factor β; ISC, intestinal stem cell.

imprinting at loci such as *IGF2*, but its pathogenic role, if any, is unclear. In mouse models, some studies suggest that global hypomethylation increases tumor susceptibility, whereas absence of the de novo methyltransferase DNMT3B slows, and DNMT3B overexpression accelerates, tumor progression. Against the backdrop of genome-wide DNA *hypo*methylation, the distinctive subset of CRCs with CIMP shows coordinate *hyper*methylation of many CpG island promoters, associated with transcriptional attenuation of tumor suppressor genes such as *HIC1* and Wnt-inhibiting secreted frizzled related protein (SFRPs).[12,18] Whole-genome methylation analyses affirm the existence of this once controversial entity,[19] revealing properties that distinguish it from *KRAS*-mutant CIN-positive disease: origin in sessile serrated adenomas; strong association with *BRAF*-mutant, right-sided MSI-hi tumors with *MLH1* gene methylation; and distinct RNA expression profiles.[9]

Insights from Mendelian Syndromes, Genome-wide Association Studies, and the Microbiome

Following clinical suspicion or diagnosis of a Mendelian risk syndrome, probands and family members should be tested for pertinent germline mutations, receive genetic counseling, and enter programs for cancer prevention and screening. The corresponding molecular defects profoundly inform understanding of sporadic CRC, revealing in particular the seminal role of Wnt signaling and early, rate-limiting effects of *APC* inactivation or *CTNNB1* activation. *STK11* and *PTEN* loss in inherited and sporadic CRCs also shed light on crucial molecular pathways, whereas HNPCC and PPAP help classify the disease and reveal the broader significance of features such as MSI-hi. The cancer spectrum in HNPCC or PPAP and the specific predilection for CRC, however, remain unexplained. Colonic, endometrial, and certain other epithelia may be especially sensitive to mutations that occur in the setting of defective DNA MMR and base-excision repair; loss of the wild-type tumor suppressor allele may occur more readily in these tissues, or they may lack repair safeguards that protect other cell types.

Even in the absence of a recognized predisposition syndrome, individuals with a history of CRC in a first-degree relative are up to four times more likely to develop CRC than those without a family history. Specific environmental factors that compound the risk of developing CRC are complex and insufficiently characterized but include obesity, excessive consumption of red meat, physical inactivity, and vitamin D deficiency. Many of these factors converge on insulin signaling, suggesting a possibly seminal role for insulin-like growth factors in CRC.[20] Longstanding inflammatory bowel disease, especially ulcerative colitis, elevates CRC risk up to 10-fold, likely reflecting increased mutation in the setting of repeated mucosal injury and repair. Colitis-associated CRCs often arise within flat adenomatous plaques and areas of nonadenomatous dysplasia. Compared to sporadic cases, *APC* inactivation is less frequent, *TP53* mutations occur earlier in the cancer sequence, and methylation of the *p16*INK4a tumor suppressor gene is more common.

Up to one-sixth of CRCs, especially MSI-hi cases, carry DNA and RNA from the invasive anaerobic microorganism *Fusobacterium nucleatum*[21,22] and a worse prognosis than cases without that DNA. *F. nucleatum* is an oral microbe that uses its Fap2 lectin to bind Gal-GalNAc moieties on colonic adenoma or cancer cell surfaces and a unique adhesin, FadA, to increase Wnt pathway activity. Although *F. nucleatum* triggers local inflammation and modulates T-cell infiltrates, its causal role remains uncertain. Notably, its DNA persists in distant metastases and experimental xenografts, raising the provocative prospect of antibiotics as adjuncts to cytotoxic or targeted agents in CRC treatment.[23]

ONCOGENE AND TUMOR SUPPRESSOR GENE MUTATIONS IN COLORECTAL CANCER PROGRESSION

Building on a foundation of constitutive Wnt activity, somatic mutations in oncogenes and tumor suppressor genes confer malignant properties. Recurring mutations provide crucial clues to decipher oncogenic signaling circuits and develop rational therapies (see Table 40.1). The high collective frequency of recurrent *KRAS*, *BRAF*, and *PIK3CA* mutations places EGFR and its downstream effectors extracellular signal-regulated kinases (ERKs, also known as mitogen-activated protein kinases [MAPKs]) at the center of research and therapeutic efforts.

The KRAS, BRAF, and PIK3CA Oncogenes

The Ras family of small G-proteins transduces growth factor signals and is aberrantly activated in a variety of cancers. *KRAS* is

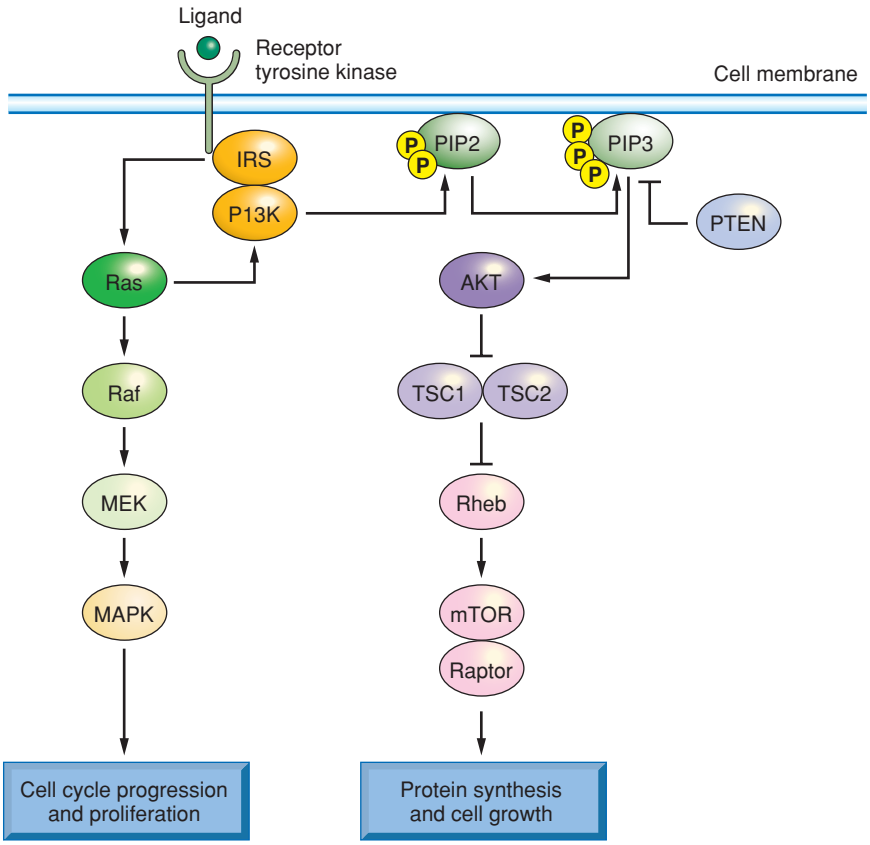


Figure 40.2 Signaling pathways, oncogenic mutations, and therapeutic opportunities in colorectal cancer (CRC). It is instructive to consider common genetic alterations in CRC in light of a common canonical outline of signaling through receptor tyrosine kinases, among which the endothelial growth factor receptor is a prime example. *KRAS*, the oncogene mutated in up to 40% of CRCs, signals receptor activation through RAF proteins (including BRAF, which is mutated in 5% to 8% of CRCs) and phosphatidylinositol 3-kinase (PI3K), whose catalytic PIK3CA subunit is mutated in 15% to 20% of CRCs. These transducers in turn activate the intracellular mitogen-activated protein kinase (MAPK) and AKT or mammalian target of rapamycin (mTOR) pathways, respectively. Hence, common mutations confer growth factor independence on cells, resulting in dysregulated proliferation, protein synthesis, and metabolism. They also represent promising targets for therapeutic interference with aberrantly activated signaling cascades.

mutated in about 40% of CRCs[24] and *NRAS* in another ~5% of cases. Mutations in both genes cluster in codon 12 or 13 and less commonly in codon 61. *KRAS* mutations appear even in lesions of low malignant potential, such as aberrant preadenomatous crypt foci and diminutive polyps, and their frequency increases with lesion size.[25] Mutant *KRAS* alone does not initiate adenomas in mice but, when combined with *Apc* mutation, accelerates tumor progression[26,27]; removing it from CRC cells or xenografts impedes cell growth. Among the many growth factor signals that KRAS transduces in diverse tissues, its activity in the colonic epithelium and CRC relates most to EGFR. Mutant *KRAS* locks EGFR signaling in the "on" state; EGFR antibodies are consequently ineffective in this subset of CRCs,[13] and targeted therapies will need to disrupt signals further downstream.

As KRAS is a plasma membrane–associated signal transducer for receptor tyrosine kinases, mutant forms potentially deregulate several effector pathways for cell survival, proliferation, and invasion (Fig. 40.2). KRAS signaling recruits RAF kinases to the plasma membrane and triggers MAPK kinase 1 (MEK1) and MAPK kinase 2 (MEK2) to activate ERK1 and ERK2. *KRAS* mutation thus induces constitutive phosphorylation of ERKs,[28] which in turn phosphorylate proteins that control the G_1 to S cell cycle transition, among other substrates.[29] Although other, non–KRAS-mediated growth factor pathways also activate the MEK/ERK cascade, signaling in CRC is most often deregulated through activating mutations in *KRAS* or *BRAF*. The latter gene is mutated in about 10% of CRCs, especially those associated with MSI and CIMP.[30,31] *V600E*, the most common *BRAF* mutation in CRC, melanoma, and other cancers, affects a residue within the activation loop of the kinase domain and constitutively activates kinase function, probably by several hundredfold and acting as a phosphomimetic.[32] Activated BRAF also leads to ERK phosphorylation, releasing intrinsic restraints on cell growth. Indeed, *KRAS* and *BRAF* mutations are mutually exclusive in CRC,[9,30] suggesting that they represent alternative routes to the same signaling end.

The prognosis in advanced *BRAF*-mutant CRC, however, is worse than in *KRAS*-mutant disease, which implies distinctive molecular or cellular features. Moreover, whereas *Kras* activation in the mouse intestine has modest consequence on its own, *Braf*V600E expression rapidly induces persistent generalized hyperplasia with a high penetrance of crypt dysplasia, serrated morphology, and invasive MSI-hi cancers with Wnt and ERK pathway deregulation[33]; in humans, *BRAF* mutation is a hallmark of nonfamilial MSI-hi CRC and occurs early in sessile serrated adenomas. Notably, whereas *BRAF*V600E mutant melanomas respond to selective, ATP-competitive BRAF inhibitors such as vemurafenib and take months to manifest secondary resistance, CRCs carrying the same mutation are intrinsically resistant. This is because BRAF inhibition rapidly induces feedback EGFR signaling through KRAS and CRAF, restoring stimuli for CRC replication; melanoma cells avoid such feedback activation because they hardly express EGFR.[34,35] In principle, BRAF inhibition should render cells newly sensitive to direct antagonism of EGFR, but in practice combined antagonism of BRAF and EGFR signaling shows limits that are unexplained. Of note, and important given the increase of immunotherapy agents that have been approved for the treatment of solid tumor malignancies, MSI-hi colon cancers appear to be more responsive to checkpoint inhibitor therapy than MSS colon cancers. Monotherapy with anti-PD1 agents have resulted in objective response rates of approximately 30% and 1-year survival rates of 76%. Treatment with anti-PD1 and anti-CTLA4 combinations improve those outcomes to response rates of 55% and 1-year survival of 85%.[36]

EPIDEMIOLOGY

Incidence and Mortality

Globally, nearly 1,200,000 new CRC cases are believed to occur, which accounts for approximately 10% of all incident cancers,

and mortality from CRC is estimated at nearly 609,000.[37] In 2021, there were an estimated 149,500 new cases of CRC and 52,980 deaths in the United States. As such, CRC accounts for 7.9% of all new cancer cases in the United States with a 5-year relative survival of 64.7%.[38] Prevalence estimates reveal that in unscreened individuals age 50 years or older, there is a 0.5% to 2.0% chance of harboring an invasive CRC, a 1.0% to 1.6% chance of an in situ carcinoma, a 7% to 10% chance of a large (≥1 cm) adenoma, and a 25% to 40% chance of an adenoma of any size.[39]

Age impacts CRC incidence greater than any other demographic factor. To that end, sporadic CRC increases dramatically above the age of 45 to 50 years for all groups. In almost all countries, age-standardized incidence rates are less for women than for men. Although CRC incidence has been steadily decreasing in the United States and Canada, the incidence is rapidly increasing in Japan, Korea, and China.[37] In the United States from 2014 to 2018, the age-standardized incidence rates per 100,000 population were 43.2 for men and 33.3 for women when combined for all races.[38] Recognizing that decreases in age-standardized CRC incidence and mortality rates are apparent in the United States over the past 10 years to 15 years, such trends may be counterbalanced by prolonged longevity.

While the incidence of CRC in the United States has decreased overall, presumably due to aggressive screening of the population over age 50, there has been a dramatic increase in younger patients. A study using data from the Surveillance Epidemiology and End Results (SEER) program found a rising incidence of CRC over the last 20 years in patients aged 20 to 49. The most pronounced growth was in the age group 40 to 44 where colon and rectal cancer increased 56% and 94%, respectively.[40,41] Based on these findings and informed by microsimulation modelling as well as the fact that CRC in younger patients tends to be more advanced, the American Cancer Society was the first national organization to recommend lowering the screening age for average-risk individuals from 50 to 45. While this announcement was initially quite controversial and the ACS was criticized for not considering the economic outcomes and potential trade-offs at the population level in reallocating resources to younger individuals from those at higher absolute risk, most other organizations have made similar changes, as more supportive data accumulate.[42,43]

Geographic Variation

The incidence rate for Alaskan Natives exceeds 70 per 100,000,[44] while that for Gambia and Algeria is <2 per 100,000.[45] Generally speaking, CRC incidence and mortality rates are the greatest in developed Western nations.[37,45] The reader is referred to the most recent detailed incidence and mortality rates in different countries over time according to gender, ethnicity, and anatomic site as established by the National Cancer Institute on their website.

As mentioned, there appears to be a recent decrease in age-standardized CRC incidence and mortality rates within the United States. From 1999 to 2006, CRC incidence and mortality both decreased.[38] Furthermore, 5-year survival improved. These trends are apparent regardless of gender, race, or ethnic group, except for Native Americans. Although at an initial glance one might invoke alterations in dietary and lifestyle factors, or the utilization of chemopreventive agents, it is clear that enhanced use of colonoscopy with polypectomy represents a significant reason for the improvements in trends in some areas.[46]

Emigration Patterns in Population Groups

Seminal studies have revealed that migrants from low-incident areas to high-incident areas assume the incidence of the host country within one generation.[47–50] For example, for Chinese who immigrate to the United States, higher CRC rates have been ascribed to greater meat consumption and diminished physical activity in contrast to controls within their original country.[48] These and other studies underscore the importance of environmental exposure in CRC incidence and provide a platform for attention to dietary and lifestyle modification as preventive measures.

Race and Ethnicity

Although dietary and lifestyle factors are of paramount importance in low-incident regions of the world, especially Asia and Africa, nonetheless there are certain trends along racial or ethnic lines. For example, an inherited adenomatosis polyposis coli (APC) gene mutation, I1307K, confers a higher risk of CRC within certain Ashkenazi Jewish families that is not apparent in other ethnic groups.[51,52] Inherited mutations in the DNA mismatch repair genes may be more common among African Americans,[53] in part accounting for anatomic variation in colon cancers among races in the United States,[54,55] an area that is receiving much attention in epidemiology- and biology-based research.

One recent study extracted data from the Adjuvant Colon Cancer ENdpoinTs (ACCENT) collaborative group database to analyze time-to-event end points for black and white patients participating in 12 randomized controlled adjuvant phase 3 trials of resected stage II and III colon cancer. In this cohort of 14,611 patients—controlling for sex, stage, age, and treatment type—both overall 5-year survival rates and 3-year recurrence-free survival rates were significantly worse for black patients (68.2% vs 72.8% and 68.4% vs 72.1%, respectively). However, recurrence-free interval was similar, arguing against a differential response to the adjuvant therapy itself. The authors concluded that poorer outcomes were more likely related to confounding factors not measured such as toxicity, comorbid conditions, and racial disparities in care for recurrent disease.[56]

Socioeconomic Factors

Generally, cancer incidence and mortality rates have been higher in economically advantaged countries.[57,58] This may be related to consumption of a high fat and high red meat diet, lack of physical activity with resulting obesity, and variations in mortality causes over a longitudinal period of time.

Anatomic Shift

Classically, colon cancer was believed to be a disease of the left or distal colon. However, the incidence of right-sided or proximal colon cancer has been increasing in North America[55,59] and Europe.[60] Similar trends have been observed in Asian countries.[61] This anatomic shift is likely multifactorial: (1) due to increased longevity; (2) as a response to luminal procarcinogens and carcinogens, which can vary between different sites of the colon and rectum; and (3) because of genetic factors, which can preferentially involve defects in mismatch repair genes with resulting microsatellite instability (MSI) in proximal colon cancers and chromosomal instability pathway predominant in left-sided colon and rectal cancers. These developments in anatomic variation will necessarily impact considerably on screening procedures, response to chemoprevention, response to chemotherapy, and, ultimately, disease-specific survival.[62–64]

ETIOLOGY: GENETIC AND ENVIRONMENTAL RISK FACTORS

Inherited Predisposition

Family history confers an increased lifetime risk of CRC, but that enhanced risk varies depending on the nature of the family history

TABLE 40.2

Etiology of Colon Cancer: Environmental Factors

Increased Incidence	Decreased Incidence
High-caloric diet	High-fiber diet?
High red meat consumption	Antioxidant vitamins
Overcooked red meat?	Fresh fruit/vegetables
High saturated fats	Nonsteroidal anti-inflammatories
Excess alcohol consumption	Coffee
Cigarette smoking	High calcium
Sedentary lifestyle	High magnesium
Obesity	Bisphosphonates
Diabetes	

(Table 40.2). Familial factors contribute importantly to the risk of sporadic CRC, depending upon the involvement of first- or second-degree relatives and the age of onset of CRC. Involvement of at least one first-degree relative with CRC serves to double the risk of CRC.[65] There is further enhancement of the risk if a case is affected prior to the age of 60. Similarly, the likelihood of harboring premalignant adenomas or CRC is increased in first-degree relatives of persons with CRC.[66,67] The National Polyp Study reveals compelling data; the relative risk (RR) for parents and siblings of patients with adenomas compared to spousal controls was 1.8, which increased to 2.6 if the proband was younger than age 60 at adenoma detection.[68]

Provocative assessments of population groups suggest a dominantly inherited susceptibility to colorectal adenomas and cancer, which may account for the majority of sporadic CRC, but this may have variable inheritance based on the degree of exposure to environmental factors.[69] What are these susceptibility factors? The answer has yet to emerge. Nonetheless, genetic polymorphisms may be of paramount importance, such as in glutathione-s-transferase,[70] ethylene tetrahydrofolate reductase,[71,72] and *N*-acetyltransferases, especially NAT1 and NAT2.[73] In fact, genetic polymorphisms can vary among different racial and ethnic groups, which may provide clues to the geographic variation of CRC as well.

The BRCA1 mutation has been linked to increased risk of CRC across a number of studies, especially for younger women, prompting recommendations for earlier screening in this population. Phelan et al. reported this in a prospective analysis of 7,105 women followed for a mean of 5.5 years (standardized incidence ratio 3.81).[74] There was no discernible impact of BRCA2 or either mutation in older women. A subsequent meta-analysis including 14 studies demonstrated that BRCA1 mutation carriers had a 1.49-fold greater risk of CRC regardless of age.[74,75]

Environmental Factors

Seminal studies have underscored the importance of environmental factors as contributing to the pathogenesis of CRC. One has to take population-based studies into the context of methodologies employed, lead-time bias, time-lag issues, definition of surrogate and true end points, and the role of susceptibility factors.

One such population-based study recently evaluated risk factors for CRC from the Women's Health Initiative, a comprehensive prospectively collected database of 150,912 postmenopausal women, in which 1,210 developed colon cancer and 282 developed rectal cancer. Eleven risk factors were independently associated with colon cancer, some of which have little or no previous support in the literature (age, waist girth, use of hormone therapy at baseline [protective], years smoked, arthritis [protective presumably due to medications used], relatives with CRC, lower hematocrit levels, fatigue, diabetes, less use of sleep medication, and cholecystectomy). Three of these factors were also significantly associated with an increased risk of rectal cancer (age, waist girth, and not taking hormone therapy).[76]

Diet

Total Calories

Obesity and total caloric intake are independent risk factors for CRC as revealed by cohort and case-control studies.[77,78] Increased body mass may result in a twofold increase in CRC risk, with a strong association in men with colon but not rectal cancer. Weight gains during early to middle adulthood have also recently been linked with increased risk of colon but not rectal cancer. This relationship too seems more prominent in men than women in a large prospective study.[79]

Meat, Fat, and Protein

Ingestion of red meat but not white meat is associated with an increased CRC risk,[80,81] and as such, per capita consumption of red meat is a potent independent risk factor. Whether the total abstinence from red meat leads to a decreased CRC incidence has not been clarified, as there are studies with opposing results.[82] Also unclear is whether the type of red meat or the degree of processing or cooking method make any difference. While Probst-Hensch et al.[83] found fried, barbecued, and processed meats to be associated with CRC risk, especially for rectal cancer, with odds ratio (OR) of 6, follow-up reports do not consistently support these claims. In the population-based Norwegian Women and Cancer cohort including 84,538 participants, highly processed meat intake (especially sausage) was associated with increased CRC risk but meat cooking methods and total meat intake were not.[84] A second study of 53,988 participants reported no difference with processed meat intake either. The authors did find that cancer risk was associated with different meat subtypes (i.e., animal of origin), which varied by tumor location—specifically, colon cancer risk was significantly elevated in the setting of high lamb intake (incidence rate ratio = 1.07) and rectal cancer risk was affected by pork (incidence rate ratio = 1.18).[85] However, McCullough et al.[86] recently reported a positive association in patients with nonmetastatic CRC between red and processed meat consumption before cancer diagnosis with higher risk of death after definitive surgery.

Coffee

Coffee contains numerous bioactive compounds that may modulate cancer risk but previous epidemiologic studies investigating its role in CRC have yielded ambiguous results. In a recent meta-analysis of 41 studies (25,965 patients), Li et al.[87] found a significant inverse association from case-control data for CRC (OR = 0.85) and colon cancer (OR = 0.79) but not rectal cancer. This was particularly true among females and in Europe.[87] Stronger evidence comes from the National Institutes of Health–AARP Diet and Health Study, a large prospective U.S. cohort including 489,706 members. In this report, both caffeinated and decaffeinated coffee drinkers had a decreased risk of colon cancer, particularly of proximal tumors (hazard ratio [HR] for more than six cups a day = 0.62), and decaffeinated coffee drinkers also had a decreased risk of rectal cancer. While known confounders such as smoking and red meat consumption were adjusted for, further investigation is warranted to confirm and clarify this association.[88]

Fiber

Classically, a high-fiber diet was associated with a low incidence of CRC in Africa,[89] with numerous studies substantiating this premise.[90] Protection was believed to be afforded from wheat bran, fruit, and vegetables.[81] A high-fiber diet was believed to dilute fecal

carcinogens, decrease colon transit time, and generate a favorable luminal environment. The European Prospective Investigation into Cancer and Nutrition is an ongoing multicenter prospective cohort study, which was one of the largest and most influential studies to initially report an inverse association between dietary fiber and CRC. More long-term data, with a mean follow-up of 11 years and a near threefold increase in CRC cases, further supports this claim while providing a more precise estimation by fiber food source as well. After multivariable adjustments, total dietary fiber was found to be inversely associated with both colon and rectal cancers (HR per 10 g/day increased in fiber = 0.87), and this did not differ by age, sex, lifestyle, or other dietary factors.[91] However, other large, well-controlled studies show no inverse relationship between CRC and fiber intake.[92] In a study of nearly 90,000 women from ages 34 to 59 who were followed for 16 years, no protective effect was noted between fiber and incidence of either adenomatous polyp or CRC.[92] This was further corroborated by two large randomized controlled trials that evaluated high-fiber diets for moderate duration and discovered a lack of effect on the number, size, and histology of polyps found on colonoscopy.[93,94] At this point, therefore, it is unclear whether dietary fiber plays any substantial role in the risk of developing CRC.

Vegetables and Fruit

A protective effect of vegetables and fruits against CRC is generally believed to be true.[80] This has been observed with raw, green, and cruciferous vegetables. Whether certain agents such as antioxidant vitamins (E, C, and A), folate, thioethers, terpenes, and plant phenols may translate into effective chemopreventive strategies requires further investigation, although the data for folate intake are sound.[95]

In addition to folate, dietary methionine (another component of one-carbon metabolism) has also recently been shown to impact risk of colorectal cancer. Zhou et al. performed a meta-analysis of eight prospective studies, 431,029 participants, and 6,331 cancer cases, reporting an inverse association between the two, particularly in studies with longer follow-up time (RR = 0.81), in Western studies (RR = 0.83), and in men (RR = 0.75).[96]

Taking this nutritional data a step further, Bamia et al.[97] recently evaluated the impact of the Mediterranean diet on CRC risk in a large European cohort. This diet, introduced in the 1960s as "health-protecting," includes a high intake of vegetables, fruits, nuts, fish, cereals, and legumes with moderate alcohol consumption and low consumption of dairy and meat. The authors found an 8% to 11% decreased CRC risk when comparing patients with the highest to lowest diet adherence rates (HR = 0.89). The association was strongest for women and colon tumors.[97]

Another approach to dietary risk assessment emphasizes the specific impact of certain foods on inflammation. The dietary inflammatory index (DII) is a literature review–based score that was developed and validated over the past several years comprising 45 food parameters with varying effects on specific inflammatory mediators such as IL-1, IL-4, IL-6, IL-10, TNF alpha, and CRP. A number of recent studies have demonstrated an association between DII and colorectal cancer risk. In a case-control study of 424 patients with colorectal cancer and 401 controls, a direct association was observed between DII score and CRC risk (OR 1.65 for highest quartile compared to lowest, $P = 0.011$), with even stronger findings for colon cancer alone (OR 2.24).[98] Tabung et al. reports similar findings in prospective analysis of the Women's Health Initiative, a population of 152,536 racially and geographically diverse postmenopausal women. When tumors were differentiated by site, however, only proximal colon cancer retained a significant risk association with elevated DII score (OR 1.35), whereas no impact was demonstrated for distal colon or rectal cancer, possibly due to the smaller number of these cases.[99]

Other dietary factors under recent investigation include calcium, magnesium, and vitamin D. Calcium has been historically implicated as having a protective effect, perhaps due to its ability to bind injurious bile acids with reduction of colonic epithelial proliferation.[100] This is supported through cell culture models as well as a 2013 case-control study of 1,556 patients from Poland in which, after adjusting for numerous dietary and lifestyle factors, the authors reported that an increase in calcium consumption was associated with decreased risk of colon but not rectal cancer (OR = 0.93). Moreover, subjects who consumed >1000 mg/day showed a 46% decreased rate of colon cancer (OR = 0.54).[101]

A recent meta-analysis evaluating the influence of magnesium intake demonstrated a modest risk reduction, with pooled RRs of 0.81 for colon cancer and 0.94 for rectal cancer. This association persisted even after results were adjusted for calcium intake in six of the analyzed studies.[102]

Vitamin D has been shown to inhibit cell proliferation and increase apoptosis in vitro, and its deficiency is considered an important risk factor for many types of solid cancers. In a meta-analysis of 18 prospective studies, vitamin D intake and blood 25 (OH)D levels were found to be inversely associated with the risk of CRC as well (RR = 0.79 and 0.62 for colon cancer, respectively; RR = 0.78 and 0.61 for rectal cancer, respectively). While this report offers only preliminary observational data, larger randomized trials for vitamin D supplementation are warranted[103] and would be needed before routine vitamin D supplementation could be recommended for the purpose of CRC prevention. It is noteworthy that the Institute of Medicine, while supporting vitamin D supplementation to maintain bone health, found the evidence insufficient to support vitamin D as being protective against colorectal or any other cancer.[104]

Lifestyle

Physical inactivity has been associated with CRC risk, for colon more than rectal cancer. A sedentary lifestyle may account for an increased CRC risk, although the mechanism is unclear. Data suggest that physical activity after the diagnosis of stages I to III colon cancer may reduce the risk of cancer-related and overall mortality and that the amount of aerobic exercise correlates with a reduced risk of recurrence following resection of stage III colon cancer.[105] More recently, positive associations have been established between increased amounts of recreational physical activity before and after CRC diagnosis and lower mortality.[106]

Most studies of alcohol have demonstrated at most a minimally positive effect. Associations are strongest between alcohol consumption in men and risk of rectal cancer. Perhaps interference with folate metabolism through acetaldehyde is responsible.[107]

Prolonged cigarette smoking is associated with the risk of CRC.[80] Cigarette smoking for >20 pack-years was associated with large adenoma risk and >35 pack-years with cancer risk. To examine the impact of smoking cessation on the attenuation of this risk, Gong et al.[108] conducted a pooled analysis of eight studies, including 6,796 CRC cases and 7,770 controls. The authors found that former smokers also remained at increased risk for up to 25 years after quitting. However, this varied substantially by cancer subsite with risk declining immediately for proximal colon and rectal cases but not until 20 years after smoking cessation for distal colon tumors.[108]

Diabetes

Type 2 diabetes has previously been implicated in the development of CRC, but it has been difficult to separate this association from other confounding lifestyle factors such as smoking and obesity. Two recent meta-analyses provide further evidence that this condition is in fact a significant independent risk factor. Yuhara et al.[109] identified 14 studies, most of which controlled for smoking, obesity, and physical exercise, and demonstrated that diabetes was associated with increased risk of both colon and rectal cancer (RR = 1.38 and RR = 1.20, respectively).[109] A second report, analyzing 24 studies, found a similar association (RR = 1.26) with even higher risk for those patients on insulin therapy (RR = 1.61).[110]

Drugs

Nonsteroidal Anti-inflammatory Drugs

Population-based studies strongly support inverse associations between use of aspirin and other nonsteroidal antiinflammatory drugs (NSAID) and the incidences of both CRC and adenomas.[111–113] As a result, NSAIDs and selective cyclooxygenase 2 (COX-2) inhibitors have been investigated intensively in hereditary and sporadic CRC.

Long-term results are now available from the CAPP2 study, the first double-blind randomized controlled trial of aspirin chemoprevention with cancer as the primary end point. In this study, 861 carriers of Lynch syndrome were randomly assigned to aspirin or placebo. With a mean follow-up of 55.7 months, the authors report a significantly decreased incidence of CRC in the treatment group as well as a trend toward reduction in extracolonic Lynch syndrome–associated cancers. Importantly, there was no significant difference in adverse events such as gastrointestinal (GI) bleeding, ulcers, or anemia during the intervention period. These data provide strong rationale for the routine use of aspirin chemoprevention in Lynch syndrome and establish a foundation for further study in sporadic neoplasia. In a combined analysis of four large randomized trials of lower-dose aspirin (75 to 300 mg per day) involving 14,033 patients, aspirin taken for 5 years or more was associated with a reduced 20-year incidence and mortality due to CRC (absolute reduction = 1.76%; 95% confidence interval [CI] = 0.61 to 2.91; $P = 0.001$). Reduction was largely confined to right-sided tumors.[114] In addition to generalized chemoprevention, the question of aspirin and other NSAIDs in patients with a diagnosis of CRC has been addressed. Liao et al.[115] have reported evidence that suggests that aspirin therapy after CRC diagnosis may be beneficial to those patients whose tumors have a PIK3CA mutation, but not in those with wild-type PIK3CA.[115] However, PIK3CA mutation status had no impact on the influence of the COX-2 inhibitor rofecoxib on cancer recurrence.[116]

Bisphosphonates

In addition to being one of the most commonly used medications for osteoporosis, bisphosphonates have been shown to have various antiproliferative, antiangiogenic, proapoptotic, and antiadhesive effects in preclinical studies. Practical impact on malignant disease, however, has been inconsistent. Singh et al.[117] performed a recent meta-analysis demonstrating a statistically significant 17% reduction in CRC incidence with bisphosphonate use. This finding was observed independently for both proximal and distal colon cancers as well as rectal cancers, highlighting another potential pathway for chemoprevention.[117]

Statins

There is longstanding debate regarding the association between statin use and cancer, with research currently underway to elucidate the specific chemopreventive properties of these agents that may include inhibiting tumor growth and angiogenesis, attenuating metastatic potential, stimulating cellular immunity, and potentiating the antitumor effects of some cytokines. Epidemiologic studies on colorectal cancer have produced very inconsistent results. Liu et al. conducted a meta-analysis of 42 studies (18 case-control, 13 cohort, 11 RCT) and found that statin use was associated with a modestly reduced risk of colorectal cancer overall (RR = 0.90). This effect was more pronounced for rectal cancer and in lipophilic statin users (e.g. simvastatin, lovastatin). Due to the heterogeneity of the included studies, the authors were unable to draw any conclusions regarding dose, duration, or side effects, and interestingly, the risk reduction was not significant in a subgroup analysis of long-term statin users. Definitive recommendations await the results of ongoing prospective prevention trials.[118]

Biomarkers

In an effort to improve screening protocols and advance understanding of colorectal carcinogenesis, investigators are focusing on a variety of biomarkers for increased risk as well.

Toriola et al.[119] evaluated the role of C-reactive protein and serum amyloid A, two common inflammatory mediators, in the Women's Health Initiative Observational Study. With over 900 case-control pairs for each marker, the authors found that elevated concentrations of both C-reactive protein and serum amyloid A conferred significantly increased risk of colon cancer (OR = 1.50, $P = 0.006$). This is not surprising given the role inflammation plays in colorectal carcinogenesis as well as the new promising data surrounding NSAID chemoprevention.[119]

Leptin, a peptide hormone produced by adipocytes, is also thought to contribute to CRC pathogenesis. A recent prospective analysis found that soluble leptin receptor levels, which may regulate leptin function, was strongly inversely associated with both CRC and colon cancer risk (RR = 0.55 and RR = 0.42, respectively). This finding was independent of leptin levels and other circulating biomarkers.[120] Chi et al.[121] performed a similar investigation of insulin-like growth factor peptides, also implicated in CRC carcinogenesis, and found that high levels of insulin-like growth factor I and insulin-like growth factor II significantly increased cancer risk (OR = 1.25 and OR = 1.52, respectively).[121] Along these lines, high circulating levels of C-peptide, a direct marker of hyperinsulinemia, may also be a predictive factor for increased CRC risk, as indicated in a recent meta-analysis.[122]

Human Papillomavirus

While human papillomavirus is well established as the critical pathogenic force behind cervical and anogenital cancer, its role in colorectal malignancy is less clear. An association between the two was first reported in 1990 and since then, a growing number of studies have detected the virus in colon adenocarcinoma specimens. In the first meta-analysis to address this topic (including 16 articles and 1,436 patients), Damin, Ziegelmann, and Damin[123] not only reported a high prevalence of human papillomavirus (31.9%) in affected patients but also found a strong correlation between human papillomavirus positivity and increased CRC risk (OR = 10.04; 95% CI = 3.7 to 27.5). These results may indicate an alternative pathway of colorectal carcinogenesis that could have vast implications for treatment and prevention.[123]

E. Coli

E. coli is another pathogen under investigation despite its ubiquity among the gut flora. Studies suggest that the mucosa-adherent or mucosa-internalized strains might play a role in carcinogenesis by inducing chronic inflammation and synthesizing toxins that affect cell proliferation, differentiation, and apoptosis. Bonnet et al. found an increased prevalence of these pathogenic strains in colonic tissue from cancer patients versus controls, as well as in more advanced versus early stage tumors. The authors also confirmed the carcinogenic properties of one of these cancer-associated strains by demonstrating a marked increase in colonic polyp burden in mice that were inoculated with this strain compared to controls.[124]

FAMILIAL COLORECTAL CANCER

Familial Adenomatous Polyposis

Familial adenomatous polyposis (FAP) constitutes 1% of all CRC incidence (Table 40.3). Hallmark features include hundreds to thousands of colonic polyps that develop in patients in their teens to 30s, and if the colon is not surgically removed, 100% of patients

TABLE 40.3

Familial and Nonfamilial Causes of Colorectal Cancer

Syndromes with Adenomatous Polyps
APC gene mutations (1%): ■ Familial adenomatous polyposis ■ Attenuated APC ■ Turcot syndrome (two-thirds of families)
MMR gene mutations (3%): ■ Hereditary nonpolyposis colorectal cancer types I and II ■ Muir-Torre syndrome ■ Turcot syndrome (one-third of families)
Syndromes with Hamartomatous Polyps (<1%)
Peutz-Jeghers (*LKB1*)
Juvenile polyposis (*SMAD4, PTEN*)
Cowden (*PTEN*)
Bannayan-Ruvalcaba-Riley
Mixed polyposis
Other Familial Causes (up to 20%–25%)
Family history of adenomatous polyps (*MYH*)
Family history of colon cancer: ■ Risk more than three times greater if two first-degree relatives or one first-degree relative <50 y with colon cancer ■ Risk two times greater if second-degree relative affected
Familial colon-breast cancer
Nonfamilial Causes
Personal history of adenomatous polyps
Personal history of colorectal cancer
Inflammatory bowel disease (ulcerative colitis, Crohn's colitis) ■ Radiation colitis ■ Ureterosigmoidostomy ■ Acromegaly ■ Cronkhite-Canada syndrome

progress to CRC. Extracolonic manifestations include benign conditions—congenital hypertrophy of the retinal pigment epithelium, mandibular osteomas, supernumerary teeth, epidermal cysts, adrenal cortical adenomas, desmoid tumors (although these tumors may lead to obstruction)—and malignant conditions—thyroid tumors, gastric small intestinal polyps with a 5% to 10% risk of duodenal or ampullary adenocarcinoma, and brain tumors.[125] The brain tumors may be of two types—glioblastoma multiforme or medulloblastoma—and the particular association of brain tumors and colonic polyposis is called Turcot syndrome.[126] The colonic polyps in Turcot syndrome are fewer and larger than in classic FAP. An attenuated form of FAP harbors up to 100 colonic polyps and has a predisposition to colorectal cancer in patients when they are in their 50s or 60s.[127]

FAP is an autosomally dominant disorder with nearly 100% penetrance. However, about 30% of patients have de novo mutations and are without an ostensible family history. Based on karyotypic analysis that reveals an interstitial deletion on human chromosome 5q and subsequent genetic linkage analysis to 5q21, the gene responsible for FAP was identified as *APC*. Patients with FAP inherit a mutated copy of the *APC* gene, thereby predisposing them to early onset polyposis. During life, patients with FAP acquire inactivation of the remaining *APC* gene copy, which accelerates the progression to CRC. Interesting genotypic–phenotypic associations exist between the location of the *APC* gene mutation and certain clinical manifestations, such as congenital hypertrophy of the retinal pigment epithelium, desmoid tumors, and classic FAP versus attenuated FAP.

The *APC* gene comprises 15 exons and encodes a protein of nearly 2,850 amino acids (310 kDa). Nearly all germline mutations in the *APC* gene lead to a truncated protein, which can be detected through molecular diagnostic assays that can be integrated into genetic counseling and genetic testing of affected patients and at-risk family members.[128,129] The functions of the APC protein and the interrelated pathways and regulatory molecules will be discussed later.

Hereditary Nonpolyposis Colorectal Cancer

Hereditary nonpolyposis CRC (HNPCC) accounts for about 3% of all CRCs. Salient features include up to 100 colonic polyps (hence the term nonpolyposis), preferentially, albeit not exclusively, in the right or proximal colon.[130] There is an accelerated rate of progression to CRC in these diminutive, at times flat, polyps with mean age of onset of CRC being 43 years. This is designated HNPCC type I. HNPCC type II is distinguished by extracolonic tumors that originate in the stomach, small bowel, bile duct, renal pelvis, ureter, bladder, uterus and ovary, skin, and perhaps the pancreas. The lifetime risk of CRC in HNPCC is 80%, up to 50% to 60% for endometrial cancer, and 1% to 13% for all other cancers.[130,131] Of note, a variant of HNPCC involves skin tumors and is designated as Muir-Torre syndrome. HNPCC is defined classically by the modified Amsterdam criteria (Table 40.4).

TABLE 40.4

Criteria for Identifying At-risk Individuals for Mismatch Repair Deficiency (High Microsatellite Instability)

Amsterdam I Criteria
At least three relatives with colorectal cancer
One relative should be a first-degree relative of the other two
At least two successive generations should be affected
At least one colorectal cancer case before age 50
FAP should be excluded
Tumors should be verified histopathologically
Amsterdam II Criteria
At least three relatives with HNPCC-associated cancer (colorectal, endometrial, small bowel, ureter, or renal pelvis)
At least two successive generations should be affected
At least one case before age 50
FAP should be excluded
Tumors should be verified histopathologically
Bethesda Criteria (for Identification of Patients with Colorectal Tumor Who Should Undergo Testing for MSI)
Cancer in families that meet Amsterdam criteria
Two HNPCC-related cancers, including colorectal or extracolonic
Colorectal cancer and a first degree relative with colorectal cancer and/or HNPCC-related extracolonic cancer and/or colorectal adenoma: one cancer before age 45 and adenoma before age 40
Colorectal cancer or endometrial cancer before age 45
Right-sided colorectal cancer with an undifferentiated pattern on histopathology before age 45
Signet-ring cell type colorectal cancer before age 45
Adenoma before age 40

FAP, familial adenomatous polyposis; HNPCC, hereditary nonpolyposis colorectal cancer; MSI, microsatellite instability.

TABLE 40.5
Genetic Testing in Inherited Colorectal Cancer

FAP	APC protein truncating testing (preferred). If APC mutation found, screen for mutation in family. Less desirable alternatives: gene sequencing, linkage testing.
HNPCC	MSI testing in tumor.[a] If MSI present, proceed to sequencing of both *hMLH1* and *hMSH2* genes. If mutation found, screen for mutation in family.

[a]Immunohistochemistry may be an option.
FAP, familial adenomatous polyposis; APC, adenomatosis polyposis coli; HNPCC, hereditary nonpolyposis colorectal cancer; MSI, microsatellite instability.

HNPCC is an autosomally dominant disorder with about 80% penetrance. Genetic and biochemical approaches led to the discovery of the involvement of human DNA mismatch repair genes in HNPCC. Recognized as the human orthologs of mismatch repair genes described in bacteria and yeast, human mismatch repair genes encode enzymes that repair errors during DNA replication that may occur spontaneously or upon exposure to an exogenous agent (e.g., ultraviolet light, chemical carcinogen). Mutations in one of these mismatch repair genes result in MSI, which creates a milieu of somatic mutations of target genes—*TGF*-β2 receptor, *bax*, *IGF* type I receptor, among others—in HNPCC-associated tumors.[132] About 60% of germline mutations in HNPCC are found in either the *hMLH1* gene or the *hMSH2* gene, but mutations in other members of this family—*hMSH6*, *hPMS1*, *hPMS2*—are rare, thereby indicating that other genes are involved but have yet to be discovered. Genetic testing is not facile for HNPCC as it is for FAP, but it involves sequencing both the *hMLH1* and *hMSH2* genes (Table 40.5). If a germline mutation is found, then the remaining at-risk family members can be genetically screened. MSI testing and hMLH1/hMSH2 immunohistochemistry (IHC) can be performed on tumor specimens as a possible prelude to genetic testing.

While these preliminary screening tests have traditionally only been applied to patients who meet the Amsterdam or Bethesda criteria for high risk family history, there has been a strong push recently towards expanding this testing to all cases of colorectal cancer. The proponents of this "universal screening" approach argue that HNPCC is underdiagnosed, with a significant number of cases being missed by selective testing. Numerous questions remain, however, as to the logistics, infrastructure, and expense of implementing such an ambitious change.

Hamartomatous Polyposis Syndromes

Hamartomatous polyposis syndromes are rare syndromes, mostly affecting the pediatric and adolescent population, and represent <1% of CRCs annually. Peutz-Jeghers syndrome involves large but few colonic and small bowel polyps that can manifest by GI bleeding or obstruction and an increased risk of CRC. The polyps are distinguished by a smooth muscle band in the submucosa. Hallmark clinical features on physical examination include freckles on the hands, around the lips, in the buccal mucosa, and periorbitally. Associated characteristics include sinus, bronchial, and bladder polyps, and about 5% to 10% of patients have sex cord tumors. Patients can also develop lung and pancreatic adenocarcinomas. The gene responsible for this syndrome is *LKB1*, a serine threonine kinase.

Juvenile polyposis have overlapping clinical manifestations with Peutz-Jeghers, but the polyps tend to be confined to the colon, although cases of gastric and small bowel polyps have been described and there is an increased risk of CRC. Extracolonic manifestations are not prevalent. This is a polygenic disease, involving germline mutations in *PTEN*, *SMAD4*, *BMPR1*, or other genes yet to be identified.

Cowden syndrome harbors hamartomatous polyps anywhere in the GI tract, and surprisingly, there is no increased risk of CRC. However, about 10% of patients will have thyroid tumors and nearly 50% of patients have breast tumors. Germline *PTEN* mutations have been reported.

It is estimated that about 20% to 30% of CRCs are compatible with an inherited predisposition, independent of known syndromes.[133] The identification of other responsible genes will have great clinical impact. Intensive approaches are being pursued through sibling-pair studies and other familial studies. As previously mentioned, patients may be predisposed to an increased risk of adenomatous polyps as well in the context of a family history of sporadic adenomatous polyps.

ANATOMY OF THE COLON

The colon and rectum make up the segment of the digestive system commonly referred to as the large bowel. Defined as the portion of intestine from the ileocecal valve to the anus, the large bowel is approximately 150 cm in length. It is divided into five segments defined by its vascular supply and by its extraperitoneal or retroperitoneal location: the cecum (with appendix) and ascending colon, the transverse colon, the descending colon, the sigmoid colon, and the rectum. The anatomy of the rectum will be discussed in detail in the chapter 41 on rectal cancer. The large bowel has a muscular wall and can be distinguished from the small intestine by its increased diameter, the presence of haustra, appendices epiploicae, and tenia coli. The tenia consist of condensations of longitudinal muscle fibers starting near the base of the appendix and continuing throughout the abdominal colon to form a continuous longitudinal muscle coat in the upper rectum. Haustra are outpouchings of bowel wall separated by folds that give a classic appearance on radiography or barium enema.

The right colon is made up of the cecum (with appendix) and ascending colon. It is anterior to the right kidney and the duodenum. Its vascular supply is from branches of the superior mesenteric artery (SMA). The SMA divides into the middle colic artery and the trunk of the SMA. The middle colic artery immediately forms two to three large arcades in the transverse mesocolon. The SMA ileocolic arterial branches then extend from the SMA. The right colic artery arises as a separate branch from the SMA in 10.7% of cases.[134] The ileocolic artery gives off a right colic artery to the upper ascending colon and forms an anastomosis with branches from the middle colic artery. The ileal branch of the ileocolic artery gives off branches to the distal small bowel and cecum, whereas the colic branch supplies the ascending colon. An anastomosis occurs between the distal SMA and the ileal branch of the ileocolic artery at the junction of the terminal ileum and cecum. The right colon is a retroperitoneal structure.

The transverse colon is supplied by branches of the middle colic artery. It is the first portion of the colon considered to be intraperitoneal, and its length can vary. Its boundaries are defined by the hepatic flexure on the right and the splenic flexure on the left. Both of these points are fixed. The hepatic flexure abuts the gallbladder fossa, while the splenic flexure lies anterior to the splenic hilum and the tail of the pancreas. The descending colon is where the colon once again becomes a retroperitoneal structure, and it is defined as the segment of colon from the splenic flexure to the sigmoid colon. The descending colon is the first segment of the left side of the colon and receives its blood supply from the inferior mesenteric artery. The inferior mesenteric artery arises from the aorta and gives off the left colic artery. It also gives off three to four sigmoidal arteries, which supply the intraperitoneal sigmoid colon. The anastomosis between the vessels of the middle colic artery and

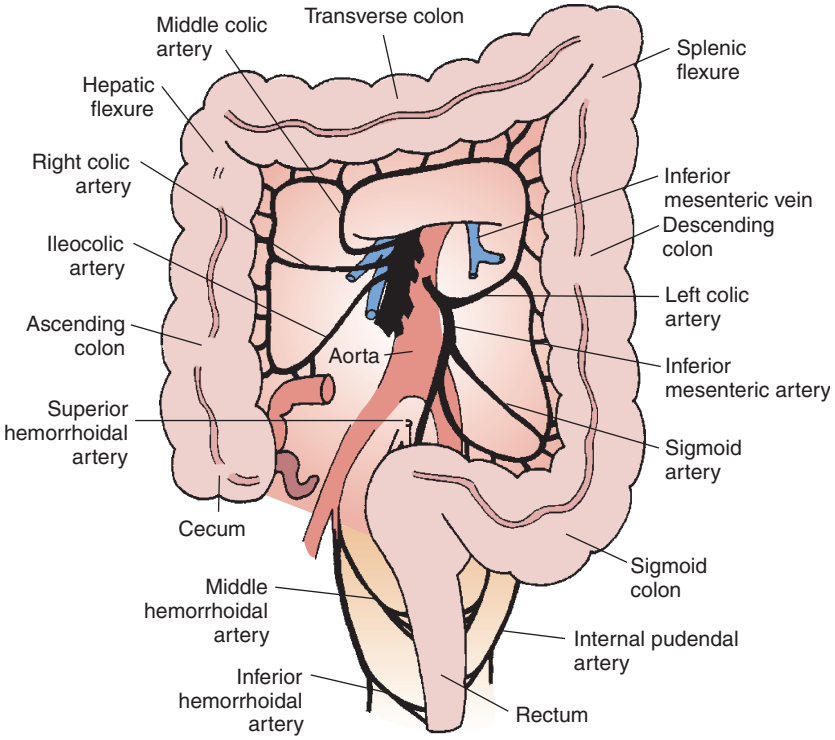


Figure 40.3 The anatomy of the colon with particular emphasis on the vascular supply.

those of the left colic artery and right colic artery is known as the marginal artery of Drummond. The arcade, which effectively connects the left and right circulations, is known as the arc of Riolan. The arterial supply to the colon is depicted in Fig. 40.3.

The venous and lymphatic drainage of the colon parallels the arterial supply, and all three vessels course and divide within the colonic mesocolon (Fig. 40.4). The mesocolon therefore contains the regional lymph nodes (LN) for the segment of colon it supplies and drains. The efferent lymphatic channels pass from the submucosa to the intramuscular and subserosal plexus of the bowel to the first tier of LNs lying adjacent to the large intestine and known as *epicolic nodes*.[135] *Paracolic nodes* lie on the marginal vessels along the mesenteric side of the colon and are frequently involved in metastases. *Intermediate nodes* are found along the major arterial branches of the SMA and inferior mesenteric artery in the mesocolon. The *principal nodes* are found around the origin of these vessels from the aorta, and they drain into retroperitoneal nodes. The drainage of the superior and inferior mesenteric veins, which drain the ascending, transverse, descending, and sigmoid colon, is to the portal vein. The rectum is drained by rectal tributaries to the vena cava.

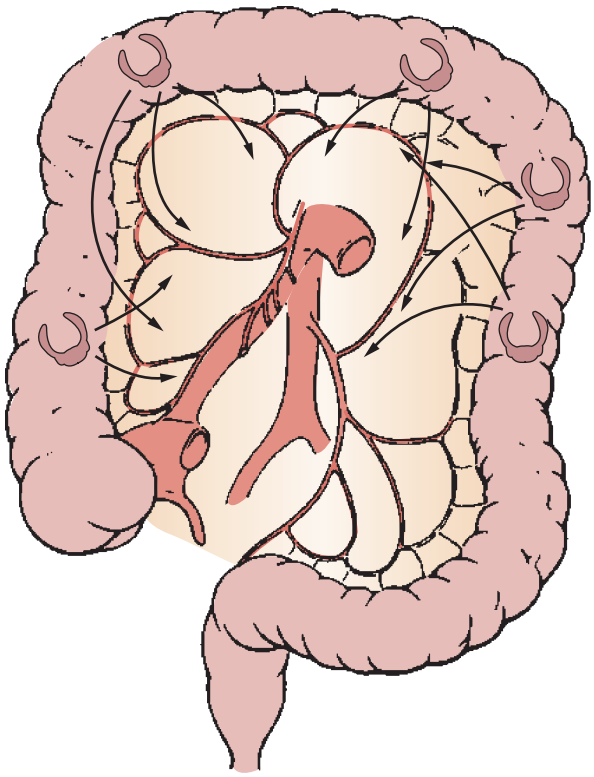


Figure 40.4 The lymphatic drainage of lesions in various anatomic locations throughout the colon.

The extent of resection of the colon is defined by the vascular supply and by the need to take the regional draining LNs.[136,137] A careful understanding of the colonic anatomy, structure, location, and vascular supply is therefore critical in order to perform a safe and effective cancer operation. The segmental resections important for removal of lesions in various locations within the colon will be described in greater detail in later sections.

DIAGNOSIS OF COLORECTAL CANCER

Symptoms associated with CRC include lower GI bleeding, change in bowel habits, abdominal pain, weight loss, change in appetite, and weakness, and in particular, obstructive symptoms are alarming.[138] However, apart from obstructive symptoms, other symptoms do not necessarily correlate with stage of disease or portend a particular diagnosis.[139]

Physical examination may reveal a palpable mass, bright blood per rectum (usually left-sided colon cancers or rectal cancer) or melena (right-sided colon cancers), or lesser degrees of bleeding (hemoccult-positive stool). Adenopathy, hepatomegaly, jaundice, or even pulmonary signs may be present with metastatic disease. Obstruction by colon cancer is usually in the sigmoid or left colon, with resulting abdominal distention and constipation, whereas right-sided colon cancers may be more insidious in nature. Complications

of CRC include acute GI bleeding, acute obstruction, perforation, and metastasis with impairment of distant organ function.

Laboratory values may reflect iron-deficiency anemia, electrolyte derangements, and liver function abnormalities. The carcinoembryonic antigen (CEA) may be elevated and is most helpful to monitor postoperatively, if reduced to normal as a result of surgery.[140]

Evaluation should include complete history, family history, physical examination, laboratory tests, colonoscopy, and pan-body computed tomography (CT) scan.[141] For rectal cancer, additional imaging techniques, such as magnetic resonance imaging or endoscopic ultrasound, are utilized to further characterize the primary tumor prior to therapy (see Chapter 41).

While transabdominal ultrasound is not routinely used in the staging of colon cancer, recent advances in this technology as well as its value as a highly cost-effective and noninvasive means for evaluating other abdominal tumors has prompted new interest. In one of the first studies to report on the role of transabdominal ultrasound in the preoperative assessment of colon cancer, Shibasaki et al. found an overall T staging accuracy of 64% which increased to 89% when a three-tier approach was used (Tis/TI, T2, and T3/4). Of the 98 biopsy proven cancers, all but two (located at the splenic flexure) were detected by ultrasonography. While still investigational, this technique could prove to be an important alternative or adjunct to existing staging protocols.[142]

Upon completion of diagnosis and staging for both colon and rectal tumors, it is essential to incorporate the expertise from medical, radiation, and surgical oncologists in order to formulate and implement an optimal treatment plan.

With the advent of molecular biologic techniques, attention has been drawn to stool-based tools and new blood-based tests. Technology now exists to extract genomic DNA or protein from stool and assay for evidence of genetic alterations.[143,144] Large-scale validation studies are in progress, including one that describes an automated multitarget sDNA assay (fecal immunochemical testing) with a 90% specificity and 98% sensitivity for the detection of CRC, as well 83% sensitivity for advanced adenoma with high-grade dysplasia.[145]

In addition, Epi proColon (Epigenomics AG, Berlin, Germany), a blood-based test, was shown to be noninferior to fecal immunochemical testing in preliminary results from a multicenter double blind comparative study (press release from Epigenomics AG, December 4, 2012). One particularly attractive pathway for stool-based diagnostics would be able to stratify patients as high, moderate, or low risk for CRC and thus influence screening modalities and frequency of screening. In a complementary fashion, functional genomics are being applied to pair-wise comparisons of normal colon and CRCs to sample the entire human genome of nearly 30,000 genes to discover those genes, known and novel, that may be upregulated or downregulated and possibly linked to detection, prognosis, and therapy.

SCREENING FOR COLORECTAL CANCER

Debate is vigorous as to the best approaches for screening, and multiple factors influence that decision: simplicity and rapidity so as to enhance patient compliance, benefit to risk ratio, sensitivity, specificity, cost effectiveness, and other economic factors. To that end, currently, optical colonoscopy likely offers the most effective approach when one considers all of these factors.

The average-risk patient is defined as a man or woman above the age of 45 without personal or family history of adenomatous polyps or CRC and absence of any occult or acute GI bleeding. Screening recommendations or guidelines for average-risk and high-risk individuals are presented in Table 40.6.

Optical colonoscopy is currently the most sensitive method for screening. Advantages include direct visualization, with the ability to remove polyps (with rate-limiting factors of size and anatomic location) and to obtain biopsies. Disadvantages involve the preparation, invasive nature of the procedure, and potential side effects that include perforation (although this is <1%).

The digital rectal examination should be part of the general physical examination. Anorectal masses may be palpated. Flexible sigmoidoscopy does not require conscious sedation and hemodynamic monitoring and will typically allow visualization of the

TABLE 40.6

Recommendations for Colorectal Cancer Screening in Average-risk and Increased-risk Patients from the Gastrointestinal Consortium Panel

Average-Risk Patient (Different Options)

1. FOBT: Offer yearly screening with FOBT using a guaiac-based test with dietary restriction or an immunochemical test without dietary restriction. Two samples from each of three consecutive stools should be examined without rehydration. Patients with a positive test on any specimen should be followed up with colonoscopy.
2. Flexible sigmoidoscopy: Offer flexible sigmoidoscopy every 5 y.
3. FOBT plus flexible sigmoidoscopy: Offer screening with FOBT every year combined with flexible sigmoidoscopy every 5 y. When both tests are performed, the FOBT should be done first.
4. Colonoscopy: Offer every 10 y.
5. DCBE: Offer every 5 y.

Increased Risk for Colorectal Cancer

1. Family history of colorectal cancer or polyps. People with a first-degree relative (parent, sibling, or child) with colon cancer or adenomatous polyp diagnosed at age <60 y or two first-degree relatives diagnosed with colorectal cancer at any age should be advised to have screening colonoscopy starting at age 40 y or 10 y younger than the earliest diagnosis in the family, whichever comes first, and repeated every 5 y.
2. FAP: Flexible sigmoidoscopy to start at ages 10 to 12 y. Genetic testing (for FAP, upper endoscopy with side-viewing scope) should be done every 1 to 3 y.
3. HNPCC: Colonoscopy every 1 to 2 y starting at ages 20 to 25 y or 10 y younger than the earliest case in the family, whichever comes first. Genetic testing (for HNPCC, consideration should be given to screening for uterine and ovarian cancer with hysteroscopy and transvaginal ultrasound, the frequency of which varies within centers).
4. Personal history of adenomatous polyps
   A. If one or more polyps that are malignant or large and sessile or colonoscopy is incomplete, then follow-up colonoscopy should be in the short term.
   B. If three of more polyps, follow-up colonoscopy in 3 y.
   C. If one or two polyps (<1 cm), follow-up colonoscopy in 5 y (or more).
5. Personal history of colorectal cancer
   A. Colonoscopy is incomplete at time of diagnosis of colorectal cancer due to obstruction, then repeat colonoscopy 6 mo after surgical resection.
   B. Colonoscopy is complete at time of diagnosis of colorectal cancer, then repeat colonoscopy in 3 y, and if that is normal, then repeat every 5 y.
6. Inflammatory bowel disease (ulcerative colitis, Crohn's colitis). Surveillance colonoscopy is recommended.

FOBT, fecal occult blood testing; DCBE, double-contrast barium enema; FAP, familial adenomatous polyposis; HNPCC, hereditary nonpolyposis colorectal cancer.

From Rex DK, Johnson DA, Lieberman DA, et al. Colorectal cancer prevention 2000: screening recommendations of the American College of Gastroenterology. American College of Gastroenterology. *Am J Gastroenterol* 2000;95:868–877; Winawer S, Fletcher R, Rex D, et al. Colorectal cancer screening and surveillance: clinical guidelines and rationale-Update based on new evidence. *Gastroenterology* 2003;124:544–560.

rectum, sigmoid colon, and descending colon to the splenic flexure. Flexible sigmoidoscopy should not be considered as a single screening measure but requires coupling with barium enema. Barium enema allows visualization of the entire colon, and experience is necessary to ensure proper visualization of the rectum. Barium enema affords advantages of ease of preparation, lack of conscious sedation and hemodynamic monitoring, and ability to visualize polyps and masses. However, small polyps may be missed. Furthermore, if a luminal polyp or mass is identified, then colonoscopy will be necessary for polypectomy or biopsies.

New noninvasive technologies, such as CT and magnetic resonance colonography, are receiving increased attention in clinical studies, which demonstrate overall feasibility, as well as some advantages.[146]

Two meta-analyses published in 2011 provide strong support for the implementation of CT colonography as a viable alternative to optical colonoscopy in both average and high-risk populations. In a review of 4,086 asymptomatic patients, de Haan et al. [147] estimates sensitivities of 82.9% and 87.9% and specificities of 91.4% and 97.6% for adenomas ≥6 mm and ≥10 mm, respectively.

In a complementary analysis looking exclusively at cancer detection, Pickhardt et al.[148] concludes that CT colonography is not only clinically equivalent to colonoscopy but perhaps even more suitable for initial investigation given consistently high sensitivity (96.1%) without heterogeneity across 49 studies and 11,151 patients, despite wide variation in technique.

Other reports suggest advantages in long-term costs and patient compliance, although these issues remain controversial.[149,150]

Lastly, CT colonography may also offer improvements in preoperative staging as one study found this technique to be highly predictive of T3-4 tumors. Whether this information will prove as clinically relevant in colon cancer as it is for the rectum remains to be seen.[151]

Another alternative to optical colonoscopy, which may be more acceptable to patients even than CT colonography (CTC), is the colon capsule, Pill-Cam Colon 2, now in its second generation, and reported to have a sensitivity of almost 90% for the detection of significant lesions. While a bowel preparation is still necessary, this technique required no intubation, insufflation, or sedation, and has minimal complication rates. In a study of patients unwilling or unable to undergo colonoscopy, Negreanu et al. reported the capsule to be effective in detecting clinically relevant lesions with a very high acceptability rate.[152] In another study, which compared CTC and the capsule to colonoscopy, Rondonotti et al. reported nearly equivalent accuracy rates for both modalities in patients with a positive fecal occult blood test and polyps >6 mm. Moreover, 78% of patients preferred the capsule over CTC or colonoscopy.[153] Further research is necessary to address issues of long-term benefit, cost, and efficacy in average risk individuals.

STAGING AND PROGNOSIS OF COLORECTAL CANCER

This discussion will focus primarily on those prognostic and predictive indicators that are best supported by available data and are appropriate for use and consideration in current practice. The reader should remain aware of the potential for rapid changes and advances in this area, however.

Staging

Although many factors have been identified that have an impact on recurrence and survival, none exceeds stage in terms of prognostic significance.[154] Staging of CRC should be done using the current TNM (tumor, node, metastasis) classification of the American Joint Committee on Cancer (AJCC)/International Union Against Cancer (UICC) staging system (Table 40.7).[155] Other systems should be regarded as of historical significance only and must be comprehended solely for the purposes of understanding the studies that were performed and reported in the past using these older classifications.

TABLE 40.7

Tumor (T) Node (N) Metastasis (M) Classification of Colorectal Cancer

Stage	T	N	M
0	Tis	N0	M0
I	T1	N0	M0
	T2	N0	M0
IIA	T3	N0	M0
IIB	T4a	N0	M0
IIC	T4b	N0	M0
IIIA	T1-T2	N1-N2a	M0
IIIB	T3-T4a	N1	M0
	T2-T3	N2a	M0
	T1-T2	N2b	M0
IIIC	T4a	N2a	M0
	T3-T4a	N2b	M0
	T4b	N1-N2	M0
IVA	T any	N any	M1a
IVB	T any	N any	M1b

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The Dukes Classification and Its Modifications

In the 1930s, Cuthbert Dukes, a Scottish pathologist working predominantly on a classification scheme for rectal cancer, developed the classification system that bears his name. The system, and the several modifications to it made by Dukes and others, is at this time of historical interest only, and the reader is referred to chapters in earlier editions of this book for further details.

Tumor, Node, Metastasis Classification

The current AJCC/UICC staging system for CRC is now the only classification system that should be used.[155] The TNM system classifies colorectal tumors on the basis of the invasiveness (not size) of the primary (T stage), the number (not size or location) of local-regional LNs containing metastatic cancer (N stage), and the presence or absence of distant metastatic disease (M stage) (see Table 40.7).

T Stage: A designation of Tx refers to the inability to describe the extent of tumor invasion due to incomplete information. In situ adenocarcinoma (Tis) includes cancers confined to the glandular basement membrane or lamina propria. The terms *high-grade dysplasia* and *severe dysplasia* are synonymous with in situ carcinoma and are also classified as Tis. T1 tumors invade into but not through the submucosa. T2 tumors invade into but not through the muscularis propria, and T3 tumors invade through the muscularis propria into the subserosa or into nonperitonealized pericolic or perirectal tissue. T4 tumors perforate the visceral peritoneum (T4a) or invade other named organs or structures (T4b). Tumors invading other colorectal segments by way of the serosa (i.e., carcinoma of

the cecum invading the sigmoid) are classified as T4b. A tumor that is adherent to other structures or organs macroscopically is classified clinically as T4b; however, if the microscopic examination of the adhesions is negative, then the pathologic classification is pT3. The V and L substaging should be used to identify the presence or absence of vascular or lymphatic invasion. The "p" prefix denotes pathologic (rather than clinical) assessment, and the "y" prefix is attached to those tumors that are being reported after neoadjuvant (presurgical) treatment. For example, the pathologic T stage of a tumor showing only penetration into the submucosa after preoperative therapy would be ypT1. Recurrent tumors are reported with an "r" prefix (rpT3).

N Stage: Because of the prognostic significance associated with increased numbers of LNs inspected (see the following discussion), the current TNM classification scheme calls for at least 12 LNs to be analyzed, and both the number of nodes that are positive for tumor and the total number of nodes inspected should be reported. The term Nx is applied if no description of LN involvement is possible because of incomplete information. A pN0 designation may be made even if fewer than the recommended number of nodes are present; however, the prognostic significance of this pN0 designation is weaker. N0 denotes that all nodes examined are negative. N1a includes tumors with metastasis in one regional LN. N1b refers to involvement of two or three nearby LNs. N1c defines the presence of cancer cells found in areas of fat near LNs but not in the LNs themselves. N2a indicates metastasis in four to six regional LNs. N2b denotes involvement of greater than seven nodes. Metastatic nodules or foci found in the pericolic, perirectal, or adjacent mesentery without evidence of residual LN tissue are regarded as being equivalent to a regional node metastasis and are counted accordingly.

Stage I disease is defined as T1-2N0 in a patient without distant metastases (M0). Stage II disease is defined as T3-4N0M0. The T-stage carries prognostic significance for stage II, and therefore T3N0 is classified as IIA and T4a-bN0 is classified as IIB and IIC, respectively.

Node positivity in the absence of M1 disease defines stage III CRC. More recently, the prognostic significance of tumor invasiveness (T stage) has been reincorporated into the assessment of risk in stage III patients. In an exhaustive review of over 50,000 patients, Greene et al.[156] demonstrated the prognostic significance of T stage within node-positive patients. Within the N1 category T stage was found to be highly prognostic, with T1-2 patients fairing significantly better than T3-4. Within the N2 population, the prognosis was worse than either subgroup of N1 patients, with T stage no longer carrying prognostic significance. Thus T stage is prognostic in patients with N0 and N1 but not N2 disease. The current TNM staging system takes these findings into account and now stratifies stage III patients into IIIA (T1-2N1), IIIB (T3-4N1), and IIIC (T any, N2, or T4, N1-2). Stages IIIA, B, and C are highly prognostic for survival.

M Stage: Patients are designated M0 if no evidence of distant metastases is present. Identification of distant metastases denotes a classification of M1. Involvement of the external iliac, common iliac, para-aortic, supraclavicular, or other nonregional LNs is classified as distant metastatic (M1) disease. The M1 category is subdivided into M1a, defined as spread of tumor to one distant organ or set of distant LNs, and M1b, where spread has occurred to more than one distant organ or sets of LNs. M1c is spread that has occurred to distant parts of the peritoneum. Although the TNM staging system is regarded as the most comprehensive tool for prognostic and predictive purposes, a major criticism of the last two revisions is that survival of stage IIIA patients continues to be superior to stage IIB. This disparity, which is actually more pronounced in the eighth edition of AJCC manual, has been attributed to inadequate LN assessment and understaging. However, a recent review of SEER data showed this problem persists even in a subset analysis of patients with >12 LNs, highlighting the need for additional refinement and perhaps the incorporation of nonanatomic prognostic factors.[157]

Residual Tumor (R Stage) at Margins of Resection: Tumors that are completely resected with histologically negative margins are classified as R0. Tumors with a complete gross resection but with microscopically positive margins are classified as R1, the positive margin indicating that at least microscopic tumor remains in the patient. Patients who have incomplete resections with grossly positive margins are classified as having had an R2 resection. The R0, R1, and R2 designations carry strong prognostic implications.

Identification of the proximal and distal margins of resection is relatively straightforward, and definitions of these margins are well understood. A more complex and often misunderstood (as well as underreported) margin of resection is the circumferential radial margin (CRM). All three margins (proximal, distal, and CRM) should be specifically commented upon in the pathology report, as all three have prognostic significance.

The CRM is, by definition, a surgically dissected surface. It is defined as the cut retroperitoneal or perineal soft tissue margin closest to the deepest penetration of tumor. It is considered positive if tumor is present microscopically (R1) or macroscopically (R2) on a *cut* radial or lateral aspect of the surgical specimen. For the ascending colon, descending colon, and upper rectum, which are incompletely encased by peritoneum, the CRM is created by dissection of the retroperitoneal aspect of the bowel. In the case of the lower rectum, which is not encased by peritoneum, the CRM is created by sharp dissection of the mesorectum.

A tumor simply penetrating into pericolonic or perirectal fat does not necessarily constitute a positive CRM, but rather is simply a description of a T3 primary. A tumor that involves a peritonealized surface of the bowel and not a surgically cut surface does not constitute a positive CRM, but rather constitutes a T4a primary. If, however, the cut surface at the deepest penetration of the tumor is positive, then the CRM is positive and the resection is staged R1 (microscopic) or R2 (macroscopic). A positive CRM is highly predictive of local recurrence and should prompt consideration of adjuvant treatment and clinical trial risk stratification as well.

Prognosis

Histologic Grade

Although histologic grade has been shown to have prognostic significance, there is significant subjectivity involved in scoring of this variable, and no one set of criteria for determination of grade is universally accepted.[154] The majority of staging systems divide tumors into grade 1 (well differentiated), grade 2 (moderately differentiated), grade 3 (poorly differentiated), and grade 4 (undifferentiated). Many studies collapse this into low grade (well to moderately differentiated) and high grade (poorly differentiated or undifferentiated). Greene et al.[156] demonstrated that this two-tiered split has important prognostic significance.

College of American Pathologists Consensus Statement: The College of American Pathologists (CAP) has published an expert panel consensus statement outlining their interpretation of the validity and usefulness of a large number of putatively prognostic and predictive factors in CRC.[158] Variables were categorized as belonging to categories I through IV. Category I was defined as those factors proven to be of prognostic import based on evidence from multiple, statistically robust, published trials and generally used in patient management. Category IIA included factors intensively studied biologically or clinically and repeatedly shown to have prognostic value for outcome or predictive value for therapy that is of sufficient import to be included in the pathology report but that remains to be validated in statistically robust studies. Category IIB included factors shown to be promising in multiple studies

but lacking sufficient data for inclusion in category I or IIA. Category III included factors felt to be not yet sufficiently studied to determine their prognostic value, and category IV included those factors that are adequately studied to have convincingly shown no prognostic significance. A number of these factors are discussed in further detail in the following.

The T, N, and M categories of the current AJCC/UICC staging system were all classified as category I. Other category I inclusions were blood or lymphatic vessel invasion and residual tumor following surgery with curative intent (the R category). Although not assessed pathologically, an elevation of the preoperative CEA level was also felt to merit category I inclusion. Factors in category IIA included tumor grade, radial margin status (for resection of specimens with nonperitonealized surfaces), and residual tumor in the resection specimen following neoadjuvant therapy. Factors in category IIB (many of which are discussed in further detail in the following) included histologic type, histologic features associated with MSI (i.e., host lymphoid response to tumor and medullary or mucinous histologic type), high degree of MSI (MSI-H), loss of heterozygosity (LOH) of 18q (*DCC* [deleted in colon cancer] gene loss), and tumor border configuration (infiltrating vs pushing border). Factors grouped in category III included DNA content, all other molecular markers except for LOH of 18q/DCC and MSI-H, perineural invasion, microvessel density, tumor cell–associated proteins or carbohydrates, peritumoral fibrosis, peritumoral inflammatory response, focal neuroendocrine differentiation, nuclear organizing regions, and proliferation. Those factors in category IV (proven to be of no significance) included tumor size and gross tumor configuration.

Total Number of Lymph Nodes

It has been well established that an adequate number of LNs must be sampled before a patient can be considered node negative, and careful pathologic technique has been demonstrated to be crucial to adequate nodal interpretation. Failure to adequately dissect and display the mesentery will lead to underreporting and understaging.[159,160] It should be noted that an insufficient number of LNs reported could be due to a suboptimal nodal dissection at operation, a less than thorough search for nodes by the pathologist, or some combination of the two. Additional patient- and tumor-related factors may also affect LN count independent of pathologist or surgeon. Belt et al.[161] found a significant association between MSI phenotype and high LN yield in both stage II and III colon cancers, with the strongest effect in the latter group. The authors postulate that this may be due to a more prominent lymphocytic antitumor response known to be exhibited by MSI-H cancers.[161] Another report suggests that low body mass index is associated with increased LN yield, although it did not affect relapse-free or overall survival in stage III cancers. Proximal tumor location, well- or moderately differentiated histology, and stage IIIC cancer were also significant variables for adequate LN recovery.[162] Finally, a multivariate analysis of two large prospective US cohort databases (121,701 women and 51,529 men) demonstrated that specimen length, tumor size, ascending tumor location, T3N0M0 stage, and year of diagnosis were positively associated with negative node count (P <0.002). Mutation of KRAS was borderline significant and requires further study. The authors recommend that these variables be taken into account when judging adequacy of LN harvest and devising individualized treatment plans in the future.[163] An analysis was reported on outcome versus nodal sampling in the patients who participated in an Intergroup trial (INT-0089), a large four-arm trial of different 5-fluorouracil (5-FU)–based adjuvant chemotherapies in patients with colon cancer. Multivariate analyses were performed on the node-positive (2,768 patients) and node-negative (648 patients) groups separately. The median number of LNs reported in the assessable patients on this trial was 11 (range, 1 to 87). Survival (overall, cancer-specific, and disease-free [DFS]) was found to decrease with an increasing number of involved LNs (P = 0.0001 for all three survival end points). However, after controlling for the number of involved nodes, survival increased with the total number of nodes (positive plus negative) reported (P = 0.0001 for overall survival, cancer-specific survival, and DFS). Even in patients who were node negative, overall survival (P = 0.0005) and cancer-specific survival (P = 0.007) were significantly increased as the number of reported LNs increased.

In a different secondary analysis of the Intergroup trial (INT-0089), a mathematical model was created to estimate the probability of a true node-negative result on the basis of the number of LNs examined in a subset of patients who had at least 10 LNs reported in their resection specimen.[164] A total of 1,585 patients with stage III or high-risk stage II colon cancer were evaluated. This model concluded that when 18 nodes are examined, there is a <25% probability of true node negativity in T1 and T2 tumors. However, examination of <10 LNs was needed in T3 and T4 tumors to achieve the same probability. The overall conclusions of this analysis were that a very significant proportion of patients are understaged and that such understaging could have important implications for decisions regarding adjuvant therapy and for overall prognosis.

The CAP consensus statement suggests that a minimum of 12 to 15 LNs should be examined in order to determine node negativity.[158] Availability of fewer nodes should therefore be regarded as a relative high-risk factor in terms of prognosis and should be factored into decisions regarding adjuvant therapy. Further support for this recommendation comes from a newly published Danish cohort study that indicates that the advantage of larger LN harvest extends beyond more accurate staging. In addition to improved outcomes for node-negative patients, the authors found a significant increase in overall survival for stage III patients with >12 LNs removed as well (58.6% vs 45.2% for <12 LNs), despite a higher prevalence of N2 disease in this group. This may be related to better surgical technique or an underlying benefit of wider lymphadenectomy in general. LN ratio was also shown to be an important independent prognostic indicator and, in fact, superior to N-stage in predicting survival for stage III patients. This finding is consistent with a number of previous reports, many of which have advocated for incorporation of this parameter into the AJCC staging system.[165–169]

Microscopic Nodal Metastases

The advent of improved pathologic techniques and sensitive methods such as IHC or polymerase chain reaction may have an impact on the number of positive LNs detected and may have important prognostic significance.[170,171] However, the prognostic value of these positive LNs, which otherwise would not be detected, remains controversial. In a recent review of 16 studies with survival data, Sirop et al.[172] found only 8 papers that reported definitely poorer outcomes, whereas the remainder were either equivocal in their conclusions or demonstrated no influence on outcome at all. Jeffers et al.[173] evaluated LNs from 77 patients who were found to have negative LNs by routine examination with immunocytochemical staining for cytokeratin AE1:AE3. Nineteen patients (25%) were found to have immunohistochemical evidence of micrometastases; however, there was no difference in survival between the microscopically positive and negative patients. A larger trial by Faerden et al.,[174] on the other hand, did demonstrate adverse prognostic impact. In this study, 39 of 126 patients with stage I/II colon cancer were noted to have micrometastases or isolated tumor cells (MM/ITC+) on IHC staining. Prospective median 5-year follow-up of MM/ITC+ compared to MM/ITC- patients revealed recurrence rates of 23% versus 7% (P = 0.010) and 5-year DFS of 75% versus 93% (P = 0.012), respectively.[174] If micrometastases are reported, the methodology by which they are detected should be specified, as it is likely that differences in reliability and reproducibility of different techniques will emerge. Although the actual TNM staging is not altered by the presence of micrometastases, many clinicians choose to regard the presence of

such a finding as a poor prognostic variable in their consideration of adjuvant treatment.

Sentinel Node Analysis

Sentinel node analysis is an approach that has received attention in the management of cutaneous melanoma and breast cancer.[175,176] This technique has been proposed as a means of increasing the yield and the diagnostic information for colon cancer.[137,138] The technique for sentinel node mapping and biopsy for colon cancer has been described by Saha et al.[177] Unlike sentinel node approaches for melanoma and breast, where the goal is to potentially limit the extent of an unnecessary formal dissection of a node basin, the goal of the sentinel node in colon cancer is to focus the pathologic analysis on fewer nodes so a more extensive study can be performed. The same extent of node dissection is performed regardless of the sentinel node procedure. The initial studies of sentinel node biopsy demonstrated it was technically feasible, with accuracy rates >80% and upstaging in 15.4% of patients according to a recent prospective trial.[178–180] In addition, Saha et al.[181] suggest that sentinel LN mapping may not just improve staging accuracy but influence the extent of nodal dissection as well. In this study, sentinel LN mapping detected aberrant lymphatic drainage in 22%, which in turn led to a change in operation (i.e., more extensive resection). In two patients, the aberrant sentinel nodes were the only positive nodes identified.[181] However, not all subsequent studies have shown positive results. False-negative rates as high as 60% have been reported, and some studies have failed to demonstrate any change in the stage determination of the lesion.[182] Based on the available data, two conclusions can be reached. First, from a technical standpoint, sentinel node dissection at the time of a colon resection can be performed and the sentinel node accurately identified. Second, the utility of this technique has not yet been established and further large-scale trials are required to establish its role in the staging of patients with CRC.

Blood or Lymphatic Vessel Invasion

Although there have been conflicting reports in the literature, the CAP consensus statement gave blood and lymphatic vessel invasion category I status, indicating that the preponderance of evidence strongly supports the reliability of these findings as indicators of poorer prognosis.[158] Unfortunately, considerable heterogeneity exists in the methodology for examining and reporting of vessel involvement. The finding of vessel involvement increases with the number of sections examined, and differentiation of postcapillary venules from lymphatics is often not possible. These aspects can make interpretation of some older data on this topic potentially problematic. Current recommendations are that at least three blocks of tumor (optimally five or more) each have a single section examined using hematoxylin and eosin stain to look for tumor invasion of vessels. Vessels not definitively interpreted as venules or lymphatics should be reported as angiolymphatic vessels.

Histologic Type

Several histologic types of CRC carry specific independent prognostic significance. Signet ring carcinomas are characterized by >50% of cells demonstrating the "signet ring" morphology in which intracellular mucin accumulation displaces the nuclei and cytoplasm toward the cellular periphery. This histology carries an adverse prognosis.[183,184] The prognostic significance of the finding of mucinous (>50% mucinous) carcinoma remains controversial. Although some reports list mucinous type as an adverse histology, this has not been consistently demonstrated. Most findings of adverse prognosis with mucinous histology are based on univariate analyses. The one finding in a multivariate analysis of a poor prognostic outcome with mucinous tumors was based on a study of tumors presenting with obstruction, a presentation that is in itself high risk. Some reports have lumped mucinous and signet cell tumors together and found this to be a negative prognostic factor; however, this may simply reflect the negative impact of the signet cell tumors, and its meaning regarding the risk of a mucinous histology is unclear. Small cell (extrapulmonary oat cell) tumors are high-grade neuroendocrine tumors with clearly adverse prognostic features. The prognostic significance of focal neuroendocrine differentiation is, however, unclear (CAP category III). Most data indicate that extensive neuroendocrine differentiation is associated with a poorer prognosis.[185] Medullary carcinoma is a subtype characterized by an absence of glands and distinctive growth pattern that previously would have been classified as undifferentiated. It is typically infiltrated with lymphocytes. This histologic subtype is tightly associated with MSI-H and carries a more favorable prognosis.[186] Histologic types other than signet ring, small cell, and medullary carcinomas are routinely designated in the pathology report; however, the majority of these other histologic types carry no established independent prognostic significance.

Microsatellite Instability

As discussed earlier in this chapter, there are two distinct mutational pathways that can give rise to CRC: the MSI pathway or the chromosomal instability pathway. Microsatellites are sections of DNA in which a short sequence of nucleotides (most commonly a dinucleotide) is repeated multiple times.[187] MSI is a situation in which a microsatellite has gained or lost repeat units and so has undergone a change in length, resulting in frame shift mutations or base-pair substitutions. Approximately 15% of CRCs display these mutations. This form of genetic destabilization is typically associated with defective DNA mismatch repair function. Studies of HNPCC tumor specimens demonstrated mutations in mismatch repair genes such as *MLH1*, *MSH*, *MSH6*, and *PMS2*. These genes encode proteins that repair nucleotide mismatches. The phenotype of tumors with this defect is termed the MSI-H–instability phenotype.

The majority (approximately 85%) of patients with CRC have cancers characteristic of the chromosomal instability pathway, typically having genetic alterations involving LOH, chromosomal amplifications, and chromosomal translocations. These are known as the microsatellite-stable (MSS) tumors. MSI-H tumors have a number of different features relative to low MSI (MSI-L) or MSS colorectal tumors.[188,189] MSI-L and MSS tumors tend to behave and present similarly. MSI-H tumors are more frequently right-sided, high grade, and mucinous type.[62,190] They are characteristically associated with increased peritumoral lymphocytic infiltration and are characteristically diploid, whereas MSS tumors are more likely to be aneuploid.[191,192] MSI-H CRCs are more likely to have a larger primary at the time of diagnosis but are more likely to be node negative. Patients with MSI-H CRCs have a better long-term prognosis than stage-matched patients with cancers exhibiting MSS.[193]

Watanabe et al.[194] evaluated MSI status as well as allelic loss from chromosomes 18q, 17b, and 8p, as well as cellular levels of p53 and $p21^{waf1/cip1}$ proteins as potential prognostic markers. Tumors were analyzed from 460 stage III and high-risk stage II patients who had been treated with 5-FU–based adjuvant therapy. A total of 62 of 298 tumors evaluated for MSI status (21%) were found to be MSI-H. Of the MSI-H tumors, 38 (61%) had a mutation of the gene for type II receptor of transforming growth factor (TGF)-β1. In this analysis, MSI-H was a favorable prognostic indicator for 5-year DFS ($P = 0.02$) and trended toward being a favorable independent prognostic indicator but did not reach statistical significance for overall survival ($P = 0.20$). However, the 5-year survival among patients with MSI-H was 74% in the presence of a mutated gene for the type II receptor of TGF-β1 and 46% in patients whose tumors lacked this mutation (RR = 2.90; 95% CI = 1.14 to 7.34; $P = 0.04$). MSI-H cells are relatively resistant to 5-FU in vitro.[195] All of the patients in Watanabe et al.'s[194] analysis received 5-FU–based chemotherapy. The TGF-β1 pathway inhibits tumor proliferation by causing a late G_1 cell cycle arrest.

Therefore, a mutated and presumably nonfunctional *TGF-β1* gene could favor increased proliferation, which would be anticipated to confer increased susceptibility to cytotoxic chemotherapy. A recent evaluation of the prognostic significance of MSI in the N0147 adjuvant trial[196] demonstrated a more nuanced result. When looking at the colon overall, MSI was not found to be predictive. However, when divided by side, MSI was found to carry a favorable prognosis for right-sided colon lesions, but was a negative prognostic factor in left-sided colon lesions. The reasons for this difference is not clear; however, the different embryologic origins of the left and right colon may play a role in these observations.

BRAF

BRAF mutation, present in 8% to 10% of CRCs, is linked to a subset of MSI-H tumors that are sporadic and generally have poorer prognosis. Ogino et al.[197] confirmed this relationship in comparative analysis of 506 stage III patients enrolled in the Cancer and Leukemia Group B (CALGB) 89803 trial. BRAF-mutated patients had significantly worse overall survival (HR = 1.66) compared to wild type, a finding that was most pronounced in the setting of MSS.[197] In a follow-up study, Lochhead et al.[198] also identified combined BRAF/MSI status as a powerful prognosticator and recommends stratification of all patients into poor (MSS/BRAF mutant), intermediate (MSS/BRAF wild-type), and favorable (MSI-H/BRAF wild-type) groups in order better inform treatment strategies. Douillard et al.[199] confirmed BRAF mutation to be a poor prognostic factor in patients with stage IV disease as well.

CpG Island Methylator Phenotype (CIMP)

Approximately 15% to 20% of colorectal cancers are characterized by widespread promoter DNA hypermethylation, known as CIMP, which is associated with aberrant silencing of tumor suppressor genes. While CIMP positive tumors are generally associated with MSI, BRAF mutations, poor differentiation, proximal location, female gender, and older age, the prognostic effect has been largely unknown. In a recent retrospective analysis of 157 patients, Kim et al. found that CIMP-high status was associated with a significantly poorer 5-year disease-free survival compared to CIMP-low/none (61.4% vs 76.3%). A more marked difference was seen in patients with colon cancer, compared to rectal, and with earlier stage disease. Another study out of Norway also found CIMP to be a poor prognostic indicator, with particularly worse outcomes among patients with MSS or MSS/BRAF mutated tumors.[200,201]

Allelic Loss of 18q (DCC Gene Loss)

Allelic LOH that involves chromosome 18q occurs in half or more of all CRCs. Allelic loss of 18q typically involves the *DCC* gene; however, other genes in this region, such as Smad2 and Smad4, may also be relevant to CRC development. DCC expression is greatly reduced or absent in many colorectal carcinomas, and loss of DCC is associated with metastasis and an adverse prognosis.[202] The specific product of the DCC gene has been shown to be the netrin-1 receptor. In the nonpathologic state, this receptor guides the migration of neuronal axons. DCC induces apoptosis in the absence of netrin-1 binding. DCC is cleaved by caspase, and mutation of the site at which caspase 3 cleaves DCC suppresses the proapoptotic effect of DCC completely. Binding of netrin-1 to DCC blocks apoptosis.[203] Loss of DCC as a result of allelic loss in 18q could therefore be anticipated to impair apoptosis, thereby resulting in greater resistance to chemotherapy. This hypothesized mechanism of action of 18q LOH is attractive; however, it should be emphasized that it is not at all clear to what extent DCC is the active moiety in the setting of 18q allelic loss. Watanabe et al.[194] evaluated allelic loss from chromosome 18q as a potential prognostic indicator in archived specimens of tumors from patients who were treated in one of two national Intergroup adjuvant trials (INT 0035 or INT 0089). MSI status was also evaluated, as were 17p, 8p, and cellular levels of p53 and $p21^{waf1/c1p1}$ proteins. Tumors were analyzed from 460 stage III and high-risk stage II patients who had been treated with 5-FU–based adjuvant therapy. Allelic loss of 18q was present in 155 of 319 cancers (49%). Allelic loss in 18q was highly prognostically significant in this analysis (Table 40.8). In the stage III patients with allelic loss of 18q, 5-year overall survival was 50%, while in those with retained 18q alleles, 5-year survival was 69% ($P = 0.005$). Other markers evaluated in this analysis were not shown to be prognostically significant.

TABLE 40.8

Loss of Heterozygosity (Allelic Loss) at 18q and Prognosis in Patients with Stage III Colon Cancer

Allelic Status of 18q	No. of Patients	Five-Y Survival (%)	*P* Value
No loss	112	69	0.005
Loss	109	50	

Data from Watanabe T, Wu TT, Catalano PJ, et al. Molecular predictors of survival after adjuvant chemotherapy for colon cancer. *N Engl J Med* 2001;344:1196–1206.

SMAD4: SMAD4, which is localized to band 18q21, is a common downstream regulator and tumor suppressor gene in the TFG-β pathway. Sporadic mutations are present in 2.1% to 20% of colorectal cancers, with limited data suggesting a poor prognostic impact. In a new study using the Cancer Genome Atlas database, investigators report that SMAD4 mutation was associated with colon cancer more than rectal cancer (OR 2.85), as well as with female sex (OR 1.71) and with shorter overall survival compared to wildtype (29 vs 56 mo). While this mutation frequently occurred with KRAS, NRAS, and BRAF mutations, it was also separately associated with poorer response to anti-EGFR receptor therapy in the setting of wildtype KRAS, NRAS, and BRAF.[204]

Host Lymphoid Response

Lymphocytic infiltration has been identified as a favorable prognostic indicator. Whether this is a truly independent predictor of outcome is not clear, however, as this finding is tightly associated with MSI-H, a favorable prognostic factor. Along these lines, the prognostic value of neutrophil-to-lymphocyte ratio has also been recently evaluated. Chiang et al.[205] found that elevated preoperative neutrophil-to-lymphocyte ratio (>3) was associated with significantly worse DFS in stage I to III colon but not rectal cancers on multivariate analysis. In another study, neutrophil-to-lymphocyte ratio (>5) was also found to be an independent risk factor for recurrence. While the direct impact of this parameter is difficult to explain, the authors of the first study postulate that it may represent a measure of innate-to-adaptive immunity under stress with relative lymphopenia, as a marker of depressed cell-mediated immunity, conferring survival disadvantage.[205,206]

Tumor Border Configuration

The configuration of the tumor border (infiltrating vs pushing border) has been shown to have independent prognostic significance. An infiltrating border, characterized by an irregular, infiltrating pattern at the tumor edge (also known as focal dedifferentiation or tumor budding), has been shown in multivariate analyses to portend a poorer prognosis than tumors with smooth, pushing borders.

Carcinoembryonic Antigen

An elevated preoperative CEA is a poor prognostic factor for cancer recurrence. Although there is variability in the available data regarding the level that denotes a prognostic cutoff, a preoperative CEA level >5 ng/ml is considered a category I poor prognostic indicator by

the CAP consensus panel.[158] Patients in whom the elevated CEA fails to normalize after a potentially curative operation are at particularly high risk. Several authors have presented evidence that indicates that CEA is an independent prognostic factor. In a report of 572 patients who underwent curative resection for node-negative colon cancer, the preoperative CEA level and the stage of disease predicted survival by both univariate and multivariate analyses.[207] Given the prognostic significance of the preoperative CEA, it is reasonable to recommend that all patients who undergo operation for CRC have a serum CEA drawn prior to operation.

No other serum markers have been demonstrated to be reliably prognostic or predictive in CRC. Cancer antigen (CA) 19-9, a factor that has become widely used for pancreas cancer, has no role at this time in the routine management of CRC.

Obstruction and Perforation

Carcinoma of the colon that is complicated by obstruction or perforation has been recognized as having a poorer prognosis. Data obtained from 1,021 patients with Dukes stage B and C CRC, who were entered into randomized clinical trials of the National Surgical Adjuvant Breast and Bowel Project (NSABP) showed that the presence of bowel obstruction strongly influenced the outcome. The effect of bowel obstruction was more pronounced when the obstruction was located in the right colon. The larger-sized tumor needed to block the ascending colon completely might allow a longer time for these tumors to grow and spread when compared with tumors located in the descending colon.

A review of the Massachusetts General Hospital records compared patients who presented with obstruction or perforation with a control group who underwent curative resection. The actuarial 5-year survival rate seen in patients who presented with obstruction was 31%, in contrast to 59% in historical controls. For patients with localized perforation, the 5-year actuarial survival rate was 44%. The Gastrointestinal Tumor Study Group (GITSG) multivariate analysis concluded that obstruction was an important indicator of prognosis, independent of Dukes stage. Bowel perforation was a poor prognostic factor only for DFS.

Category III Factors

Multiple factors, while of investigational interest, are at this time not appropriate for routine clinical use and have so been designated as category III (defined as not sufficiently studied to prove their prognostic value) by the CAP consensus panel. These include DNA content, or ploidy, and proliferation indices. Also included in category III are all molecular markers other than MSI and 18q deletions, such as thymidylate synthase (TS), dihydropyrimidine dehydrogenase (DPD), and p53 mutational status. Perineural invasion, microvessel density, tumor cell–associated proteins or carbohydrates, peritumoral fibrosis, peritumoral inflammatory response, and focal neuroendocrine differentiation are also category III. The area of molecular prognostic markers is one of particular activity, however, and it is anticipated that clinical trials that are now ongoing will shed light on these important areas.

Perineural Invasion

The ability of CRCs to invade perineural spaces as far as 10 cm from the primary tumor has long been described. Early reports suggest an increased disease recurrence rate and worse 5-year survival. Multivariate analyses have failed to show the prognostic significance of this finding. The CAP consensus panel classified perineural invasion as category III (insufficient evidence of determine prognostic significance).

Tumor Size and Configuration

Studies have consistently shown that both the size and configuration of the primary tumor in CRC do not carry prognostic significance (CAP category IV). In a review of 391 patients, the mean diameter of Dukes stage B2 tumors was actually greater than the mean diameter of stage C2 tumors ($P < 0.001$) and D tumors ($P < 0.05$). The size of the primary tumor showed no relationship to 5-year adjusted survival. These results were confirmed by the NSABP experience.[208] Tumor configuration is described as exophytic (fungating), endophytic (ulcerative), diffusely infiltrative (linitis plastica), or annular. The vast majority of studies have failed to show any of these configurations to have consistent independent prognostic significance. Linitis plastica has been related to a poor prognosis; however, this may be due to the signet cell and other high-grade features of the tumors that are typically associated with this morphology.

Hemorrhage or Rectal Bleeding

It has been speculated that tumors that present with bleeding might be found earlier and therefore might be associated with a better prognosis. This has not been confirmed by data. In the GITSG multivariate analysis, the presence of melena or rectal bleeding showed a trend as a prognostic factor for prolonged survival but failed to reach statistical significance ($P = 0.08$). One large study found bleeding to be a favorable prognostic indicator on univariate analysis; however, this finding disappeared on multivariate analysis. Bleeding at presentation does not appear to carry any significance.

Primary Tumor Location

Large retrospective reviews of data from the NSABP suggest that right-sided colon cancers carry a worse prognosis than left-sided ones. However, poorer prognosis for patients with disease in the left colon has also been reported. Several investigators report no difference based on the location of the primary tumor. The large GITSG colon cancer experience showed that tumor location (left, right, and rectosigmoid or sigmoid) was of low prognostic value. A recent analysis of SEER-Medicare data by Weiss et al.[209] provides additionally ambiguous results. Of 53,801 patients, 67% had right-sided colon cancer and were more likely to be older, women, and diagnosed with more advanced stage and with more poorly differentiated tumors. However, on multivariate analysis, there was no significant difference in mortality for all stages combined or for stage I. Compared to left-sided lesions, right-sided cancers were associated with a lower mortality within the stage II subgroup (HR = 0.92; $P = 0.001$) but higher mortality within stage III (HR = 1.12; $P = 0.001$). Critics of this report point out that a less aggressive treatment approach was likely employed in this older study population, as at least 40% of stage III cases did not receive adjuvant therapy and nearly half underwent inadequate LN harvest. Regardless, these results further dispel the notion of a straightforward relationship between tumor location and mortality.[209]

Body Mass Index

While obesity is known to be a risk factor for the development of colon cancer, the prognostic impact of body mass index on long-term outcomes is controversial. In a cohort study conducted within a large randomized trial of 3,759 patients with high-risk stage II or III colon cancer (INT-0089), obese women had significantly worse overall mortality (HR = 1.34; 95% CI = 1.07 to 1.67); however, this finding was not apparent in men.[210] Sinicrope et al.[211] found the opposite gender correlation using the ACCENT database, a pooled resource of 25,291 participants in national and international adjuvant chemotherapy trials. On multivariate analysis, with a median follow-up of 7.8 years, obese and underweight men, but not women, had significantly poorer survival compared to overweight and normal weight patients.[211] And in another prospective cohort of 913 patients with stage II and III colon cancer, Alipour et al.[212] found no association between obesity (as measured by either body mass index or body surface area) and oncologic

outcomes. Evidently, this topic warrants further study before any conclusions can be drawn.[212]

Diabetes Mellitus

The influence of diabetes mellitus on outcome is also unclear. In the INT-0089 cohort, diabetes conferred a strong disadvantage with affected patients experiencing a significantly worse DFS (48% vs 59%; P <0.0001), overall survival (57% vs 66%; P <0.0001), and recurrence-free survival (56% vs 64%; P = 0.012) at 5 years. Median survival for diabetics was 6 years, whereas for nondiabetics it was 11.3 years.[210] Other reports, however, have generated less consistent results. Among 2,278 subjects from the Cancer Prevention Study-II Nutrition Cohort, patients with CRC and type 2 diabetes were at higher risk of all-cause mortality (ACM; RR = 1.53), but only those without insulin use were at higher risk for CRC-specific mortality. These results are in line with previous evidence that hyperinsulinemia (as in poorly controlled diabetes) plays an important role in tumorigenesis and metastasis of CRC.[213] Another population-based study did not find any such an association in 6,974 patients with colon cancer. Disease-specific mortality was only significantly increased for patients with rectal cancer (n = 3,888, 10% of whom were diabetic; HR = 1.30). While hyperinsulinemia is again implicated, the authors call for additional study to clarify specific pathways responsible for these rectum-specific findings.[214]

Gender

Female gender has generally been considered a favorable prognostic factor, but data are limited and inconclusive. In the first study to examine the impact of gender in the era of oxaliplatin-based therapy, Cheung et al.[215] performed a prospectively planned, pooled analysis of 33,345 patients participating in the ACCENT database of randomized trials. The authors found a significant but very modest survival advantage for women with early stage disease that persisted across all ages, stages, and types of adjuvant therapy. Gender was not a predictive factor for treatment efficacy, however, suggesting that chemotherapy regimens should be not be altered based on this parameter.[215]

Smoking

As discussed earlier, prolonged cigarette smoking appears to be a moderate risk factor for CRC with continued effect even after smoking cessation. Increasing evidence indicates that this association differs not just by tumor site but also by molecular features, such as the presence of MSI-H and BRAF mutations, which cumulatively seem to confer the strongest risk. Impact on survival has now also been reported in a recent study analyzing data from a large multicenter randomized adjuvant chemotherapy trial (N0147). The authors found that smokers experienced significantly shorter 3-year DFS (74% vs 70%; HR = 1.21) that was most evident in BRAF wild-type and KRAS-mutated tumors.[216]

Blood Transfusions

Considerable controversy has surrounded the question of an association between perioperative blood transfusions and the recurrence rate of CRC. Some investigators have reported worse DFS in patients who require transfusions. By multivariate analysis in a large prospective study, however, no negative influence of transfusion on survival could be detected, and it does not appear that perioperative blood transfusions carry negative prognostic value. A retrospective analysis evaluating 1,051 patients treated with curative surgery for stage II or III colorectal adenocarcinoma at the Mayo Clinic demonstrated that the use of blood components probably had no impact on disease recurrence, and the documented adverse impact of transfusions is more likely due to other variables or to the underlying illness necessitating the transfusion.[217]

Steroids

It is well established that certain perioperative factors such as anesthetic technique and drug choice can impact cell-mediated immunity and potentially long-term oncologic outcomes after cancer surgery. Dexamethasone, which is commonly used by anesthesiologists for prophylaxis against postoperative nausea and vomiting, was recently evaluated in a randomized, placebo-controlled study of 43 patients who underwent curative resection for Stage I-III colon cancer. While short-term data demonstrated an improvement in postoperative fatigue and reduction in early peritoneal inflammatory response, 5-year follow-up analysis was notable for a significantly higher rate of distant recurrence in the intervention group (6 vs 1, P = 0.04). There was no difference in overall or disease-free survival. While this is a small study, the implication that this medication, widely used in anesthetic practice, might have a strong enough anti-inflammatory and immunosuppressive effect, even in a single preoperative dose, to promote long-term tumor proliferation and metastasis is quite concerning and certainly warrants further investigation.[218]

Oncogenes and Molecular Markers

At present, none of the markers under investigation has achieved adequate validity to permit routine clinical use. However, the study of molecular markers continues to progress and continues to advance the understanding of the development and treatment of CRC. TS continues to be a major area of investigation. Data are conflicting on its prognostic significance; however, preliminary studies suggest that high TS levels may be predictive for resistance to 5-FU–based therapies.[219] At present, there is no role for TS determinations in routine clinical practice. The *p53* gene located on chromosome 17p is a well-known tumor suppressor gene. The abnormal *p53* appears to be a late phenomenon in colorectal carcinogenesis. This mutation may allow the growing tumor with multiple genetic alterations to evade cell cycle arrest and apoptosis. In a retrospective review of 141 patients with resected stage II and III colon carcinoma, a *p53* mutation increased the risk of death by 2.82 times in patients with stage II disease and by 2.39 times in patients with stage III colon carcinoma. The Southwest Oncology Group assessed the prognostic value of p53 in 66 patients with stage II and 163 stage III colon cancer. *p53* expression was found in 63% of cancers and was associated with favorable survival in stage III but not stage II disease. Seven-year survival with stage III disease was 56% with *p53* expression versus 43% with no*p53* expression (P = 0.012).[220] Overall, the data are conflicting on the utility of *p53* as a prognostic variable, and it does not have a use at this time in standard practice.

Epidermal growth factor receptor (EGFR) is an important molecular target for antibody-based therapy in various cancer types and is ubiquitous in colonic tissue. The prognostic impact of this biomarker was recently addressed in a meta-analysis demonstrating worse postoperative survival in patients with high compared to low EGFR expression (HR = 2.34).[221]

Genetic Polymorphisms

Extensive preliminary work is indicating that genetic polymorphisms can potentially have important predictive implications in terms of both efficacy and toxicity with chemotherapy. For example, the UGT1A1 polymorphism has been correlated with CPT-11 toxicity, and TS and XRCC1 polymorphisms may predict efficacy for oxaliplatin or 5-FU combinations.[222] Although a commercial assay is currently available for measurement of UGT1A1 polymorphisms, it is not, at this time, clear how, or if, this assay should be used in routine practice. Currently, there are no specific guidelines for dose modifications on the basis of UGT1A1polymorphism, and the 7/7 mutation, associated with higher toxicity, has also been associated with greater antitumor activity. These approaches will require considerable more validation and exploration before they can be considered for standard management.[223]

APPROACHES TO SURGICAL RESECTION OF COLON CANCER

The management of colon cancer is best understood as a multimodality approach tailored to the stage of disease. However, there are certain basic tenets of surgical management for the resection of the primary lesion that can be applied across various pathologic stages. Therefore, in order to provide a clear description of these techniques, they will first be described based on the type of surgical resection. These procedures will then be referred to throughout the discussion of stage-specific treatment.

Colonoscopic Resection of Polyps

Many lesions of the colon are first detected during endoscopic procedures. These lesions can range from small hyperplastic polyps to large fungating invasive carcinomas. The appearance of these lesions often indicates their relative potential for malignancy. However, the only definitive way to make a diagnosis is through a pathologic examination of the tissue. Therefore, the goal of a colonoscopic biopsy or resection is to, whenever feasible, remove the lesion in its entirety and preserve a tissue architecture in order to achieve both a therapeutic resection and an accurate pathologic diagnosis. Various techniques can be employed for the removal of lesions in the colon depending on their size and location. Biopsy forceps and snares are the two most commonly employed instruments used during a colonoscopy. These devices are fashioned from flexible coated wires that can conform to the shape of the colonoscope and can also conduct electrical current in order to achieve coagulation and hemostasis.

Bleeding and perforation, while uncommon, are seen at an increased frequency during a therapeutic as opposed to a diagnostic colonoscopy.[224,225] Small polypoid lesions (up to 5 to 8 mm) that are found during the course of a colonoscopic examination can often be removed in their entirety along with a small amount of normal mucosa using a biopsy forceps. Bleeding is usually minimal but can be controlled by electrocautery if persistent. Larger well-pedunculated polyps can often be removed using a technique employing a snare and electrocautery. The snare is placed over the polypoid lesion and cinched down at the base of the polyp. Once tightened, an electrical current is applied and the polyp is resected. If the lesions are too large to be retrieved through the working port of the colonoscope, they can be held in place with a snare just beyond the tip of the colonoscope where they can be kept in view and withdrawn with the scope from the patient. It is important, when sending these specimens to pathology, to properly orient the polyp so as to indicate the base where the resection took place as well as the other positions of the lesion. This will allow the pathologist to provide important information as to the margin status for the resection. Carcinoma in situ as well as stage I invasive carcinomas found in a well-pedunculated polyp can be treated with colonoscopic resection, as described previously, and no further surgical management is needed as long as there is a negative margin >2 mm and the tumor is well-differentiated without lymphovascular invasion or extension of malignant cells beyond the stalk (Haggitt levels 1 to 3) as per consensus guidelines. If these criteria are not met, further therapy is required. It is for this reason that it is often helpful to mark the site of the polyp resection with an agent that will leave a "tattoo" to guide additional intervention.

Larger lesions with a broad base or sessile lesions are best biopsied to make a diagnosis rather than resected using the colonoscope. The risk of perforation or inadequate resection margins is greatly increased with broad-based and sessile lesions. Multiple biopsies should be taken in order to determine whether the lesion harbors an invasive cancer, and further resection decisions are made based on the pathologic findings. In cases where there is low suspicion for malignancy, an endoscopic mucosal or submucosal resection may be attempted, usually by a gastroenterologist with advanced interventional endoscopic expertise. However, if such a lesion is left behind, it is of critical importance to note the position of the lesion in order that it might be more easily found if a subsequent procedure is required. In addition to determining the depth of insertion of the scope, which can be highly inaccurate with flexible instruments, other landmarks, including the appendiceal orifice or ileocecal valve in the cecum and the liver/splenic shadows at the flexures, should be noted. The most important step, however, is to properly mark the polyp site with 1 mL of tattoo injected submucosally in each of the four quadrants for definitive intra- and extraluminal recognition at a later date.[226]

For lesions that cannot be resected through the scope or are found to be invasive carcinomas that are sessile or broad based, a variety of surgical resections can be employed depending on the position of the lesion and its T stage. It is important to keep in mind, however, that the formal staging of the lesion does not occur until after the resection is completed; therefore, if there is any suspicion of an invasive carcinoma being present, a definitive oncologic resection should be performed.[227]

Bowel Preparation

An important part of the preoperative regimen for a colon resection is the proper cleansing of the bowel in order to reduce the risk of postoperative complications as well as to allow for easier visualization during the procedure, particularly with the laparoscopic approach. A variety of regimens have been described, and there are many that have demonstrated efficacy.[228,229] Although there are several choices described in the literature, the basic components of a bowel preparation are a mechanical cleansing of the bowel using a cathartic or volume-displacing agent and appropriate antibiotic prophylaxis.[230,231] Recently, some studies have suggested that mechanical bowel preparation may be unnecessary; however, this remains controversial.[232,233]

For rectal and low sigmoid tumors, a number of surgeons also perform distal rectal washout prior to resection, with the professed intention of eliminating exfoliated intraluminal cancer cells that may increase local recurrence risk. There has been little evidence to support this theory, and washout has not been routinely recommended as standard practice. However, a recent meta-analysis of nine studies and 5,395 patients is the first to demonstrate a significant benefit to this maneuver with a nearly twofold reduction in local recurrence rates (5.79% vs 10.05%; P <0.00001). While the lack of randomized controlled trials limits the strength of this data, the authors conclude that distal washout should be reconsidered in all patients given the minimal cost, time, and risk it entails.[234]

Anatomic Resection

For invasive carcinomas of the colon, stages I through III, the surgical approach will be dictated by the size and location of lesions in the colon.[235] The location will determine what region of bowel is removed, and the extent of its resection is dictated by its vascular and lymphatic supply.

Resection of the Right Colon

Lesions in the cecum and ascending colon are managed with a right hemicolectomy (Fig. 40.5A,B). The right colon is mobilized from the retroperitoneum by incising its retroperitoneal attachments, taking care to avoid injury to the ureter, inferior vena cava, duodenum, and gonadal vessels. The colon is mobilized from the ileum to the transverse colon, taking care at the hepatic flexure not to injure the gallbladder or duodenum. The ileocolic, right colic, and right branch of middle colic vessels are then ligated and divided. A proximal ligation in order to allow for the removal of colonic mesentery along with LNs is performed for staging purposes. Once the vascular supply is divided and the intervening

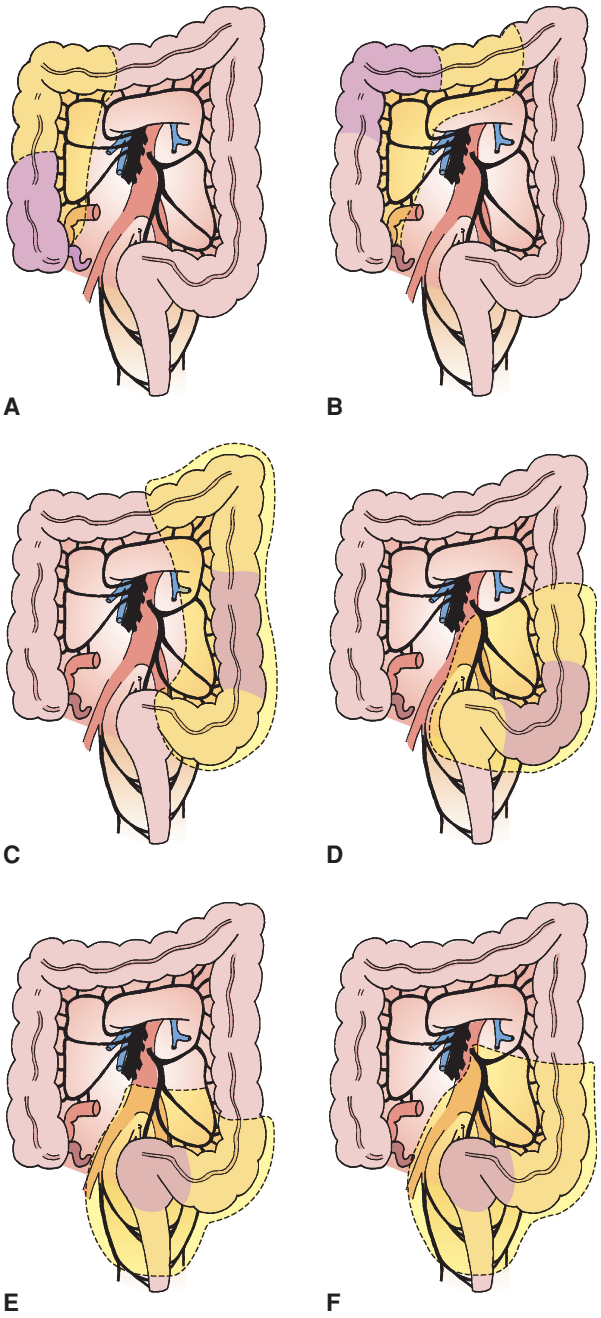


Figure 40.5 **A:** Surgical resection for a cecal or ascending colon cancer. **B:** Surgical resection for a cancer at the hepatic flexure. **C:** Surgical resection for a descending colon cancer. **D:** Preferred surgical procedure for cancer of the middle and proximal sigmoid colon. In poor-risk patients, the inferior mesenteric artery and the left colic artery may be preserved. **E:** Surgical resection for cancer of the rectosigmoid. **F:** A more radical surgical resection for cancer of the rectosigmoid.

mesenteric tissue ligated and divided, attention can be addressed to the resection of the colonic tissue.

There are a variety of techniques for dividing the colon. This can be done between clamps using scalpel or a variety of stapling devices. One method would be to use a linear GI anastigmatic stapler. After making a small hole just below the colonic wall though the mesentery at the point chosen for resection, the stapler can be positioned across the colon and fired, thus dividing the tissue. This is then repeated across the ileum just proximal to the ileocecal valve. Once divided, all remaining mesenteric tissue is carefully ligated and divided, and the colonic specimen can be removed. Although a no-touch "technique" has been advocated in the past, studies have demonstrated that this has no influence on recurrence or seeding of distant disease.[236] Once the right colon has been removed, intestinal continuity can be re-established by creating an anastomosis between the terminal ileum and the remaining transverse colon using either a hand-sewn or stapled technique.

Resection of the Transverse Colon

For lesions located in the transverse colon, a variety of approaches can be undertaken. Those lesions that are proximal and near the hepatic flexure can be resected with an extended right hemicolectomy. This extension should encompass up to and include the middle colic vessel. The advantage of such a resection over a true transverse colectomy is that the anastomosis performed to restore intestinal continuity involves an anastomosis between the ileum and the remaining colon. Due to the improved blood supply delivered by the small bowel mesentery, there is a decreased risk of an anastomotic leak in an ileocolic as opposed to a colocolic anastomosis.[227] Likewise, a lesion in the distal transverse colon can be resected with an extended left hemicolectomy, which will be described in more detail in the following section. For those lesions that are in the midportion of the transverse colon, however, a transverse colectomy can be performed. This procedure requires mobilization of the right colon in order to allow this tissue to be brought over for an anastomosis following the resection.

The omentum is divided from the greater curvature of the stomach up to and including its attachments at the splenic helium. The omentum can often be a source of micrometastatic disease and therefore its resection at the time a transverse colectomy is indicated. After dividing the omentum and mobilizing the right and transverse colon up to and including the splenic flexure, the middle colic artery is ligated at its trunk and smaller vessels from the right and left colic artery branches can be ligated and divided as required. A linear stapler can once again be used to divide the colonic tissue, and then the mobilized right colon can be anastomosed to the descending colon in an end-to-end fashion using a hand-sewn anastomosis or using a side-to-side stapled technique. Depending on the size of the transverse colon, however, it is often safer and easier to resect the right and transverse colon and connect the ileum to the descending colon. This allows enough colonic reserve for water absorption and normal bowel movements.

Resection of the Descending and Sigmoid Colon

For lesions in the proximal descending colon, the splenic flexure is mobilized and the left colic artery can be ligated and divided with the portion of colon removed by mobilizing the splenic flexure and dividing the omentum (Fig. 40.3C,D). The transverse colon can be brought over to the region of the sigmoid colon for anastomosis. For lesions in the midportion of the descending colon, a left hemicolectomy can be performed, taking care to ligate the left colic vessel along with some sigmoidal branches and taking an adequate portion of mesentery for staging purposes.

For lesions that involve the sigmoid colon, a sigmoid colectomy can be performed with margins of resection on either side of the lesion. The descending colon is mobilized (together with the splenic flexure as needed) and connected to the rectum using either a hand-sewn anastomosis or stapling device. The mesentery can be divided either at the level of the sigmoidal branches with preservation of the left colic artery or at the origin of the inferior mesenteric pedicle (Fig. 40.3E,F). While the latter approach is preferred by some to achieve greater mobilization and higher lymph node counts, neither this nor a more extensive left hemicolectomy has resulted in improved survival.

The approaches to the resection of lesions below the peritoneal reflection will be discussed in the chapter 41 on rectal cancer.

Total Abdominal Colectomy

For patients with ulcerative colitis or familial polyposis syndrome who either have evidence of invasive carcinoma or are at significant risk for the development of invasive carcinoma, a total abdominal colectomy may be required. This can be performed by mobilization of the right colon, transverse colon along the omentum, taking the omentum as part of the resection, the hepatic and splenic flexures, as well as the complete mobilization of the descending colon down to the peritoneal reflection. Ligation of the ileocolic, right colic, middle colic, left colic, and sigmoid branches will allow for removal of the colon down to the peritoneal reflection. For ulcerative colitis and familial polyposis syndromes without evidence of carcinoma below the peritoneal reflection, the operation can be terminated at this point with ileorectal anastomoses and careful surveillance of the remaining rectum via proctoscopy. However, in order to remove all tissue at risk for further lesions, a total protocolectomy is often advocated.[237,238] Although this procedure can be performed as an abdominal perineal resection with a permanent end ileostomy, most surgeons now advocate one of many continent pull-through procedures in order to preserve fecal continence in a patient population that is often very young. Such procedures provide very good control of continence and a relatively normal lifestyle.[239]

Complete Mesocolic Excision

Recently, and primarily in European centers, there has been a movement towards more radical lymphadectomy in the treatment of colon cancer. Proponents of this technique argue that the same principles of total mesenteric excision (TME), which is now standard of care for rectal cancer and associated with indisputable oncologic benefit (see Chapter 41), should be applied to colon resection as well. Prospective randomized data are unavailable at this time and prior results have been inconsistent, but a number of new studies lend support to this approach. Storli et al. analyzed a cohort of patients with Stage I-II tumors from the cecum to the rectosigmoid, 89 of whom underwent complete mesocolic excision (CME) with high vascular ligation while 105 underwent the standard technique. Three-year OS and DFS were significantly better in the CME group (88.1% vs 79% and 82.1% vs 74.3%, respectively), which suggests an advantage to more radical resection even when no lymph node metastases are removed.[240] Galizia et al. reported similarly positive results when right-sided cancers were examined alone.[241] Compared to historical controls, patients undergoing CME experienced over a 50% reduction in the risk of cancer-related death as well as increased detection of tumor deposits, allowing them to benefit from chemotherapy for upstaged N1c disease. In addition, for patients with node-positive tumors undergoing CME, the disease-specific survival rate was significantly increased (88% vs 50%, $P = 0.05$). More recently, Gao et al. performed a prospective, nonrandomized, double-blind, controlled trial of 330 patients surgically treated for both right & left-sided colon cancer.[242] In 220 cases, the predefined criteria for CME were met and 110 were considered non-CME. While 3-year overall survival and disease-free survival were statistically similar between the two groups, the study's primary outcome of local recurrence-free survival was improved with CME for Stage II and III disease (100% vs 90.2%, $P < 0.001$). There was no difference in postoperative complications.

Although further validation of these results is necessary, the authors advocate urgent standardization and dissemination of this approach, especially for more advanced tumors.

SURGICAL MANAGEMENT OF COMPLICATIONS FROM PRIMARY COLON CANCER

Patients with primary lesions of the colon can present with obstruction, bleeding, and perforation. The surgical management of these patients can be complex, requiring intraoperative decisions tailored to the situation encountered. Blood per rectum can be one of the most frightening experiences for patient and physician alike. Bleeding from a CRC can occur anywhere from the cecum to the distal rectum. Although bleeding can be temporized with endoscopic fulguration and the patient supported with transfusion, definitive management of the lesion with either surgery or radiation therapy will ultimately be required. Other maneuvers such as angiographic embolization may provide only a temporary solution. Fortunately, life-threatening hemorrhage due to a colon cancer primary is a rare occurrence. More often, these lesions lead to a chronic blood loss, resulting in anemia.

Colonic obstruction due to a primary tumor is not uncommon. Obstructing colon lesions present several important issues. First, the acute obstruction must be managed. Ideally, an exploration with resection of the tumor and primary anastomosis with or without a diversion is ideal. However, given the fact that the operation will be performed on unprepared bowel and the patient's physical condition may be less than optimal, resection without an anastomosis and an end colostomy should be considered. In some instances, the obstructing lesion may present significant technical hurdles for resection in the setting of an acutely dehydrated and ill patient. In these circumstances, a decompression maneuver that can be performed rapidly and with minimal morbidity such as a transverse loop colostomy or a colostomy and mucous fistula can be performed to temporize the situation and allow the patient to be prepared and resuscitated adequately for a definitive resection at a second exploration.

Bypass operations should be reserved only for the most extreme circumstances as complications following these procedures due to repeat obstructions and leakage with abdominal sepsis are not insignificant.

Another option is to place an endoscopic stent either for temporary decompression or for definitive palliation of unresectable lesions. Multiple studies over the past 10 years have demonstrated the feasibility and safety of this maneuver in selected patients.[243–248] As a bridge to surgery, stenting can provide a minimally invasive means for converting an emergency situation into an elective one, allowing time for resuscitation, bowel prep, and adjuvant therapies. In a small randomized, controlled trial from 2009, Cheung et al.[249] reported additional advantages including significantly reduced rates of perioperative morbidity and stoma creation.

While short-term data seem to support the use of self-expanding metallic stents (SEMS) as a bridge to surgery, long-term results are conflicting.[250,251] Sabbagh et al.[252] performed a head-to-head, intention-to-treat analysis of 87 patients undergoing either stenting or emergency surgery, using a propensity score to correct for selection bias. Overall survival at 3 and 5 years was significantly better in the surgery group (66% vs 44%, $P = 0.015$, and 62% vs 25%, $P = 0.0003$, respectively) and remained superior even when patients with perforation and metastatic disease were excluded (74% vs 51%, $P = 0.02$, and 67% vs 30%, $P = 0.001$, respectively). Five-year cancer-specific mortality was significantly higher in the SEMS group (48% vs 21%, $P = 0.02$) and there were trends toward worse 5-year DFS and increased recurrence as well as mean time to recurrence. Based on these findings, the authors have markedly changed their management of left-sided malignant obstruction, now reserving SEMS strictly for palliative indications and patients with high postoperative mortality risk.[252] On the other hand, a more recent study with 10-year follow-up reports more positive outcomes. In this prospective analysis of 97

patients from Belgium, technical success of stenting was 94.8%, with similar survival at 1 year, 5 years, and 10 years compared to national data across the same time period.[253]

Additional long-term data continue to support the use of SEMS as a bridge to surgery in left-sided malignant obstruction, demonstrating similar perioperative results and equivalent oncologic outcomes when compared to emergency resection (ER). In a 2019 large systematic review of 24 articles and 2,508 patients, including 5 RCTs, Cao et al. reported no significant differences in 3-year or 5-year survival rates, 5-year disease-free survival rates, or in overall or local recurrence rates between the two surgical strategies.[254] Interestingly, a more traditional bridge to surgery, the diverting stoma (DS), has garnered renewed attention in recent studies with some data suggesting it is not only superior to emergent resection but may have important advantages over the more novel SEMS technique.[255] Veld et al. used propensity-score matching to compare 236 patients who underwent DS to 472 patients undergoing ER.[256] Significantly more laparoscopic resections and primary anastomoses were performed in the DS group with lower 90-day mortality that was mainly attributed to the subgroup over 70 years of age. DS was also associated with improved 3-year overall survival and fewer permanent stomas. When compared to SEMS by the same authors, the data on DS are more opaque, with SEMS requiring fewer post-resection stomas and subsequent interventions but diverting stoma leading to higher rates of primary anastomosis and fewer major complications.[257] In a meta-analysis of 33 studies and 15,224 patients, Jain et al. also found increased rates of stoma creation in the DS group but higher rates of ileus and reoperation leading to longer hospital stays in the SEM patients.[258] Both publications reported similar intermediate-term oncologic outcomes and highlighted the need for randomized trials to determine the optimal approach.

Along these lines, two other clinical scenarios are under investigation as potential beneficiaries of a bridge to surgery strategy. Obstructing right-sided colon cancer has nearly always been managed by upfront emergent resection (ER) but little is known about alternative options. In a recent systematic review, ER was compared to staged treatment, defined as SEMS or ileus tube decompression followed by primary tumor resection.[259] The authors found that the staged treatment group had lower mortality rates, fewer complications, and fewer anastomotic leaks and stoma creations suggesting possibly a new paradigm for this familiar clinical presentation. Another potential role for the staged approach is to facilitate neoadjuvant chemotherapy in obstructive disease. Ishibe et al. addressed this possibility in the first prospective clinical trial on the subject, which enrolled 46 patients, 45 of whom underwent R0 resection of their primary tumors.[260] Objective response rate was 67.4%, with a downstaging rate of 48.9%, and no marked increase in surgical morbidity or mortality. More importantly, the 3-year relapse-free and overall survival rates were 76.5% and 95.4%, respectively, which is better than most prior outcome data for obstructive colorectal cancer. The authors attribute this directly to the efficacy of planned perioperative chemotherapy, although further large-scale confirmation is warranted.

LAPAROSCOPIC COLON RESECTION

Since its introduction to the field of general surgery for gallbladder resection, the use of laparoscopic surgery has found increasing applications.[261] Laparoscopic surgery has become a particularly important addition to the armamentarium of the surgical oncologist. The use of laparoscopy for the staging of the extent of disease for peritoneal malignancies, pancreatic cancer, colon cancer, and gastric cancer is now widely accepted.[262–264] Laparoscopic resection has also found a niche for the removal of adrenal tumors, the spleen, and distal pancreas.[265,266] The use of laparoscopic approaches for the resection of malignant lesions in the colon is now becoming more common.

With the increasing application of laparoscopic techniques to colon cancer surgery, concerns ranging from inadequacy of resection margins, inadequacy of LN sampling, and the potential seeding of port sites with malignant cells have been raised.[267–269] Although these concerns are important, there are several potential advantages for laparoscopic approaches to the surgical management of colon cancer. Issues regarding length of incision, patient recovery time, and return to bowel function are often cited as justification for a laparoscopic approach. However, just as important are the technical advantages of surgery utilizing laparoscopic systems. The improved visualization due to magnification provided by video laparoscopy allows much more intricate and careful dissections in the deep pelvis, which could potentially reduce postoperative morbidity from low anterior resections that utilize a mesorectal excision technique. The ability to carefully trace vessels in the mesentery under magnification could improve the ability to perform high ligations in order to retrieve a greater number of LNs for sampling.

The technical difficulties faced during laparoscopic resection of the colon relate, in general, to the size of the specimen being removed and the need to perform an anastomosis. Each of these can be overcome through careful placement of incisions for specimen removal as well as a judicious use of stapling devices in order to perform both intracorporeal as well as a combination of intracorporeal and extracorporeal anastomotic techniques. A number of studies have examined the relative risks and benefits of the laparoscopic resection of colon cancer.[267,270–272] The three most significant randomized trials to definitively address this issue over the past decade include the COST study, the CLASICC trial, and the European COLOR trial.

The Clinical Outcomes of Surgical Therapy Study Group (COST) examined both the oncologic outcomes with respect to DFS and overall survival as well as the impact of laparoscopic versus open surgery on patient recovery, pain management, and time to return of bowel function. An initial report on quality of life showed only a modest short-term benefit for laparoscopic resection versus a conventional open procedure,[273] but the overall results of the trial with respect to oncologic outcomes demonstrated equivalence between the laparoscopic and open approach.[274]

Long-term follow-up from the corresponding UK randomized study (CLASICC Trial Group), which was similarly designed to compare laparoscopic to conventional surgery for colon and rectal cancer, and that initially reported noninferiority results in 2007, lends further support to the laparoscopic approach. With a median follow-up of 62.9 (range, 22.9 to 92.8) months, the authors found no statistically significant differences in overall survival (82.7% vs 78.3%), DFS (77% vs 89.5%), or local recurrence.[275,276]

Finally, long-term data from the COLOR (Colon cancer Laparoscopic or Open Resection) study also uphold the conclusions of the COST and CLASICC studies.[277] In keeping with previously published 3- and 5-year results showing similar outcomes, 10-year follow-up of the 329 Dutch patients included in this trial demonstrated similar rates of disease-free survival, overall survival, and recurrence between the two approaches.

A case-matched comparison of clinical and financial outcomes following laparoscopic and open colorectal surgery has been performed.[268] The group at the Cleveland Clinic studied patients from a prospective database who had undergone laparoscopic or open colectomy and were matched for age, gender, and disease-related groupings. A group of 150 patients undergoing laparoscopic colectomy was compared to a matched group of patients undergoing open colectomy. There was no difference found between the two groups for diagnosis, complications, or 30-day readmission rate. Although operating room costs were significantly higher after laparoscopic colectomy, this was offset by a decrease in the length of hospital stay with an overall significant reduction in total costs. This is attributed mainly to a lower cost for pharmacy, laboratory, and ward nursing expenses.

As laparoscopic colectomy is becoming more widely adopted in the setting of increasingly robust and supportive data, exploration of other minimally invasive approaches to resection is ongoing. Early studies report the feasibility and safety of single port surgery, or "SILC" (single-incision laparoscopic colectomy), as well as equivalent oncologic outcomes at 2 years, and robotic colectomy has also been performed although its greatest potential is thought to lie with rectal dissection.[278–285]

Natural orifice surgery is another area of interest with data accumulating on transvaginal specimen extraction as well as a pure transrectal approach.[286–288]

Growing evidence, including early randomized data, highlights at least short-term benefits of NOSE for anterior resection of sigmoid and upper rectal tumors, while continuing to support the safety and feasibility and oncologic equivalence of this technique. A recently published large propensity score-matched comparative study of 188 patients with median follow-up of 50.3 months demonstrated statistically similar 5-year local recurrence risk and disease-free survival for the NOSE and conventional extraction groups (2.3% vs 3.5% and 87.3% vs 82.0%, respectively).[289] Moreover, a meta-analysis of 10 studies compromising 1,787 patients showed significantly reduced postoperative complication rates (OR 0.65), incision-related complication rates (OR 0.13), postoperative pain scores, and analgesic usage, as well as earlier recovery of GI function and shorter hospital stay.[290] Operating time was longer (7.4 min, $P = 0.04$) but other surgical parameters, severe complication rates, and long-term outcomes were comparable. Early randomized data further substantiates these findings. Zhu et al. randomized 344 patients with upper and middle rectal cancer to NOSE, laparoscopic non-NOSE, and open surgery.[291] While there was no significant difference in disease-free survival between the three study arms, the NOSE group had superior quality of life and postoperative pain scores as well as shorter time to bowel recovery and length of stay. The postoperate complication rate was also significantly lower in this group compared to non-NOSE (11.5% vs 16.8%). Similar results were reported in a second randomized study by the same group where there was no open surgery arm.[292] The authors attributed most of these findings to the fact that NOSE maintains the integrity of the abdominal wall and substantially reduces exposure of the abdominal cavity compared to traditional laparoscopy. While there is wide variability in the NOSE technique and substantial room for further optimization, these results certainly pave the way for further study and more widespread acceptance.

POLYPS AND STAGE I COLON CANCER

The management of polyps and stage I colon cancer is through surgical resection. Most cancer in polyps is not diagnosed until after the polypectomy is performed. Therefore, with respect to pedunculated lesions, care should be taken to resect the stalk completely, down to its base. Invasive early stage I cancers found in a polyp managed by polypectomy do not require further resection if there is a negative margin >2 mm and the tumor is well differentiated without lymphovascular invasion or extension of malignant cells beyond the stalk (Haggitt levels 1 to 3).[293] Sessile lesions that are biopsied and shown to harbor an invasive cancer should be managed with a segmental colon resection. Large polypoid lesions may also require a segmental resection.

Because the stage of the lesion will not be determined until after the resection, all colon cancer lesions managed with a segmental resection should be approached the same way. The type of resection will be dictated by the location of the lesion, as has been described. Following a complete resection of a stage I lesion, no further adjuvant therapy is required. Patients managed in this way can expect a 5-year survival of over 95%.[295] Those that recur are most likely improperly classified stage II or III lesions.

STAGE II AND STAGE III COLON CANCER

Adjuvant Chemotherapy Considerations

The earliest clinical trials of adjuvant chemotherapy in colon cancer were conducted in the 1950s, utilizing the limited arsenal of anticancer agents that were available at that time. Many of these agents are now known to have no meaningful activity in metastatic CRC and thus would not be studied in the adjuvant setting today.

The adjuvant trials of the 1950s through the mid-1980s tended to be small by current standards. Based perhaps on an unrealistically optimistic expectation of what magnitude of benefit might be achieved from the use of available chemotherapies, the size of the trials did not allow evaluation of more modest clinical benefits. A large meta-analysis of controlled randomized trials of adjuvant therapy published through 1986 indicated a nonsignificant trend toward an overall survival benefit, with a mortality OR of 0.83 in favor of therapy (95% CI = 0.70 to 0.98).[294] This sobering analysis suggested that substantially larger trials would be needed to detect the modest advantages that available chemotherapies might afford.

Large-Scale Randomized Trials

The large-scale 5-FU trials have been well summarized previously, and the reader who is interested in the details is referred to subject-relevant chapters in the previous edition of this book.[295] The outcome of numerous trials performed largely in the 1990s can be briefly summarized as follows. Trials comparing 5-FU–based therapy to surgery only demonstrated a clear benefit in terms of 5-year DFS (essentially, an increased cure rate) for stage III patients who received chemotherapy.[296,297] Six months of chemotherapy was sufficient, and no further benefit was provided by extending treatment to either 9 or 12 months. Levamisole, an agent initially thought to be active, was, in fact inactive, and high-dose leucovorin did not confer superior efficacy over low-dose leucovorin, so comparisons of various 5-FU/leucovorin schedules did not demonstrate clear superiority of one schedule over the other in terms of efficacy. However, the Mayo Clinic daily times five schedule was substantially more toxic than either weekly bolus of biweekly infusion schedules. Alfa interferon conferred substantial toxicity and provided no benefit.[298–302]

Oral Fluoropyrimidine Therapies

Oral administration of 5-FU proved to be problematic secondary to erratic bioavailability. This was likely due in large part to variable effects of DPD, the rate-limiting enzyme in catabolism of 5-FU, on the first pass clearance of oral 5-FU by the liver. Two oral 5-FU prodrugs, capecitabine and uracil/tegafur (UFT), have demonstrated efficacy in metastatic disease that is comparable to the Mayo Clinic schedule of parenteral 5-FU/leucovorin. Both of these agents have now been studied in the adjuvant setting in comparison to the now defunct Mayo Clinic 5-FU schedule. In a study designed to assess for noninferiority in 3-year DFS, Twelves et al.[303] randomly assigned 1,987 patients with resected stage III colon cancer to receive either oral capecitabine (1,004 patients) or Mayo Clinic bolus 5-FU plus leucovorin (983 patients). Each treatment was planned for 24 weeks. DFS in the capecitabine group was at least equivalent to that in the 5-FU/leucovorin group (in the intention-to-treat analysis [$P < 0.001$] for the comparison of the upper limit of the HR with the noninferiority margin of 1.20), and capecitabine resulted in significantly fewer adverse events than Mayo Clinic bolus 5-FU/leucovorin ($P < 0.001$). Overall, this trial demonstrates that capecitabine is a reasonable alternative to intravenous 5-FU/leucovorin in the adjuvant treatment of colon cancer in reliable, motivated patients who are able to comply with a complex schedule of oral medication.

However, as discussed in the following, while possibly appropriate for some stage II patients, 5-FU/leucovorin alone is no longer the standard postsurgical adjuvant treatment for stage III colon cancer. As such, the role of single-agent capecitabine in the adjuvant management of resected colon cancer remains limited at this time. Data supporting its use with concurrent intravenous oxaliplatin are discussed subsequently.

The NSABP C-06 trial assessed the use of oral UFT plus oral leucovorin in the treatment of stage II and III colon cancer.[304] A total of 1,608 patients with stage II (47%) and stage III (53%) colon cancer were randomly assigned to receive either oral UFT with leucovorin or intravenous 5-FU with leucovorin. With a median follow-up of 62.3 months, there were no significant differences in DFS or overall survival between the treatment groups. Toxicity and primary quality of life end points were similar in the two groups. As such, similar to the situation with capecitabine, the combination of oral UFT with leucovorin is an acceptable alternative to parenteral 5-FU/leucovorin; however, use of fluoropyrimidine plus leucovorin alone is no longer routine standard practice (see the following) in the adjuvant treatment of at least stage III disease. Furthermore, UFT is not commercially available in the United States.

S-1 is another oral fluoropyrimidine used primarily in Japan for the treatment of gastric and pancreatic cancers, similar to UFT. An open-label randomized controlled trial comparing the two drugs in the adjuvant setting for stage III colon cancer confirmed the non-inferiority of S-1 in terms of disease-free survival.[305]

Combination Adjuvant Therapies

Clinical trials in the metastatic setting have established the antitumor activity of combinations of agents, including irinotecan, oxaliplatin, bevacizumab, cetuximab, and panitumumab (see discussion of treatment of metastatic disease for more details). Although it had been assumed that activity in the metastatic setting would translate into an increased cure rate in the adjuvant setting, this assumption has turned out to be overly simplistic and often untrue. Of the agents listed previously, only the addition of oxaliplatin to fluoropyrimidines has resulted in benefit in the adjuvant setting.

Oxaliplatin

Oxaliplatin plus biweekly infusional 5-FU/leucovorin was first evaluated in the adjuvant setting in the Multicenter International Study of Oxaliplatin/5-Fluorouracil/Leucovorin in the Adjuvant Treatment of Colon Cancer (MOSAIC) trial.[306] The results of this trial are summarized in Table 40.9. A total of 2,246 stage II and III patients were randomized to the LV5FU2 regimen, a biweekly infusional and bolus 5-FU/leucovorin regimen that has been demonstrated to have comparable efficacy to the Mayo Clinic daily times five bolus schedule in the adjuvant setting, or to the FOLFOX-4 regimen, which is LV5FU2 plus oxaliplatin on day 1.[298] For the combined stage II and III study population, the 5-year DFS rates were 73.3% and 67.4% in the FOLFOX-4 and LV5FU2 groups, respectively (HR = 0.80; 95% CI = 0.68 to 0.93; $P = 0.003$).[306] Six-year overall survival rates were statistically significantly improved by 2.5% (78.5% vs 76.0% in the FOLFOX-4 and LV5FU2 groups, respectively; HR = 0.84; 95% CI = 0.71 to 1.00; $P = 0.046$). For the stage III population, the 6-year overall survival rates were improved by 4.2% (72.9% vs 68.7%, respectively; HR = 0.80; 95% CI = 0.65 to 0.97; $P = 0.023$), whereas for the stage II population, the addition of oxaliplatin conferred no survival benefit (6-year survival 85.0% and 83.3%, respectively; $P = 0.65$). A more recent update showed that even amongst the stage II patients with high-risk factors, an improved outcome with the addition of oxaliplatin was not evident. While toxicity was regarded as manageable, the FOLFOX-4 regimen resulted in 41% grade 3 or 4 neutropenia versus 5% in the control arm and 11% grade 3 or 4 diarrhea versus 7%. All-cause mortality in the first 60 days was 0.5% in each arm. Peripheral sensory neuropathy, a toxicity not present in the LV5FU2 control arm, was a frequent occurrence on the FOLFOX-4 arm. Grade 2 neuropathy was reported in 32% of the patients, and grade 3 occurred in 12%. In some cases, the duration of the neuropathy was substantial. One year after completion of therapy, 30% of patients still experienced some grade of neuropathy (0.8% grade 2 and 1.3% grade 3). Four years after completion of therapy, 15.4% still had some degree of neuropathy, and 0.7% still had grade 3 neuropathy. It is reasonable to assume that the toxicity still present at 4 years out from the last treatment is essentially permanent.

TABLE 40.9

Results of The Mosaic Trial: Biweekly Infusional Fluorouracil/Leucovorin versus 5-Fluorouracil/Leucovorin Plus Oxaliplatin in Patients with Stage II and III Colon Cancer

	FOLFOX (%)	5FULV2 (%)	*P* Value
Five-y disease-free survival (stage II+III)	73.3	67.4	0.003
Six-y overall survival (stage II+III)	78.5	76.0	0.046
Six-y overall survival (stage III only)	72.9	68.7	0.023
Six-y overall survival (stage II only)	85	83.3	0.65
Grade 3–4 neutropenia	41	5	
Grade 3–4 diarrhea	11	7	
Grade 3 neuropathy	12	0	
Grade 2 neuropathy	32	0	

FOLFOX, fluorouracil/leucovorin plus oxaliplatin; 5FULV2, fluorouracil/leucovorin.

Long-term data from the MOSAIC trial support earlier conclusions with regard to the benefit of oxaliplatin in the adjuvant treatment of patients with resected Stage III colon cancer. In fact, the gain in absolute overall survival with the FOLFOX4 regimen was noted to increase from the 6-year follow-up data, from 2.2% to 4.6%. Specifically, 10-year overall survival rates with and without oxaliplatin were 67.1% versus 59%, respectively, for Stage III disease ($P = 0.016$). This benefit was seen regardless of MMR status or BRAF mutation. As for Stage II disease, the FOLFOX4 regimen still was not associated with significant survival advantage over LV5FU2, although a trend towards improved 10-year overall survival was demonstrated for high-risk cases (75.4% vs 71.7%, $P = 0.58$).[307]

Oxaliplatin has also been combined with a weekly bolus 5-FU regimen in an adjuvant trial. The NSABP C-07 trial studied the FLOX regimen of oxaliplatin given on weeks 1, 3, and 5 plus weekly bolus 5-FU/leucovorin on weeks 1 through 6, repeated at 8-week cycles, versus the standard weekly Roswell Park regimen of 5-FU/leucovorin.[308] A total of 2,409 patients were randomized to FLOX or to 5-FU/leucovorin. A total of 29% of patients had stage II disease and 71% had stage III. With a median follow-up of 8 years, FLOX showed a superior DFS, with 69.4% versus 64.2% alive and free of disease at 5 years (HR = 0.82; 95% CI = 0.72 to 0.93; $P = 0.002$). However, the overall survival difference was not statistically significantly different between the two arms. Treatment-related deaths were 1.3% versus 1.1% in the FLOX and 5-FU/leucovorin arms, respectively. Grade 2 or higher neurotoxicity was reported in 30.4% of patients on the FLOX arm versus 3.6% on 5-FU/leucovorin. Grade 3 diarrhea was 38.1% and 32.4% in the two arms, respectively, reflecting the higher incidence of serious diarrhea with the weekly bolus 5-FU regimen. Kidwell et al.[309] published long-term data regarding the persistent

neurotoxic side effects of oxaliplatin beyond 4 years from this same NSABP C-07 trial and found that there was a statistically but not clinically significant increase in total neurotoxicity for those who received the agent, with initial differences between the two groups dissipating by 7 years. However specific symptoms of numbness and tingling in the hands and feet did remain substantially elevated over time.[309]

Another more recent study examined the effect of diabetes and other comorbidities on oxaliplatin-induced neuropathy. With symptoms identified in 65% of patients, hypertension, smoking, and diabetes were associated with higher trends although not statistically significant differences in severe neuropathy. Additionally, patients with diabetes developed oxaliplatin-induced neuropathy at a significantly lower cumulative dose, highlighting the importance of tailoring patient-specific regimens to minimize toxicity.[310]

In an exploratory analysis, however, the authors noted significant age-related differences in response to oxaliplatin, finding statistically improved overall survival in patients <70 years old, whereas older patients actually fared worse with increased grade 4–5 toxicity (OR = 1.59) and a 4.7% decrease in 5-year overall survival.

This age–treatment interaction has also been supported by a 2012 pooled analysis of 5,489 patients >75 years old from four large data sets that demonstrated minimal benefit of oxaliplatin in this group.[311] Moreover, post hoc analysis of MOSAIC data as well as revised findings from the ACCENT study (which initially showed no age-related difference) further indicate the limited clinical utility of this agent in older patients.[311,312] Taking all this into account, consensus guidelines since 2013 have recommended individualizing the decision to add oxaliplatin to adjuvant regimens in the elderly.

More recently, a 1,866-patient study comparing capecitabine plus oxaliplatin (Cape/Ox) with bolus 5-FU/leucovorin in the adjuvant treatment of stage III colon cancer has been reported.[313] The Cape/Ox regimen had a statistically significant DFS advantage over 5-FU/leucovorin, with 66.1% of patients alive and disease-free at 5 years with Cape/Ox versus 59.8% with 5-FU/leucovorin. The difference between arms in overall survival at 5 years favored the Cape/Ox arm by 3.4%; however, this difference did not reach statistical significance at the time of this analysis ($P = 0.15$).

The authors of this trial just published their final survival data as well as a biomarker analysis after a median follow-up of almost 7 years.[314] Both DFS and OS were significantly improved in the Cape/Ox arm compared to FU/FA (7-yr DFS 63% vs 56%, $P = .004$ and 7-yr OS 73% vs 67%, $P = 0.04$). In the 498 participants who consented to biomarker analysis, low tumor expression of dihydropyrimidine dehydrogenase (DPD) was found to be predictive of Cape/Ox efficacy. The authors postulate that this leads to reduced catabolism of FU and therefore higher and more effective intracellular concentrations of this agent. However, no such association was seen in the FU/FA arm for unclear reasons. Further investigation may clarify these differential findings as well as the utility of this biomarker in patient selection for oxaliplatin-based therapy.

The efficacy results of the FOLFOX and Cape/Ox studies appear more similar than different at this point in time and appear to justify interchangeability of these regimens in the adjuvant setting. Data for FLOX regimens appear to show higher rates of serious or life-threatening diarrhea, and the lack of a statistically significant survival benefit at 6 years is notable. The higher degree of severe and life-threatening diarrhea seen with FLOX would appear to be a potential reason for favoring FOLFOX or Cape/Ox over FLOX, although in the absence of a head-to-head comparison, the relative safety and efficacy when comparing one of these regimens to the other is impossible to know with certainty. The Cape/Ox regimen is a reasonable consideration only in highly reliable, motivated patients who can be expected to comply with taking multiple pills of capecitabine orally (typically three to five pills, twice daily) for 2 weeks on, 1 week off, in the setting of concurrent emetogenic intravenous chemotherapy.

More recently, a collaboration of six parallel trials, reported together as the International Duration Evaluation of Adjuvant Chemotherapy (IDEA) Collaboration, asked whether 3 months of oxaliplatin-containing therapy was noninferior to 6 months. Patients and physicians were permitted to select CapeOx or FOLFOX and then were randomized to either 3 or 6 months of total treatment. The prespecified primary end point was disease-free survival. A total of 12,834 patients were included in the intention to treat analysis. Although the two arms appeared to perform quite similarly, with the disease-free survival hazard ratio at 1.07 (95% confidence interval, 1.00 to 1.15) and the 3 year disease-free survival being 74.6% for the 3-month arm versus 75.5% for the 6-month arm, this study in fact failed to meet the prespecified definition of noninferiority, which had been established with a noninferiority margin of 1.12. Thus this large trial did not demonstrate noninferiority of 3 months versus 6 months. The differences were most notable in the subset of patients who were treated with FOLFOX, with a 3 year disease-free survival of 76% versus 73.6% for the 6-month and 3-month arms, respectively. For the subset treated with CapeOx, 3 months of treatment did appear to be noninferior to 6 months. In a post hoc consensus analysis, the IDEA investigators separated the patient populations into high-risk and low-risk groups, with high risk being either T4 or N2 disease and low risk being T1-3, N1 disease. In this analysis, 3 months of treatment appears to be noninferior to 6 months. It should be noted that the study was not designed either to compare high- versus low-risk groups or to compare Cape Ox to FOLFOX, and the comparison of Cape ox to FOLFOX is nonrandomized.

The IDEA trial does not provide straightforward, simple answers to the optimal duration of therapy for patients with stage III colon cancer. Taken at its simplest, the study missed its prespecified primary end point and we could therefore conclude that 3 months of therapy is not noninferior to 6. However, the more recently published final analysis of the IDEA trial André et al.[316] demonstrated no statistically significant difference in 5 year survival. Five-year disease-free survival differed by less than 1%, and there was virtually no numeric differences between 3 months and 6 months of Cape Ox. As such, the authors concluded that 3 months of Cape Ox is appropriate therapy for most stage III colon cancer patients. We must keep in mind the considerable increased toxicity, especially neurotoxicity, that is associated with longer duration of oxaliplatin therapy. It is also important to recall that Cape Ox requires a motivated patient who is both willing and able to comply with the complexities of the intravenous plus oral chemotherapy schedule. Within these contexts, the selection and duration of therapy will need to be individualized based on the overall clinical scenario and upon a detailed discussion between the patient and her treating physician.

Irinotecan

Based on improved overall survival in the first- and second-line metastatic settings, it was widely assumed that irinotecan would be beneficial to patients in the adjuvant setting.[316–318] This assumption has turned out to be incorrect, however, and the results of the adjuvant trials with this agent underscore the importance of both performing trials in the adjuvant setting and waiting for the results of those trials before adopting changes in practice.

The CALGB studied the weekly schedule of irinotecan plus bolus 5-FU and leucovorin (IFL). Early safety analysis of this trial identified an alarming elevation in early mortality for the experimental arm on this trial, with 18 deaths within the first 4 months of treatment on the IFL arm versus 6 deaths within the same time period on the control arm ($P = 0.008$).[319] At a median follow-up of 2.1 years in each arm, futility boundaries for both DFS and overall survival had been crossed; thus, the final result of this trial is that the addition of irinotecan provided no benefit, while increasing toxicity, including lethal toxicity.[320]

Results of adding irinotecan to biweekly infusional 5-FU/leucovorin (LV5FU2 vs FOLFIRI) were also negative. In the ACCORD 02 trial, 400 patients with high-risk stage III disease (defined as four or more positive nodes or perforated or obstructed primary tumors) were randomly assigned to LV5FU2 versus FOLFIRI. In this high-risk population, there was no benefit seen in the FOLFIRI group, and in fact the study trended insignificantly in favor of the nonirinotecan-containing arm (3-year DFS 60% for LV5FU2 vs 51% for FOLFIRI).[321]

A second, larger trial of LV5FU2 versus FOLFIRI was conducted by the PETACC-3 investigators.[322] The prespecified primary efficacy analysis of this trial was based on 2,094 patients with stage III disease. At a median follow-up of 6.5 years, there was no statistically significant difference in the 5-year DFS (56.7% vs 54.3% for FOLFIRI vs LV5FU2, respectively; $P = 0.1$) or in 5-year overall survival (73.6% vs 71.3%, respectively; $P = 0.94$). FOLFIRI was associated with an increased incidence of grade 3 or 4 GI events and neutropenia.

Taken together, the results of these three trials to evaluate irinotecan in the adjuvant setting clearly establish that despite having substantial activity in the metastatic setting, irinotecan has no meaningful activity, and no role, in the adjuvant treatment of colon cancer. Of interest, an analysis from the CALGB trial suggested that patients with MSI-H showed a benefit from inclusion of irinotecan in their adjuvant treatment; however, a similar, substantially larger analysis from the PETACC-3 trial contradicted this and showed no benefit from adding irinotecan in patients with MSI-H.[323,324]

Bevacizumab

As detailed subsequently, bevacizumab has demonstrated the ability to favorably augment standard chemotherapy for metastatic disease and has become a part of standard management in that arena. This led to evaluation of this agent in the adjuvant setting. In the NSABP C-08 trial, 2,672 patients, 25% with stage II and 755 with stage III colon cancer, were randomized to receive modified FOLFOX-6, either alone or with bevacizumab.[325] A design imbalance that could have been problematic had this been a positive trial, the FOLFOX was given for 6 months in each arm, while the bevacizumab was given both with the FOLFOX and then for an additional 6 months, for a total of 1 year of bevacizumab. This was, however, a fully negative trial, so the issues regarding the design of the trial are moot. With a median follow-up of 5 years, the addition of bevacizumab to FOLFOX did not improve either the DFS or the overall survival. DFS was 77.9% versus 75.1% ($P = 0.35$) for the study overall, and DFS was 73.5% versus 71.7% ($P = 0.55$) for the stage III patients. Five-year overall survival was 82.5% versus 80.7% ($P = 0.56$) for the entire study and 78.7% versus 77.6% for the stage III patients. There was a separation between the curves at the 1-year mark; however, this began to diminish a few months later and was all but absent by year 3. This finding suggests that bevacizumab did delay progression of micrometastases in some patients, but only for as long as it was continued. Bevacizumab did not contribute to the eradication of micrometastases and thus did not improve the cure rate in the adjuvant setting. Although some might choose to interpret these data to suggest that if bevacizumab were continued indefinitely an improved survival *might* be seen, the long-term consequences of lifelong suppression of vascular endothelial growth factor (VEGF), as well as the psychological, social, and economic considerations involved, render such an approach inappropriate, especially considering that only a very small percentage of patients so treated would actually have the potential to benefit, if there is a benefit. Another trial, termed the AVANT trial, also explored the use of bevacizumab in the adjuvant treatment of stage III colon cancer and also found no benefit.[326] This study randomized 3,451 patients, 2,876 of whom had stage III disease, to FOLFOX4, FOLFOX4 plus bevacizumab, or Cape/Ox plus bevacizumab. The bevacizumab-containing arms did not achieve a statistically significant improvement in DFS. After a minimum of 60 months follow-up, overall survival data suggested a possible detriment with the addition of bevacizumab, as the survival in the two bevacizumab-containing arms trended toward inferior to the FOLFOX-alone control arm; however, these differences did not reach statistical significance. More recently the QUASAR 2 trial evaluated the addition of bevacizumab to adjuvant capecitabine in 1,941 patients with stage II (38%) or stage III (62%) colorectal cancer.[327] There was no benefit, and a trend in the direction of negative outcome, for the addition of bevacizumab to capecitabine adjuvant chemotherapy.

Thus, the available evidence suggests that bevacizumab is not beneficial in the treatment of colon cancer in the adjuvant setting and might, in fact, be harmful. Until and unless data to the contrary emerge, bevacizumab should not be used in the adjuvant treatment of stage II and III colon cancer.

Cetuximab

As outlined in detail in the following, cetuximab has demonstrated clinical activity in metastatic CRC, prompting investigation of its usefulness in the adjuvant setting. Intergroup trial N0147 randomized patients with stage III colon cancer to modified FOLFOX-6 with or without cetuximab.[328] Once investigators became aware that the study included only patients whose tumors lacked mutations in the *KRAS* gene (see the following), the study was modified to obtain KRAS genotyping on all patients and to only enroll those with wild-type KRAS. Despite this selection, this large, adequately powered, randomized phase 3 trial showed no benefit for the addition of cetuximab to FOLFOX in the adjuvant treatment of patients with KRAS–wild-type stage III colon cancer. DFS and overall survival curves trended insignificantly in favor of the FOLFOX-alone control arm, and the addition of cetuximab was overtly harmful in patients >70 years of age. Cetuximab should therefore not be used in the treatment of stage III colon cancer.

Panitumumab

Panitumumab, like cetuximab, is a monoclonal antibody that blocks ligand binding to the EGFR. Although no investigations have been reported to evaluate panitumumab in the adjuvant setting, results in the metastatic setting suggest that panitumumab and cetuximab are extremely similar in terms of target, mechanism of action, mechanisms of resistance, and clinical activity. It is therefore extremely unlikely that these agents would differ in the adjuvant setting, and statements regarding cetuximab in this setting may be reasonably applied to panitumumab.

TREATMENT OF STAGE II PATIENTS

The optimal management of patients with stage II colon cancer remains undefined. Although the role of adjuvant therapy in patients with stage II colon cancer has not been firmly established, it is interesting to see what practice patterns have been emerging. Using the SEER-Medicare linked database, Schrag et al.[329] identified 3,151 patients age 65 to 75 with resected stage II colon cancer and no adverse prognostic features. Using Medicare billing records, they identified those patients who did or did not receive chemotherapy within 3 months of operation. Their review identified that 27% of patients received chemotherapy during the 3-month postoperative period. Younger age, white race, unfavorable tumor grade, and low comorbidity were associated with a greater likelihood of receiving treatment. The 5-year survival was 75% for untreated patients and 78% for those patients who received therapy in this nonrandomized comparison. After adjusting for known between-group differences, the HR for survival associated with adjuvant treatment was 0.91 (95% CI = 0.77 to 1.09). Thus, despite the lack of proven benefit, a substantial percentage of Medicare beneficiaries have received adjuvant chemotherapy for stage II disease.

Because stage II patients as a group have a relatively favorable prognosis, benefits from treatment could only be expected if either a highly efficacious therapy were used or if extremely large trials were done to detect very subtle differences. The International Multicentre Pooled Analysis of B2 Colon Cancer Trials meta-analysis provides one of the largest samples of stage II patients.[330] A total of 1,016 stage II patients were randomized between 5-FU/leucovorin and surgery alone. The surgery-only arm had a long-term overall survival rate of 81% versus 83% for those stage II patients who received adjuvant 5-FU/leucovorin. This absolute difference of 2% closely approached, but did not reach, statistical significance.

More recently, published studies have not generated any consensus on this topic. In a large retrospective, population-based analysis of 3,716 patients undergoing surgery for stage II disease with or without adjuvant chemotherapy, there was a statistically significant survival advantage for the adjuvant group (12 years vs 9.2 years). However, this is not a randomized trial, and patients receiving chemotherapy were more likely to be younger, with left-sided lesions and a higher LN yield, all of which are now known to be favorable prognostic factors. As such, it is difficult to draw conclusions from this study.[331] Another study that evaluated data from 24,847 patients, 75% of whom had one or more prognostic features, found no survival benefit from an adjuvant regimen.[332] Finally, Wu et al.[333] performed a systematic review of 12 randomized controlled trials including both colon and rectal cancers that suggested some improvement in 5-year overall survival and DFS for both tumor sites as well as a significant reduction in recurrence risk for stage II colon cancers. While the review has substantial flaws that limit interpretation of these results, it does indicate that larger, higher-quality trials are warranted.[333] One such study the SACURA trial, failed to show any superior outcomes with addition of a 1-year adjuvant chemotherapy regimen compared to surgery alone. In this multicenter, randomized phase 3 study of 1982 patients with stage IIA-C disease, there was no significant difference in 5-year DFS (80.2% vs 78.4%) or 5-year OS (94.5% vs 94.3%) between the two study groups. Furthermore, subgroup analysis of "high-risk" cancers also showed no significant benefit from adjuvant therapy. Two of the traditionally recognized high-risk factors, lymphovascular invasion and poor differentiation, actually had no impact on recurrence-free survival at all in this trial, leading the authors to question whether they should be eliminated from stratification schemes moving forward. A major weakness of this study, however, was the use of a single oral chemotherapy agent, which may not be powerful enough to prevent recurrence especially in high-risk disease. Follow-up studies that incorporate improved risk stratification and addition of other agents such as oxaplatin are certainly warranted.[334]

Several prognostic indicators have been identified that correlate with a higher risk for subsequent failure in stage II patients. These include obstruction or perforation of the bowel wall as well as other less-established risk factors, such as elevated preoperative or postoperative CEA, poorly differentiated histology, and tumors not demonstrating high levels of MSI, or an 18q deletion in colorectal tumors, which may correlate with a poor prognosis.[194,335–337] Additionally cited poor prognostic factors include macroscopically infiltrating-type tumors, high serum CA 19-9 levels, extensive venous invasion, male gender, age >50 years old, and <12 dissected LNs.[338] It appears that stage II patients with one or more of these risk factors have a poorer prognosis, one closer to patients with stage III disease. Whether adjuvant chemotherapy can provide similar benefits in these patients as it does in stage III patients remains a matter of conjecture, and in the absence of definitive data, definitive recommendations on this topic cannot be made at this time. In fully informed high-risk stage II patients, it is reasonable to consider adjuvant treatment. All reports form the MOSAIC trial have shown no benefit for the addition of oxaliplatin to 5-FU/leuvocorin in terms of overall survival of stage II patients. Unequivocally, stage II patients lacking high-risk factors have not been shown to benefit from oxaliplatin, and considering the short- and long-term toxicities of this agent, oxaliplatin should not be used in the management of good-risk stage II patients. A recent update of the outcomes for stage II patients on the MOSAIC trial demonstrates no benefit to use of oxaliplatin, even in patients on the trial with one or more high-risk factors.[312] Whereas patients with exceptionally poor risk stage II tumors may still seem reasonable for consideration of oxaliplatin-containing regimens, these negative data suggest that routine use of oxaliplatin in most patients with stage II colon cancer is unlikely to be appropriate.

While perineural invasion (PNI) and tumor size do not seem to carry any prognostic significance in colorectal cancer overall, recent reports suggest that they do affect outcomes in Stage II disease and therefore may be important additional factors to consider when risk stratifying these tumors for selective postoperative chemotherapy. In a SEER database analysis of 7,719 patients, smaller tumors (<5 cm) were more likely to result in worse cancer-specific survival (HR 0.775) and, as a continuous variable, decreasing tumor size was also associated with decreasing CCS, particularly in Stage IIA and IIC tumors.[339] And in a study of 507 patients with stage I-II disease, PNI positivity was found to be associated with poorer disease-free survival at 5 years compared to PNI negative tumors (73.5% vs 88.6%).[340] Interestingly, adjuvant chemotherapy seemed to neutralize this negative impact.

Impact of Microsatellite Instability on Treatment in Stage II and III Colon Cancer

Ribic et al.[341] investigated the usefulness of MSI status as a predictor of benefit from 5-FU–based adjuvant chemotherapy in 570 stage II and III patients from five randomized trials in which no treatment control arm was used. MSI-H was exhibited in 95 patients (16.7%), MSI-L in 60 patients (10.5%), and MSS in 415 patients (72.8%). In the 287 patients who did not receive adjuvant chemotherapy, those with tumors exhibiting MSI-H had superior 5-year survival compared to patients with MSI-L or MSS tumors (HR = 0.31; $P = 0.004$). In the population of patients who received adjuvant chemotherapy, there was no difference in survival between the patients with MSI-H and patients without MSI-H ($P = 0.8$). In the patients with MSI-L or MSS tumors, chemotherapy resulted in improved survival versus no chemotherapy (HR = 0.72; $P = 0.04$). However, 5-FU/leuvocorin did not improve survival in the patients with MSI-H tumors (Table 40.10).

TABLE 40.10

Microsatellite Instability versus Outcome with 5-Fluorouracil-Based Adjuvant Chemotherapy

	No. of Patients	Five-Y Disease-Free Survival (%)	*P* Value
All Patients			
Adjuvant chemotherapy	285	70	0.06
No adjuvant chemotherapy	287	62	
Patients with MSI-L/MSS			
Adjuvant chemotherapy	230	70	0.01
No adjuvant chemotherapy	245	59	
Patients with MSI-H			
Adjuvant chemotherapy	53	69	0.11
No adjuvant chemotherapy	42	83	

MSI-L, low level of microsatellite instability; MSS, microsatellite stable; MSI-H, high level of microsatellite instability.

Data from Ribic CM, Sargent DJ, Moore MJ, et al. Tumor microsatellite-instability status as a predictor of benefit from fluorouracil-based adjuvant chemotherapy for colon cancer. *N Engl J Med* 2003;349:247–257.

5-FU/leuvocorin was associated with improved outcome in both stage II and III patients with MSS or MSI-L, with an HR of 0.67 (95% CI = 0.39 to 1.15) in stage II patients and 0.69 (95% CI = 0.47 to 1.01) in patients with stage III cancer. In contrast, in patients with MSI-H tumors, treatment did not improve survival and in fact was associated with a trend toward worse outcome for both stage II (HR for death = 3.28; 95% CI = 0.86 to 12.48) and stage III cancers (HR = 1.42; 95% CI = 0.36 to 5.56). Of note, an analysis of MSI status from the NSABP failed to corroborate the results of the Ribic et al.[341] study, and the authors concluded that in their trial there was no interaction between MSI status and treatment effect and that their data do not support the use of MSI-H as a predictive marker for chemotherapy benefit.[342] However, an updated expansion on the data from the Ribic et al.[341] report appears to corroborate the original findings, leading the authors of that study to suggest that MSI determinations should be performed on all stage II patients and that stage II patients with MSI-H should not be treated with fluoropyrimidines alone.[343] The Eastern Cooperative Oncology Group I is leading a trial (ECOG 5202) in which patients with MSI-H and absence of LOH in chromosome 18q are being selected for observation, on the hypothesis that these patients will have a highly favorable prognosis with observation alone, whereas others are being assigned to FOLFOX chemotherapy and randomized with or without bevacizumab. This trial has accrued and data are pending at the time of this writing.

An analysis of data indicates that patients with MSI-H stage III tumors, while having a more favorable prognosis than MSS tumors, do nevertheless achieve benefit from, and in the absence of contraindications, should receive, 5-FU–based adjuvant therapy.[344] This recommendation is further supported when considering oxaliplatin-based chemotherapy in stage III disease. This would be reasonable to expect, as platinum-DNA adducts are not removed by the mismatch repair enzymes that are deficient in MSI tumors. A more recent analysis of MSI impact in stage III disease in the N0147 trial suggested a more complex and nuanced impact, with MSI-H tumors having a favorable prognostic impact when arising in the right side of the colon but actually having a negative prognostic impact when occurring in the left colon.[196] The reasons for this difference remain unclear; however, the different embryologic origins of the left and right colon, and the resultant different venous drainages, may contribute to this.

The incidence of MSI has more recently been shown to vary with stage of disease presentation, with 22% of stage II patients, 12% of stage III, and only 3.5% of stage IV patients found to exhibit MSI, consistent with the data that MSI-H is a favorable *prognostic* factor. Data from the PETACC-3 trial suggest that it is a strong prognostic factor in stage II disease; however, it was less so in stage III.[345–347] As will be discussed below, immune checkpoint inhibitors have shown meaningful activity in the treatment of metastatic MSI-H colorectal cancer. Their safety and efficacy in stage III patients, however, has not been established, and use of these agents in stage III MSI-H colorectal cancer patients, until data become available to the contrary, should be regarded as investigational and is not recommended outside of a clinical trial.

Immune checkpoint blockade (ICB) shows promise in microsatellite instable patients in the metastatic setting (see below); however, thus far it has not shown activity in microsatellite stable (MSS) patients with metastatic colorectal cancer. While several small studies have suggested some degree of activity, these have so far failed to be confirmed when larger confirmatory trials have been undertaken. Further investigation is needed to identify if there are subgroups, if any, of MSS that will benefit from treatment with immunotherapy.

Circulating Tumor DNA (ctDNA)

Tumor-specific genomic alterations can be detected in the bloodstream of almost all patients with advanced colorectal cancer as well as other solid tumors. Emerging research suggests that these mutations, known as ctDNA, may not only serve as an important prognostic tool but might also enable more effective decision making and tailored treatment strategies where established protocols fall short. CtDNA has already outperformed many existing clinicopathologic risk factors as a surrogate biomarker for minimal residual disease and recurrence risk in smaller studies, with many large randomized trials now underway to further evaluate its validity, particularly for risk stratification of stage II tumors and selective use of adjuvant treatment. Other areas of investigation involve serial monitoring of ctDNA during chemotherapy, to inform on efficacy and the possibility of early discontinuation or escalation, as well as after therapy completion as a tool for detecting recurrence much earlier than conventional surveillance methods. Researchers also believe this technology could substantially improve the ease and efficiency of clinical trials if ctDNA clearance can be used as a primary end point, replacing survival metrics and thereby allowing for smaller sample sizes and shorter follow-up. Despite encouraging preliminary data, there are a number of limitations to broader clinical implementation of ctDNA-based testing including the limited sensitivity of assays and lack of standardization. However it is impossible to ignore the remarkable impact this technology could have on both practical and investigative cancer management once these challenges are addressed.[348–351]

Other Molecular Markers

As the majority of stage II, and even stage III, patients do not benefit from adjuvant therapy, it would be highly desirable to be able to identify those patients who are both at risk for recurrent disease (i.e., harbor micrometastases) and those whose micrometastases are sensitive to, and will be eradicated by, a particular chemotherapy. At present there are no such validated markers, with the possible exception of MSI, as discussed previously. KRAS or NRAS mutations have no prognostic value in the adjuvant setting, and as there is no role for use of anti-EGFR agents in the adjuvant setting, the expense of genotyping patients with less than stage IV disease is difficult to justify.[347,352] BRAF mutations appear to be prognostic of poorer overall survival in stage II disease that is not MSI-H; however, this appears to be regardless of therapy, and so again it provides no information to inform treatment decision making.[347] At present, no molecular test has been validated as useful for making adjuvant treatment decisions, and none should be utilized outside of a clinical trial. A genetic profiling assay utilizing 21-gene signature analysis has become available that has been shown to provide risk stratification with stage II patients who are classified as having from 8% to 22% chance of recurrence.[345] However, there was no interaction with treatment, meaning that the test is prognostic, identifying relatively lower or higher risk individuals, but it provided no guidance on whom to treat. Thus, despite the interesting data outlined in the following, it would appear to be of little value in decision making at this time.

Despite this limitation, research efforts have been increasingly focused on developing and refining such gene signatures over the past few years, with three that stand out presently, promising to improve and possibly replace current risk stratification models.[353] Unfortunately, none of these approaches are able to predict who will or will not benefit from adjuvant chemotherapy. While these prognostic scores may be of scientific interest, at present they are not recommended for routine clinical use, as they appear to offer little if any guidance in terms of treatment decisions.

The Oncotype DX Recurrence Score, which is the only externally validated assay available for commercial use, is based on a 12-gene signature that separates 3-year recurrence risk into low, intermediate, and high (12%, 18%, 22%, respectively), independent of clinicopathologic features and with enhanced accuracy. It has been shown to be highly prognostic in stage II and III patients

receiving 5-FU/leucovorin and enables better discrimination of absolute oxaliplatin benefit as a function of risk. A recent analysis of long-term costs and outcomes in "average-risk" stage II colon cancers (T3, MSS) concluded that this signature, when applied appropriately, could potentially reduce adjuvant chemotherapy use by 17% with increased quality-adjusted life expectancy of 0.035 years and cost savings of $2,971 per patient.[354]

This score has now been independently validated on colon cancer patients enrolled in the CALGB 9581 and NSABP C-07 trials, as well as on stage II and III rectal cancer patients from the Dutch TME trial.[355–357]

In addition, the newly published SUNRISE study provides the first validation of the Oncotype DX score in Stage III cancers that were not treated with adjuvant therapy.[358]

Another interesting signature, ColoPrint (Agendia Inc. USA, Irvine, CA), is an 18-gene prognostic classifier developed on fresh-frozen tumor tissue to identify 5-year metastasis-free survival. Separating patients into two risk categories, this assay is particularly precise in identifying low-risk stage II cases that can be managed without chemotherapy irrespective of MSI status. It also appears to better classify high-risk patients than clinicopathologic factors alone. While the signature has been validated twice in retrospective trials, it is not yet available for commercial use. A large prospective clinical validation study is under way (PARSC: Prospective Analysis of Risk Stratification by ColoPrint) that may better clarify the role of ColoPrint specifically in the management of stage II disease.[359]

Colorectal Cancer DSA (Almac Diagnostics, Craigavon, United Kingdom) is the most recently reported signature, based on 634 genes, developed to identify stage II patients at higher risk of recurrence (HR = 2.53) and cancer-related death (HR = 2.21) at 5 years. It has been shown to perform independently of known prognostic factors. While this assay is also not available for commercial use at the time of this writing, a prospective validation trial in stage II patients is being planned.[360]

Considering the high cost and number of genes that need to be examined in these expression profiles, an alternative strategy involves developing a prognostic model based on microRNAs. Studies have shown that these molecules, which are small, noncoding RNAs, play a significant role in modulating cellular pathways and have a more distinctive and reliable signature than mRNA. Zhang et al. has recently described a six microRNA classifier that demonstrated superior prognostic value compared to both clinicopathological risk factors and MMR status in stage II patients (HR 4.24 for 5-yr DFS in high-risk compared to low-risk groups).[361] It also predicted more favorable response to adjuvant therapy. While this study focused on microRNA expression from harvested tissue, Shivapurkar et al. reported a similar prognostic role for circulating microRNAs in serum samples, a panel of which were predictive of recurrence in patients with early stage, conventionally low risk disease.[362]

The lack of clear direction on the matter of treatment of good-risk stage II patients is reflected in a current consensus statement, which, while not recommending therapy for all stage II patients, does recommend a medical oncology consultation for the purpose of discussing the pros and cons of chemotherapy for all stage II patients.[363]

TREATMENT OPTIONS FOR STAGE III PATIENTS

It is clear that in the absence of medical or psychiatric contraindications, patients with node-positive colon cancer should receive postoperative chemotherapy. At the very least, a 5-FU–based regimen would appear to be appropriate, and approximately 6 months of therapy in fluoropyrimidine-only regimens would be supported by the majority of trials. The daily times five Mayo Clinic schedule or a variant has been shown to be more toxic than other 5-FU/leucovorin schedules; therefore, daily times five schedules should not be used. Oral capecitabine or oral UFT/leucovorin are acceptable alternatives if a fluoropyrimidine-only approach is selected. In the absence of medical contraindications, incorporation of oxaliplatin into the routine adjuvant treatment of colon cancer should be regarded as standard practice. Cape/Ox is the preferred option in appropriately motivated and reliable patients, and as noted above, based on the IDEA trial 3 months of total treatment is appropriate for most stage III patients. FLOX is an alternative when FOLFOX or Cape/Ox is not feasible; however, nonrandomized comparisons suggest that FLOX may carry a higher risk of serious diarrhea than FOLFOX and long-term efficacy data are less compelling. Although the pivotal adjuvant study was done with FOLFOX-4, in practice this regimen is rarely used, and the modified FOLFOX-6 regimen, which has been the basis for all FOLFOX-based National Cancer Institute Intergroup adjuvant and metastatic trials, is routinely used due to its greater convenience. The risk of peripheral neuropathy and the possibility of long-term neuropathy must be considered in the selection of therapy. At the time of this writing, FOLFOX or Cape/Ox are the regimens of choice in treatment of all patients with stage III colon cancer in the absence of specific contraindications. The long-term morbidity of oxaliplatin treatment has become more appreciated; however, it is anticipated that risk stratification strategies may become available in the near future to identify those patients who are likely to benefit from oxaliplatin treatment. Data have failed to show definitive noninferiority of 3 months as compared to 6 months of therapy; however, consideration of 3 months of oxaliplatin-containing treatment should be considered in most stage III patients following a discussion of the risks and benefits of shorter versus longer exposure.

Oxaliplatin is associated with side effects of peripheral neuropathy (PN) that can adversely affect patient function and quality of life leading to reduced duration of treatment. At present, no interventions have been convincingly shown to either prevent, mitigate, or treat oxaliplatin neuropathy. Zhang et al. performed a double-blind randomized study of 179 patients receiving Cap Ox for six cycles. Patients were randomized to take w-3 PUFAs (polyunsaturated fatty acids) 600 mg 3 ×/day or placebo during chemotherapy and 1 month afterwards. Significant reductions in PN incidence and severity were found in the treatment group as well as improved motor and nerve conduction. Further validation of these findings and research into similar preventative therapies is certainly warranted.[364]

Irinotecan-based regimens should not be used in the adjuvant setting, as randomized data have shown increased toxicity and no long-term benefit. Bevacizumab, cetuximab, and panitumumab should also not be used in the adjuvant setting, as they add toxicity and expense and do not add benefit.

Timing of Treatment

Adjuvant chemotherapy is usually started once the patient has adequately recuperated from surgery. Traditionally, adjuvant chemotherapy is commenced between 3 and 8 weeks of surgery. However, this time frame is somewhat arbitrary, based largely on what has been mandated in clinical trials. A 2010 meta-analysis pooled data from 13,158 patients with stage II or III disease and concluded that delaying treatment beyond this interval was associated with inferior survival (RR = 1.20).[365] However, these data are retrospective, nonrandomized, and fail to adequately consider the collateral implications of whatever medical and surgical conditions might have contributed to the delay in start of therapy. As such, these findings should be regarded as hypothesis generating only, and certainly not definitive. Another study that included only stage III cancers found no such clear correlation. Secondary analysis did demonstrate a trend toward poorer outcomes after 8 weeks, particularly in younger patients (<66 years) as well as significantly inferior survival for patients beginning chemotherapy after 9 and 10 weeks (HR of death = 1.68 and 1.67, respectively).[366]

As outlined above, the majority of studies on adjuvant chemotherapy support 6 months as the ideal duration for therapy regardless of agent or regimen, with the IDEA trial suggesting a possible role for 3 months duration in selected good risk patients. However, due to substantial side effects, many patients are unable to complete the full treatment course and it is unclear whether this is associated with a true negative impact on local disease control and survival. Retrospective data suggest that early discontinuation of FOLFOX (<10 cycles) does not impact survival outcomes.

INVESTIGATIONAL ADJUVANT APPROACHES

Portal Vein Infusion

The NSABP C-02 trial randomized 1,158 patients with Dukes A, B, or C colon cancers to either a 7-day portal vein infusion of 5-FU (600 mg/m^2 per day) or to surgery alone.[367] A modest, albeit statistically significant, advantage in DFS (74% vs 64% at 4 years) was demonstrated for the group who received intraportal chemotherapy; however, no difference was seen in the incidence of hepatic recurrences.

Similar findings were reported from a 533-patient trial performed by the Swiss Group for Clinical Cancer Research.[368,369] In this trial, intraportal chemotherapy included 10 mg/m^2 mitomycin C by 2-hour infusion followed by a 7-day infusion of 5-FU at a dose of 500 mg/m^2 per day. The 5-year DFS and overall survival were modestly improved in the intraportal treatment versus surgery-only groups (57% vs 48%, and 66% vs 55%, respectively).

Subsequently, a large meta-analysis of intraportal chemotherapy trials involving over 4,000 patients in 10 randomized studies revealed only a 4% improvement in 5-year overall survival for the patients who received portal infusion. At present, intraportal adjuvant chemotherapy has not been accepted as routine practice and remains limited to clinical investigations.

The OCTREE regimen is a new variation on portal vein infusion found to confer a 34% risk reduction in tumor recurrence for patients with Stage II and III disease in a newly published randomized trial. In contrast to prior studies of portal vein infusion, where 5-FU was commonly administered alone via the intraportal route and without any subsequent oxaliplatin-based adjuvant therapy, this novel regimen entails intraoperative bolus administration of 5-FU and oxaliplatin followed by adjuvant mFOLFOX6 for six months. The authors report a significant improvement in the primary end point of 3-year DFS with this regimen compared to standard mFOLFOX6 (85.2% vs 75.6%, $P = 0.30$). There was no difference in grade 3 or 4 toxicity. Five-year overall survival data are still pending.[370]

Intraperitoneal Chemotherapy

Hyperthermic intraperitoneal chemotherapy has been explored as a possible means of providing a benefit in patients at high risk for developing peritoneal metastases. Sammartino et al.[371] explored this hypothesis in a small case control study of patients with T3/4 lesions and mucinous or signet ring cell histology who underwent either standard resection or extended surgery (including omentectomy, bilateral adnexectomy, hepatic round ligament resection, and appendectomy) followed by hyperthermic intraperitoneal chemotherapy. The experimental group experienced statistically significantly decreased rates of peritoneal recurrence (4% vs 22%) as well as improved DFS (36.8 months vs 21.9 months; $P < 0.01$). Large randomized trials are necessary, however, before nonresearch use of this highly aggressive and potentially toxic treatment strategy could be considered.

Results from the Dutch COLOPEC trial, the first of these studies, have been disappointing however. Two hundred four patients who underwent curative resection for T4 or intra-abdominally perforated cM0 colon cancer were randomized to receive adjuvant HIPEC with oxaliplatin followed by routine systemic chemotherapy or systemic chemotherapy only. In the intention-to-treat analysis, there was no significant difference in the primary end point of peritoneal disease-free survival at 18 months (80.9% HIPEC vs 76.2% control arm, $P = 0.28$). Postoperative complications occurred in 14% of patients receiving HIPEC, including one case of encapsulating peritoneal sclerosis.[372,373]

The primary downfall of aggressive loregional procedures such as CRS+HIPEC is distant or extra-abdominal tumor spread and therefore it has become increasingly important to identify prognostic factors that will aid in patient selection. In addition to clinical factors such as the peritoneal carcinomatosis index, tumor grade, response to standard chemotherapy, and the peritoneal surface disease severity score, genetic, and molecular factors may be also be taken into consideration. Melero et al. recently reported on the prognostic significance of circulating tumor cells (CTC) in a study of 14 patients eligible for CRS+HIPEC. The authors found that detection of CTC in the peripheral blood correlated with distant disseminated disease and was a significant predictor of progression after treatment. This finding was not surprising as the presence of CTC is considered indicative of hematogenous spread and has been proven a poor prognostic factor for survival in many cancers. Pending larger studies, CTC detection and characterization may prove a useful tool for improving selection of patients who would most benefit from CRS+HIPEC.[374]

While the cytoreductive techniques of this surgery are becoming increasingly standardized across the world, there exists tremendous variation in the HIPEC component, including most importantly the choice of infusional drug. The American Society of Peritoneal Surface Malignancies published a comparison of cytoreductive surgery and HIPEC using oxaliplatin versus mitomycin C in patients with peritoneal carcinomatosis from colorectal cancer. When data were stratified by Peritoneal Surface Disease Severity Score (PSDSS), mitomycin C was associated with significantly improved survival for patients with lower disease burden (54.3 vs 28.2 months in PSDSS I/II, $P = 0.012$). No difference was found for patients with PSDSS III/IV. The authors cite the need for larger prospective trials to guide definitive recommendations.[375]

Vaccines

Vaccination strategies endeavor to stimulate the patient's immune system to recognize and eradicate the patient's tumor cells. An ideal immunologic target molecule would be a highly antigenic epitope that is always expressed on the tumor and never expressed on normal tissue. Such an ideal target has yet to be identified; however, a number of approaches have been explored.

CEA is a commonly expressed antigen in colorectal carcinomas. Unfortunately, CEA does not appear to be particularly immunogenic. Several approaches have been pursued in an attempt to increase immune recognition of CEA. Thus, a number of avenues of investigation are being pursued; however, at this time the use of vaccine therapy for treatment of resected colon cancer remains highly investigational.

One tumor-associated antigen that may be a more promising candidate is the MUC1 glycoprotein, which is abnormally expressed on neoplastic cells in a hypoglycosylated form that induces humoral and cellular response. A vaccine based on this antigen was found to be highly immunogenic in the premalignant setting, inducing long-term memory responses and no significant toxicity when administered to patients with advanced colonic adenomas. Subsequent studies will determine whether these results translate into meaningful clinical outcomes.[376]

Active Specific Immunotherapy

Irradiated cancer cells maintain their immunogenicity; however, they are unable to proliferate. Active specific immunotherapy is a maneuver in which patients are immunized with a preparation of

their own irradiated tumor cells plus an immunostimulant such as bacillus Calmette-Guérin. This technique has been explored for some time now as a potential adjuvant immunotherapy for CRC. Overall, trials have failed to show a benefit for the use of active specific immunotherapy in the management of colon cancer, and its use should remain limited to investigational settings.

Preoperative Chemotherapy

Investigators are currently exploring the role of preoperative chemotherapy in the management of nonmetastatic disease. A pilot phase of the first randomized trial to address this topic (FOxTROT) has recently been completed, demonstrating the feasibility and safety of this approach in locally advanced, operable colon cancer. The 150 patients who underwent meticulous radiologic staging were randomly assigned to receive 6 weeks of preoperative oxaliplatin/5-FU/l-folinic acid followed by surgery and 18 weeks of postoperative chemotherapy or upfront resection followed by 24 weeks of treatment. There were no significant differences in postoperative morbidity between the two groups. While a small proportion of the patients receiving preoperative therapy had apparent progression during the time between staging and surgery, there were no tumor-related complications during this interval. Overall, preoperative therapy resulted in significant downstaging, including reductions in apical node involvement and incomplete resections as well as two pathologic complete responses. Whether these results will translate into improved survival and potentially change the accepted pathway for management of nonmetastatic disease remains to be seen. A number of studies are currently under way to address this matter as well as the potential role of biologic agents.[377] The PRODIGE 22–ECKINOXE trial, for example, incorporated a third arm to include cetuximab in kras–wild-type patients, comparing FOLFOX-4 alone for four cycles and FOLFOX + cetuximab to the control arm of resection followed by adjuvant treatment.[378] While the results from this study reinforced the feasibility and tolerability of perioperative chemotherapy, the primary end point of improved major pathological response rate (TRG1 8%) was not reached. The authors hypothesized that this could have been due to the limited number of treatment cycles, the short time interval between the last cycle and surgery, as well as the high percentage of low risk stage II patients in the experimental arm as a result of CT overstaging. When moderate and major regression rates were pooled, the experimental arm did show a trend to tumor downstaging with significant regression comparable to results from the FOXTROT pilot study (TRG1-2 44% vs 8%, $P < 0.001$). Of note the FOLFOX-cetuximab arm was stopped at interim analysis due to lack of efficacy.

Despite this setback and based on the positive results from phase II trials, consensus guidelines added neoadjuvant chemotherapy as a treatment option for patients with clinical T4b disease. While additional long-term results are still pending, Dehal et al. examined utilization and outcomes for all patients with nonmetastatic clinical T3 and T4 disease from the National Cancer Data Base between 2006 and 2014. Of 27,575 patients, 97% received standard adjuvant therapy, while only 3% received neoadjuvant treatment. Patients with stage T4b disease demonstrated significantly improved survival with neoadjuvant therapy, with a 23% lower risk of death at 3 years compared to those receiving a postoperative regimen only. This benefit was not seen in T3 or T4a disease. The authors hope that their study will encourage increased awareness and utilization of this new treatment paradigm in appropriate patients, and especially in those with T4b tumors, while prospective data are awaited.[379]

Patient selection for neoadjuvant chemotherapy remains challenging, particularly when current imaging techniques are notoriously unreliable for the assessment of lymph nodes and hence the identification of Stage III disease. Norgaard et al. tackled this recently, focusing on the utility of CT in differentiating high risk features of the tumors themselves, including T stage, extramural tumor and venous invasion (ETI and EVI), and distance to the nearest retroperitoneal fascia (DRF). While EVI and DRF were not consistently identified, CT accurately predicted 81% of T4 tumors and 71% of T3 tumors with extramural invasion >5 mm, with only 7% understaged, indicating that this may be a reasonable modality for selecting patients who would benefit from neoadjuvant therapy.[380]

Other imaging modalities that may prove even more useful than CT include MRI and endoscopic ultrasound. In a retrospective study of 55 patients with tumors throughout the colon, MRI demonstrated a high sensitivity (72% to 91%) and specificity (84% to 89%) in detecting T3/T4 tumors with a low sensitivity (43% to 67%) but high specificity (75% to 88%) for T3cd/T4 tumors. MRI also had a high sensitivity and moderate specificity for detecting serosal involvement and extramural vascular invasion and was moderately accurate for detecting nodal metastases. Another report compared the accuracy of EUS to CT scan specifically in left-sided tumors. The authors found that while the two modalities had comparable sensitivities for T stage evaluation, EUS had significantly higher specificity in the evaluation of "low risk" tumors (T1-3 with <5 mm extramural invasion). Results for more advanced lesions were difficult to interpret due to small sample size and both EUS and CT were poor in predicting N stage.[381,382]

CT colonography is another option for preoperative staging but with widely variable accuracy. A recently published retrospective study that focused on the FOxTROT trial inclusion criteria for the identification of high-risk tumors found that the specificity for T stage was higher compared to prior data on CT alone, which may be directly explained by the differences in technology, i.e., smaller section thickness and lack of underdistension with CTC.[383] On the other hand, CTC did not confer any additional advantage in N-staging over standard CT or other modalities described above. Further validation studies are still necessary to determine the optimal approach.

FOLLOW-UP AFTER MANAGEMENT OF COLON CANCER WITH CURATIVE INTENT

Follow-up after definitive management has two primary goals. First, patients with a history of CRC are at higher risk than the general population for a second colon cancer primary.[384,385] A colonoscopic screening may benefit in the early detection of a second primary malignancy or detection of a benign polyp, which can then be resected, potentially preventing the development of an invasive cancer.

Second, surveillance may increase the chance of identifying local regional or distant recurrence that is potentially curable by surgery. It should be noted that it is this detection of potentially curable recurrent or second primary disease that justifies routine postoperative surveillance. To date, there are no compelling data that indicate that early detection of unresectable asymptomatic metastatic disease is of benefit to the patient. In other words, if recurrent disease is unresectable and therefore incurable, there is no urgency to identify it; there is no compelling evidence that the early initiation of palliative chemotherapy is of benefit in the asymptomatic, incurable patient. Although the choice of follow-up routine and which studies to include in that follow-up have been the subject of much debate in the colon cancer literature,[386–389] subsequent analyses have shown that most interventions that have been considered are not value added and not appropriate for routine use. The American Society of Clinical Oncology (ASCO) has recently updated its guidelines for postsurgical follow-up.[390]

These recommendations are for physical examination and blood CEA monitoring every 3 months to 6 months for the first

3 years and every 6 months for years 4 and 5. CT scans of the chest and abdomen are recommended once a year for the first 3 years. Positron emission tomography (PET) scans are specifically not recommended for routine screening and surveillance. Consensus guidelines differ only slightly in that they recommend annual CT scanning for up to 5 years and include CT scans of the pelvis. Repeat colonoscopy should be performed 1 year after surgery or immediately after adjuvant therapy if no preoperative exam was completed and then every 3 to 5 years depending on findings. CT scans and CEA monitoring should not be continued beyond 5 years. In addition, the authors recommend secondary prevention measures, such as maintaining a healthy body weight and active lifestyle. For patients who are not candidates for further surgery or systemic therapy, surveillance tests should be discontinued. Of note, older studies advocating routine monitoring of complete blood count, liver function studies, lactate dehydrogenase, chest X-rays, and fecal occult blood monitoring have not been supported by subsequent data, and none of these are recommended for routine monitoring of patients at this time. The role of CEA measurement in patients following definitive management of CRC has been controversial.[140,391,392] ASCO carefully reviewed its utility in an additional panel guided in 1996.[393] Their recommendation for CEA monitoring then, which was confirmed in the surveillance guideline panel review, was postoperative serum CEA testing to be performed every 3 months in patients with stage II or III disease for up to 3 years after diagnosis. An elevated CEA level, if confirmed by retesting, warranted further evaluation for metastatic disease.

This further workup of an elevated CEA typically consists of a colonoscopy and a CT scan of the chest, abdomen, and pelvis. If these studies are negative, the clinician is faced with a dilemma. The question of what to do in the face of a rising serum CEA level in the absence of imageable disease by conventional imaging modalities is one that has been addressed in clinical trials.[394–399] Strategies to image CEA expression might improve upon the detection capability of standard imaging studies. Several studies were performed using immunoscintigraphy with an antibody directed against CEA or Tag72, a CEA-like glycoprotein (CEA scan and OncoScint [Cytogen Corp, Lonza Biologics, Princeton, NJ] scan, respectively).[400–402] The results of these studies using antibody-directed immunoscintigraphy were variable, and at this time, CEA scintigraphy is no longer considered in standard care.

Based on these findings, it appears that serum CEA surveillance following definitive management of a primary CRC is a reasonable surveillance technique. If the CEA level is elevated on repeat testing, imaging studies should be performed consisting of CT scans and a thorough evaluation of the colon with colonoscopy performed as well. If no recurrence or second primary is detected, watchful waiting and repeat of CT imaging at approximately 3-month intervals versus a PET scan with FDG can be considered. If disease is discovered, it should be managed as indicated. If no disease is detected, then continued surveillance is warranted with repeat CEA levels and CT or magnetic resonance imaging at intervals.[397,403]

The role of physical examination has also been evaluated.[404] The ASCO panel noted that no formal examination of the contribution of physician's history and physical examination to help outcomes of CRC has been performed. However, data from the larger studies of surveillance showed that 80% of recurrences were found by CEA testing, whereas only 20% were found by routine history and physical examination done at the same time.[405] This has been confirmed by other studies.[406,407] Although no direct effects were shown on history or physical examination about the impact and detection or outcome in the surveillance period, a physician–patient encounter provides a vital link for other studies that may influence outcome. Therefore, while not in itself substantiated by the data in the literature, it is felt that routine postresection visits be performed every 3 to 6 months for the first 3 years following resection and every 6 months during years 4 and 5.

The role of liver function tests as a means for detecting colorectal recurrence has also been carefully evaluated. No studies that were reviewed by the ASCO panel demonstrated any benefit for the routine use of liver function test measurements in the post-surveillance period.[404,408] In fact, studies suggest that other routine blood tests such as CEA detected recurrence far earlier than liver function test abnormalities.[140] Therefore, the 2005 ASCO consensus panel did not recommend the routine use of liver function test measurements in the postresection surveillance period.

Routine fecal occult blood testing, routine complete blood counts, and routine chest X-rays were all not thought to be of benefit in postoperative surveillance. Although the panel was not in uniform agreement with respect to chest X-ray, it was thought that all three of these modalities should be reserved for the evaluation of the patient with evidence of recurrence such as a rising CEA level or a positive endoscopy. Each of these modalities in and of itself was not found to be useful.

The panel recommended that annual CT of the chest and abdomen (with pelvis for rectal cancer) be performed for 3 years in those patients at higher risk for recurrence who could be candidates for curative-intent surgery of a recurrence. With respect to colonoscopy and flexible proctosigmoidoscopy, the panel, after reviewing the literature, recommended that all patients have a colonoscopy for the pre- or perioperative documentation of the cancer and to ascertain that the remainder of the colon is free from polyps. Further, the panel agreed that the data were sufficient to recommend colonoscopy at 3 years to detect new cancers and polyps and then every 5 years if normal. However, they did not recommend routine annual colonoscopy as follow-up following definitive management of patients with CRC. Further, the panel concluded that colonoscopy was superior to flexible proctosigmoidoscopy and therefore should be performed as previously discussed for patients following both colon and rectal cancer surgery. Other studies have also supported the routine use of colonoscopic examination following definitive management of CRC.[409,410]

A meta-analysis and systematic review of randomized trials to address the impact of close postoperative surveillance on overall survival following definitive management of CRC was also performed by Renehan et al.[411] A total of five randomized trials that met their inclusion criteria were reviewed, representing 1,342 patients. For four of the studies, intensive follow-up consisted of blood work including serum CEA, colonoscopy, physical examination, abdominal ultrasound, and CT scans. In one study, no CEA measurements or CT scans were performed. Follow-up in the intensive arm was performed every 3 months for 2 years and then every 6 months thereafter up to 5 years, with yearly CT scans and endoscopy. All five studies had a control arm subjected to a less aggressive follow-up regimen, which varied from study to study ranging from no specific follow-up to interval laboratory tests and plain X-rays or ultrasound. They found that there was an absolute reduction in mortality of 9% to 13% by employing an aggressive follow-up regimen, consisting of serum CEA measurements and CT scans. Two studies in particular showed the greatest impact on survival.

In summary, a rational postoperative surveillance program would include CEA measurements every 3 to 6 months and a yearly CT scan of the chest, abdomen, and pelvis (for rectal cancer) for the first 3 years. Colonoscopy can be performed every 3 years to 5 years following the resection. At the time of CEA measurements, a physician encounter should be scheduled where a discussion of patient symptoms and a physical examination can be performed. If a rising serum CEA is detected on two consecutive measurements in the absence of imageable disease by CT scan, a PET scan with FDG can be considered. Lesions found on colonoscopy should be managed appropriately either with colonoscopic resection or surgical management. These surveillance guidelines should allow for the early detection of either resectable recurrence or second primary lesions and therefore the potential to impact patient outcome.

SURGICAL MANAGEMENT OF STAGE IV DISEASE

For a select group of patients with metastatic CRC, complete surgical resection of stage IV disease (discussed in detail in Chapter 41) may be an option and may provide a long-term survival advantage. This is especially true with respect to metastatic sites in the liver and lung. Resection of locoregional recurrence can also benefit the patient with respect to local control and overall outcome. Numerous regional approaches have also been explored for the treatment of stage IV colon cancer depending on the organ or body cavity involved. Organ-specific infusional therapy, isolated or continuous perfusion therapy, radiofrequency ablation or cryotherapy, surgical debulking, and radiation are all technical approaches that have been performed. Many of these regional strategies as well as surgical metastasectomy will be discussed in separate chapters and therefore will not be specifically addressed here.

Some patients with initially unresectable metastatic disease may also become candidates for curative procedures based on their response to chemotherapy.[414]

MANAGEMENT OF UNRESECTABLE METASTATIC DISEASE

Unresectable metastatic CRC is generally not curable with current technology. Management centers around palliation and control of symptoms, control of tumor growth, and attempts to lengthen progression-free and overall survival. Given the palliative nature of such treatments, extreme care must be taken to adequately assess each individual's potential for both benefit and harm from chemotherapy. Care must also be taken with surgical interventions as well. Quality-of-life issues must be frankly and objectively discussed with patients and their caregivers so that informed decisions can be made and expectations can be contained within a realistic framework. The issue of whether patients presenting with unresectable stage IV disease should have their primary tumors resected has been the matter of some debate. Although such resections had been routinely performed in the past, more recent data suggest that such interventions are not required and may in fact be counterproductive. Temple et al.[413] reviewed the linked SEER-Medicare databases and found that of 9,011 elderly patients presenting with synchronous stage IV disease, 72% had undergone resection of the primary and the 30-day mortality for these resections was 10%. In a retrospective review, Poultsides et al.[414] demonstrated that 93% of 233 patients never required surgical intervention on their primary tumors. More recently, the NSABP C-10 trial prospectively addressed this question, treating 90 patients with unresectable stage IV disease and intact, asymptomatic primary tumors with initial medical management with FOLFOX-6 plus bevacizumab. Twelve patients (14%) experienced major morbidity due to the intact primary tumor. This study met its prespecified end points for acceptability of initial nonoperative management, and the investigators concluded that good performance status patients with asymptomatic primaries can be spared initial noncurative resection of their primaries.[415]

Surgical intervention can be a very effective method of palliation and is often indicated in cases of impending obstruction, perforation, bleeding, or pain. However, it can also be associated with high postoperative morbidity and mortality. Stillwell et al.[416] identifies several adverse prognostic factors that can guide clinicians considering surgical palliation versus other less aggressive maneuvers. In a retrospective analysis of 379 patients who underwent palliative resection, elderly (≥70 years) patients with advanced local disease or extrahepatic metastases were at greatest risk for postoperative mortality. Other independently associated factors included emergency operation and medical complications. Advanced nodal disease (N2) and poor tumor differentiation were significant predictors of decreased long-term survival.[416]

Li et al. also reported that age over 70 years was negatively associated with overall survival in a more recent series of 110 patients who underwent primary tumor resection in the setting of metastatic disease.[417] Additional poor prognostic factors included alkaline phosphatase (ALP) level over 160 IU/I, ascites, and a platelet/lymphocyte ratio greater than 162. The authors further combined these parameters into an "AAAP scoring system" to stratify the study patients into low-, high-, and, medium-risk groups. Two-year overall survival was significantly different between the groups, indicating that this might be a useful tool for surgical decision making in this population moving forward. The role of primary tumor resection in metastatic disease should be more definitely addressed, however, in a new randomized controlled trial currently underway.[418] The authors aim to recruit 480 patients with colon or upper rectal cancer and asymptomatic, synchronous, unresectable metastases. Patients will be randomized to receive chemotherapy with or without primary tumor resection and followed for a primary end point of 2-year overall survival. Secondary end points will include primary tumor-related complications, quality of life, surgery-related morbidity and mortality, chemotherapy-related toxicities, and total cost of care. The authors hypothesize that the group undergoing resection will show a 10% improvement in 2-year overall survival. Regardless of outcome, the findings of this report will be instrumental in defining the optimal treatment approach for this challenging subset of colorectal cancer patients.

While liver resection is the only currently considered curative option for patients with colorectal liver metastases (CRLM), 5-year survival rates are low, ranging from 25% to 50%, and only about 20% of patients are candidates for this approach, with the majority developing further recurrence. Liver transplantation has been investigated as an alternative option, but early reports of <20% survival at 5 years were discouraging. However, a 2013 pilot study from Oslo (SECA-I) revived interest in this strategy, demonstrating a 5-year overall survival rate of 60 percent in 21 patents with unresectable CRLM who underwent liver transplantation following neoadjuvant therapy. Negative predictive factors for short OS included maximum tumor size greater than 5.5 cm, progressive disease on chemotherapy, interval from primary tumor resection to transplant less than 24 months, and pretransplant CEA level >80 ug/mL. Based on these findings, the SECA-II study established more stringent inclusion criteria for enrollment, with interval from diagnosis to transplant of at least 1 year and chemotherapy response rate of at least 10%. The authors recently published the best oncologic outcomes ever reported for this patient population with a 5-year Kaplan-Meier-estimated OS rate of 83% at a median follow-up of 36 months in the first 15 patients. Enrollment is ongoing for both studies as well as an abundance of other newer investigations into this approach, including the SECA-III that will compare 2-year OS in 30 patients randomized to liver transplantation or palliative chemotherapy +/- other locoregional treatments.[419–421]

The chemotherapy options available and the developmental work that supports their utility are outlined subsequently. It is of paramount importance to keep in mind that virtually all of the clinical trials done in patients with metastatic disease were performed by design on patients who were in good overall general medical condition. Entry criteria for most trials require a favorable performance status and acceptable bone marrow, renal, and hepatic function, and they often specify evidence of reasonable nutritional intake.

It is not reasonable to extrapolate the results of these trials to patients who do not conform to these entry criteria. The likelihood of benefit in a poor performance status patient is substantially diminished, and the likelihood of a serious adverse event is greatly increased. Patients with hepatic or renal dysfunction may be particularly prone to additional toxicity if the drug is cleared or metabolized by these organs. Patients with marginal nutritional intake may have their nutritional deficiencies further exacerbated by drugs that produce nausea or anorexia, and patients with partial or complete bowel obstruction or other causes of prolonged GI transit time may have increased toxicity from those drugs that undergo an enterohepatic recirculation.

TABLE 40.11

Commonly Used 5-Fluorouracil Regimens[a]

Name of Regimen	Author (Ref.)	Schedule (All Agents Administered Intravenously)
Roswell Park	Haller et al., 1998[299]	LV 500 mg/m^2 over 2 hr; 5-FU 500 mg/m^2 by bolus 1 hr into LV infusion. Treatments given weekly for 6 consecutive wk, repeated every 8 wk.
Low-dose weekly LV	Jager et al., 1996[423]	LV 20 mg/m^2 over 5–15 min, followed by bolus 5-FU 500 mg/m^2; treatments given weekly for 6 consecutive wk, repeated every 8 wk.
Protracted venous infusion	Lokich et al., 1989[424]	5-FU 300 mg/m^2/d by continuous infusion.
AIO (weekly 24-hr infusion)	Kohne et al., 1998[425]	LV 500 mg/m^2 over 2 hr, followed by 5-FU 2,600 mg/m^2 over 24 hr, repeated weekly.
LV5FU2	de Gramont et al., 1997[441]	LV 200 mg/m^2 over 2 hr days 1 and 2, followed by bolus 5-FU 400 mg/m^2/day 1 and 2, followed by 5-FU 600 mg/m^2 over 22 hr days 1 and 2; cycle repeated every 14 d.
Simplified LV5FU2	Adapted from Andre et al., 1999[435]	LV 400 mg/m^2 over 2 hr, followed by bolus 5-FU 400 mg/m^2, followed by 5-FU 1,200 mg/m^2/d times 2 d (2,400 mg/m^2 over 46–48 hr); cycles repeated every 14 d.

LV, leucovorin; 5-FU, 5-fluorouracil; AIO, Arbeitsgemeinschaft Internistische Onkologie (Oncology Working Group, Germany).
[a]Doses listed are recommended starting doses for good performance status patients with normal renal, hepatic, and bone marrow function. Individual dose adjustments may be required.

Thus, chemotherapy for patients with incurable metastatic disease should be approached with appropriate caution. Good performance status in well-motivated patients with good bone marrow reserve and good organ function portend a significant potential for substantial benefits from chemotherapy and should be strongly considered for aggressive therapy. Patients with poor performance status and significant comorbidities should be considered for either less aggressive therapies or for supportive care only.

Fluorouracil

Virtually the entire history of chemotherapy for CRC has revolved around the use of 5-FU. Developed by Heidelberger et al.[422] and patented in 1957, it is a source of frustration and humility for investigators working to move beyond it that over 50 years later this agent remains at the very core of most chemotherapeutic approaches to CRC.

5-FU must be metabolized before it can exert cytotoxic activity. The details of 5-FU metabolism are covered in a separate chapter. The history of investigations of 5-FU in CRC treatment has been well summarized in previous editions of this book.[295] 5-FU remained the only drug available to treat CRC for almost four decades, during which time numerous agents were studied for their ability to "biomodulate" 5-FU. Of these, only leucovorin remains in use today, and it is debatable whether this reduced folate truly contributes to the efficacy of 5-FU. Most studies that evaluate the same dose of 5-FU with or without leucovorin find that both activity and toxicity are increased in the leucovorin arm, whereas studies that evaluate single-agent 5-FU versus an equitoxic schedule of a lower dose of 5-FU plus leucovorin find equivalent activity. Nevertheless, use of leucovorin persists in most standard regimens today. Data comparing bolus versus infusional schedules of 5-FU show a slight benefit for infusions. These infusional schedules achieved widespread acceptance in Europe sooner than in the United States. It was not until the advent of combination schedules of 5-FU plus other active agents that the benefits of infusional schedules, especially in terms of improved toxicity, asserted themselves in North American practice. A selection of commonly used 5-FU regimens is outlined in Table 40.11.[423–425]

Capecitabine

Capecitabine is a 5-FU precursor that is administered orally. It is absorbed intact through the gut and then activated by a series of enzymatic alterations. Some data suggest that thymidine phosphorylase levels are higher in tumor than in normal tissue. This could, in theory, provide a degree of preferential intratumoral activation; however, clinical trials do not appear to support a substantially better therapeutic index than 5-FU.[426] Phase 2 studies demonstrate that this agent has substantial activity in CRC, with an acceptable toxicity profile.[427] Because the addition of leucovorin did not appear to show any benefit, clinical development went forward without additional biomodulation. Phase 3 randomized clinical trials, performed both in the United States and Europe, have now shown that this orally administered agent is at least as effective as intravenous 5-FU/leucovorin, and the side effect profile of capecitabine is superior to the daily times five Mayo Clinic schedule of 5-FU/leucovorin.[428,429] However, as this bolus daily times five schedule is now known to be unacceptably toxic and is not used anymore, the relevance of the toxicity comparison is questionable. Head-to-head studies of modern infusional 5-FU/leucovorin regimens versus capecitabine have never been done; however, a reasonable extrapolation from available data would suggest that these two approaches are extremely similar in efficacy and tolerability. The dose used in these pivotal trials was 1,250 mg/m^2 given twice daily for 14 days followed by a 7-day rest. The major side effects of capecitabine appear to be palmar-plantar erythrodysesthesia, commonly called hand-foot syndrome, and to a lesser extent diarrhea. The hand-foot toxicity is frequently a dose-limiting side effect, and although the approved starting dose is 1,250 mg/m^2 twice daily, this dose is based on trials conducted mainly in Europe. For unclear reasons, possibly related to higher serum folate levels, American patients tolerate capecitabine less well than European patients, and clinicians in the United States often choose to initiate therapy at a lower dose and escalate if little or no toxicity is seen.

Irinotecan

Irinotecan (CPT-11) is a semisynthetic derivative of camptothecin, a plant alkaloid extracted from the wood of the Asian tree *Camptotheca acuminate*.[430] CPT-11 possesses a bulky dipiperidino side chain linked to the camptothecin molecule via a carboxyl-ester bond. This side chain provides solubility but greatly decreases anticancer activity. Carboxylesterase, a ubiquitous enzyme with primary activity in the liver and gut, cleaves the carboxyl-ester bond to form the more active metabolite 7-ethyl-10-hydroxycamptothecin (SN-38).[431] SN-38 is as much as 1,000-fold more potent in inhibiting topoisomerase I than irinotecan and is thus the predominant

TABLE 40.12

Commonly Used Irinotecan/5-Fluorouracil Combination Regimens[a]

Name of Regimen	Author (Ref.)	Schedule (All Agents Administered Intravenously)
FOLFIRI	Douillard et al., 2000[317]	Irinotecan 180 mg/m^2 over 2 hr; LV 200 mg/m^2 concurrently with irinotecan (can be given in same line through "Y" connector); followed by 5-FU bolus 400 mg/m^2, followed by 5-FU 600 mg/m^2 infusion over 22 hr. Irinotecan given day 1 only. All other meds given days 1 and 2. Cycle repeated every 14 d.
FOLFIRI (simplified)	Andre et al., 1999[433]	Irinotecan 180 mg/m^2 over 90 min; LV 400 mg/m^2 concurrently with irinotecan (can be given in same line through "Y" connector); followed by 5-FU bolus 400 mg/m^2, followed by 5-FU 1,200 mg/m^2/d times 2 d (2,400 mg/m^2 infusion over 46–48 hr). Cycle repeated every 14 d.
FUFIRI	Douillard et al., 2000[317]	Irinotecan 80 mg/m^2, then LV 500 mg/m^2, followed by 5-FU 2,300 mg/m^2; all drugs given weekly for 6 wk, repeated every 7 wk.

[a]Doses listed are recommended starting doses for good performance status patients with normal renal, hepatic, and bone marrow function. Individual dose adjustments may be required.
FOLFIRI, irinotecan, 5-FU, LV; LV, leucovorin; 5-FU, 5-fluorouracil; FUFIRI, 5-FU, LV.

active form of the drug. CPT-11 is often considered to be a prodrug for SN-38; however, this concept may be a bit too simplistic, as achieved CPT-11 concentrations may be several logs higher than those of SN-38.

Camptothecin, CPT-11, and SN-38 function as inhibitors of topoisomerase I (topo I). Topo I is a nuclear enzyme that aids in DNA uncoiling for replication and transcription. When topo I binds to DNA, it causes a reversible single-stranded break in the DNA, allowing the intact strand to pass through the break to relieve torsional stress on the coiled helix, and then reseals the break. CPT-11 and SN-38 stabilize these single-stranded breaks. Although the stabilized breaks do not cause irreversible damage, the collision of replication forks with open single-stranded breaks results in double-stranded breaks, leading to lethal DNA fragmentation. The early development of irinotecan and its single-agent schedules have been well documented in earlier editions of this textbook.[295]

CPT-11 in First-line Combination Regimens

Numerous phase 1 combinations of 5-FU, usually with leucovorin, plus CPT-11, were tried. Saltz et al.[432] reported a phase 1 trial built on the weekly CPT-11 schedule that had been selected for phase 1 development in North America. A low dose of weekly leucovorin was utilized in order to reduce the potential for 5-FU/leucovorin–induced diarrhea. The phase 1 trial showed that the full single-agent dose of 125 mg/m^2 of CPT-11 could be given with 500 mg/m^2 of 5-FU and 20 mg/m^2 leucovorin, with all drugs given weekly for 4 consecutive weeks followed by a 2-week break. This and other CPT-11/5-FU/leucovorin regimens are summarized in Table 40.12. This combination of IFL was compared to the Mayo Clinic schedule of 5-FU/leucovorin in a multicenter, multinational phase 3 trial.[273] For regulatory reasons, a single-agent CPT-11 arm was included as well. The IFL arm was found to be superior to Mayo Clinic 5-FU/leucovorin in terms of response rate, time to tumor progression, and overall survival. The CPT-11–alone arm appeared to be comparable in efficacy to the 5-FU/leucovorin arm. The overall incidence of severe toxicity was similar in all arms of this trial. More serious diarrhea and vomiting were seen with IFL, while more neutropenia, neutropenic fever, and stomatitis were seen with 5-FU/leucovorin. Treatment-related deaths occurred in 1% of patients in each arm of this trial. Although this IFL schedule represented a step forward over the Mayo Clinic 5-FU/leucovorin schedule, neither of these are recommended for current use. As outlined in the following, the infusional 5-FU schedules have a superior safety and efficacy profile and are preferred for use, especially in combination regimens.

In Europe, a parallel study to investigate the benefit of adding irinotecan to a 5-FU–based schedule was undertaken.[317] Two high-dose intermittent infusional schedules were developed. In France, a biweekly treatment for 2 consecutive days was explored, while German investigators, building on their experience with weekly 24-hour high-dose infusions of 5-FU, combined CPT-11 with this schedule. A randomized phase 3 trial was performed in which a participating center chose which of these two schedules would be used, and then the patients were randomized to that 5-FU/leucovorin schedule plus or minus CPT-11. Again, response rate, progression-free survival, and overall survival were superior in the CPT-11–containing arm of the trial. Of note, only the cohort treated with the biweekly schedule demonstrated a statistically superior survival over the 5-FU/leucovorin control arm, and the biweekly combination schedule is the only one registered for use in the United States.

More recently, the biweekly schedule of LV5FU2 plus irinotecan has been studied with a simplified LV5FU2 infusion schedule.[433] This schedule, known as FOLFIRI (FOL for folinic acid, F for 5-FU, and IRI for irinotecan), was initially studied as a salvage regimen; however, this has now gained widespread acceptance as a first-line treatment option, based on the data discussed subsequently.

The BICC-C (Bolus, Infusional, or Capecitabine with Camptosar-Celecoxib) trial is the only trial to directly compare weekly bolus IFL to FOLFIRI.[434] This trial utilized a modified bolus IFL schedule, giving treatment on days 1 and 8, repeated on a 3-week cycle. This modified IFL was compared to FOLFIRI as well as to capecitabine/irinotecan. The first phase of this trial (430 patients) confirmed the superior safety and efficacy of FOLFIRI over IFL (median progression-free survival 7.6 months for FOLFIRI vs 5.8 months; $P = 0.007$) and over capecitabine/irinotecan (progression-free survival 5.7 months; $P = 0.003$). The trial was halted when bevacizumab (see the following) became commercially available and a second phase randomized 117 patients to modified IFL plus bevacizumab versus FOLFIRI plus bevacizumab (capecitabine/irinotecan was dropped from the second phase of this trial). This second phase showed a significant overall survival advantage for FOLFIRI/bevacizumab over modified IFL/bevacizumab ($P = 0.002$). The BICC-C trial also had a second randomization of all patients to celecoxib versus placebo. Celecoxib was found to provide no benefit in terms of either safety or efficacy and does not appear to have a role as part of standard chemotherapy of this disease.

Oxaliplatin

Oxaliplatin (1,2-diaminocyclohexane (trans-l) oxalatoplatinum) is a third-generation platinum compound of the diaminocyclohexane family. Initial single-agent phase 1 studies established that

TABLE 40.13

Selected Commonly Used Oxaliplatin/5-Fluorouracil Combination Regimens[a]

Name of Regimen	Author (Ref.)	Schedule (All Agents Administered Intravenously)
FOLFOX-4	de Gramont et al., 2000[446]	Oxaliplatin 85 mg/m^2 over 2 hr; LV 200 mg/m^2 concurrently with oxaliplatin (can be given in same line through "Y" connector); followed by 5-FU bolus 400 mg/m^2, followed by 5-FU 600 mg/m^2 infusion over 22 hr. Oxaliplatin given day 1 only. All other meds given days 1 and 2. Cycle repeated every 14 d.
FOLFOX-6	Tournigand et al., 2004[449]	Oxaliplatin 100 mg/m^2 over 2 hr; LV 400 mg/m^2 concurrently with oxaliplatin (can be given in same line through "Y" connector); followed by 5-FU bolus 400 mg/m^2, followed by 5-FU 1,200 mg/m^2/d times 2 d (2,400 mg/m^2 infusion over 46–48 hr). Cycle repeated every 14 d.
Modified FOLFOX-6	Widely used in current phase 3 trials, Wolmark et al. 2009[446]	Oxaliplatin 85 mg/m^2 over 2 hr; LV 400 mg/m^2 concurrently with oxaliplatin (can be given in same line through "Y" connector); followed by 5-FU bolus 400 mg/m^2, followed by 5-FU 1,200 mg/m^2/d times 2 d (2,400 mg/m^2 infusion over 46–48 hr). Cycle repeated every 14 d.
FUFOX	Grothey et al., 2002[445]	Oxaliplatin 50 mg/m^2 over 2 hr, followed by LV 500 mg/m^2, followed by 5-FU 2,000 mg/m^2 over 24 hr, weekly for 5 wk, repeated every 6 wk.

[a]Doses listed are recommended starting doses for good performance status patients with normal renal, hepatic, and bone marrow function. Individual dose adjustments may be required.
FOLFOX, folinic acid, 5-FU, oxaliplatin; LV, leucovorin; 5-FU, 5-fluorouracil; FUFOX, 5-FU, folinic acid.

oxaliplatin could be safely administered, with evidence of clinical activity.[435,436] No significant nephrotoxicity was seen. Nausea and vomiting, minimal leucopenia, and rare thrombocytopenia were observed. Extra et al.[435] were the first to describe in detail the most notable toxicity encountered with oxaliplatin: neurotoxicity. This neurotoxicity manifested as paresthesias and dysesthesias of the hands, feet, perioral region, and throat. Pharyngolaryngeal dysesthesia, a sensation of choking without overt airway blockage, was described as well. These neurologic toxicities were induced or worsened by exposure to cold. Early single-agent explorations with oxaliplatin have been well outlined in the prior edition of this textbook.[437]

Oxaliplatin/5-Fluorouracil/Leucovorin Combination Trials

Based on a series of phase 2 trials by Levi et al., Giachetti et al.[438] from the same group reported a phase 3 trial of chronomodulated 5-FU/leucovorin alone or with oxaliplatin.[438–440] Two hundred patients were randomly assigned to receive a 5-day course every 3 weeks of chronomodulated 5-FU and leucovorin (700 and 300 mg/m^2 per day, respectively; peak delivery rate at 4 a.m.) with or without oxaliplatin on the first day of each course (125 mg/m^2, as a 6-hour infusion). The group who received oxaliplatin had a superior response rate (53% vs 16%; P <0.001). Progression-free survival was also superior, just reaching statistical significance (8.7 months vs 7.4 months; P = 0.048). There were no differences in median overall survival (19.4 months and 19.9 months, respectively). Survival outcomes in this trial are somewhat difficult to interpret as extensive use of resection of metastatic disease was applied in both arms.

Most of the combination 5-FU/leucovorin/oxaliplatin trials have used flat (nonchronomodulated) administration of agents and have centered on variants of the FOLFOX regimen. The acronym FOLFOX (FOL for folinic acid [leucovorin], F for fluorouracil, OX for oxaliplatin) refers to a series of combinations of these agents. These are biweekly (every other week) regimens using 2 days of infusional 5-FU on a 14-day cycle (LV5FU2).[441] The FOLFOX-1, 2, and 3 regimens employed various alterations in dosing of each oxaliplatin, 5-FU, and leucovorin.[442,443] They are of historical interest but were never evaluated in randomized trials. FOLFOX-3 and FOLFOX-4 were reported in a combined series to have a response rate of 21% in a population of patients who had progressed on the same 5-FU/leucovorin schedule without oxaliplatin.[442] The FOLFOX-4 regimen had a modestly higher response rate and lower toxicity than FOLFOX-3 (which used higher doses of 5-FU and leucovorin), and FOLFOX-4 appeared to be better tolerated. The more commonly used oxaliplatin/5-FU/leucovorin combinations are outlined in Table 40.13.[444,445]

A randomized phase 3 trial was undertaken to evaluate FOLFOX-4 versus the LV5FU2 schedule in patients with previously untreated metastatic CRC (essentially a trial of LV5FU2 with or without oxaliplatin).[446] Patients treated with FOLFOX-4 had a statistically significantly superior outcome in terms of response rate (51% vs 22%; P = 0.001) and progression-free survival (9 months vs 6.2 months; P = 0.0003). The FOLFOX arm had a 1.5-month improvement in median overall survival; however, this did not reach statistical significance (16.2 months vs 14.7 months; P = 0.12). The number of patients experiencing grade 3 or 4 neutropenia was increased with FOLFOX-4 over LV5FU2 (42% vs 5% of patients). Grade 3 or 4 diarrhea (12% vs 5%) was also increased in the FOLFOX arm. Neurotoxicity, virtually absent in the LV5FU2 arm, was frequent in the FOLFOX arm, with 18% of patients experiencing grade 3 neurosensory toxicity.

The FOLFOX-4 regimen has also been evaluated in a multicenter randomized trial in second-line therapy following failure of first-line IFL chemotherapy.[447] Patients were randomly assigned to one of three arms: FOLFOX-4, LV5FU2, or single-agent oxaliplatin. Response rates were 10% for FOLFOX, 0% for LV5FU2, and 1% for oxaliplatin alone (P <0.0001 for FOLFOX vs LV5FU2). Time to tumor progression was also superior for FOLFOX-4 (4.6 months) versus LV5FU2 (2.7 months) and oxaliplatin alone (1.6 months). These data confirm initial clinical impressions that oxaliplatin/5-FU combinations have superior activity to single-agent oxaliplatin, even in 5-FU–refractory disease. FOLFOX-4 has activity in IFL-refractory disease; however, single-agent oxaliplatin essentially does not.

Further modifications have been made to the FOLFOX schedule. FOLFOX-5 was designed with an increased dose of oxaliplatin to 100 mg/m^2 every 14 days; however, this regimen was never tested in clinical trials. FOLFOX-6 utilized this 100 mg/m^2 oxaliplatin dose with a simplified 5-FU/leucovorin schedule.[448] Oxaliplatin 100 mg/m^2 is given over 2 hours, with leucovorin 400 mg/m^2 given concurrently via a "T" connector. These are then followed by a 400 mg/m^2 bolus of 5-FU and then a 46-hour infusion of 5-FU at 2,400 to 3,000 mg/m^2. More recently, the FOLFOX-7 regimen has been reported, utilizing a 130 mg/m^2 dose of oxaliplatin every 14 days. The simplified leucovorin/5-FU administration of FOLFOX-6

is maintained, with deletion of the bolus 5-FU. Oxaliplatin is discontinued after 3 months, with planned reintroduction after 12 weeks or sooner if clinical progression occurs.[449] This rationale appears promising, both for treatment of metastatic disease and for potential use in the adjuvant setting. Given the similar response rates after 12 weeks, it would appear that the increased dose of oxaliplatin in FOLFOX-7 is unnecessary. A reasonable approach to standard use of FOLFOX in the metastatic setting is to use a modified FOLFOX-6, with 85 mg/m^2 of oxaliplatin and simplified LV5FU2 at a dose of 2,400 mg/m^2 over 46 to 48 hours (1,200 mg/m^2 per day for 2 consecutive days). As discussed in the following, the OPTIMOX trial data support cessation of oxaliplatin after 12 weeks and reintroduction of the oxaliplatin at a later date upon disease progression. It had been previously hypothesized that administration of high doses of calcium and magnesium with oxaliplatin would be protective against neurotoxicity, but a definitive trial has now demonstrated that this is not the case.[450]

Comparisons of Oxaliplatin- and Irinotecan-based Combinations

With both oxaliplatin- and irinotecan-based regimens showing encouraging activity, the question of which agent to use first was addressed by a number of investigators. Tournigand et al.[449] reported a phase 3 trial of FOLFOX-6 versus FOLFIRI. This trial utilized identical simplified LV5FU2 schedules, with the only variable being oxaliplatin or irinotecan. All patients were planned to crossover to the other regimen at time of progression, and the primary end point was time to tumor progression after *both* chemotherapy regimens. Results are shown in Table 40.14. Although the study is somewhat underpowered at a total of 226 patients, the results show a striking consistency between regimens, suggesting that use of either FOLFOX-6 or FOLFIRI in first-line treatment is acceptable. A somewhat larger trial of 360 patients randomized to FOLFOX-4 versus the equivalent FOLFIRI schedule, utilizing the same LV5FU2 dose and schedule in each arm, again shows comparable efficacy data, with differing and predictable toxicity profiles.[451]

The North Central Cancer Treatment Group–led US Intergroup study N9741, a complex and important trial that underwent many iterations before its completion, initially opened as a four-arm trial comparing the Mayo Clinic 5-FU/leucovorin control arm to three different CPT-11/5-FU/leucovorin regimens: weekly bolus IFL as reported by Saltz et al.,[318] a "Mayo II" schedule of CPT-11 on day 1 and bolus 5-FU/low-dose leucovorin on days 2 to 5, or the biweekly infusional schedule of LV5FU2 plus CPT-11 as reported by Douillard et al.[317] and Goldberg.[452] After accruing a small number of patients, the trial was closed to incorporate three oxaliplatin-containing arms: FOLFOX-4, IROX (a once every 3 weeks combination of irinotecan and oxaliplatin, without 5-FU), and a modified Mayo Clinic schedule of bolus 5-FU plus low-dose leucovorin days 1 through 5, with oxaliplatin given on day 1. The infusional LV5FU2 plus CPT-11 arm was dropped. This created a six-arm trial. In March 2000, the trial was again halted, based on presentation of evidence that the combination of CPT-11/5-FU/leucovorin, using either bolus or infusional schedules, was superior to 5-FU/leucovorin. The Mayo Clinic control arm of N9741 was now dropped, and weekly bolus IFL became the control arm. At the same time, ongoing real-time monitoring of fatal toxicities identified unacceptably high rates of treatment-related mortality in the oxaliplatin plus Mayo Clinic 5-FU/leucovorin and in the CPT-11 plus 5-FU/leucovorin arms. These schedules were also dropped from the trial and from further development, leaving a three-arm trial of CPT-11 plus bolus 5-FU/leucovorin (IFL), oxaliplatin plus infusional LV5FU2 (FOLFOX-4), and oxaliplatin plus CPT-11 (IROX).

The trial was stopped a third time in April 2001, when monitoring of the trial indicated what appeared to be a higher than expected early mortality in the IFL control arm.[319] This observation, however, was based on utilization of a new metric, the 60-day ACM. This metric records death from *any* cause within 60 days of initial therapy. The 60-day ACM of the IFL arm was initially noted to be 4.5%. Because this was a new metric, however, there were no readily available historical controls; no one had ready access to data to say what the 60-day ACM had been in previous trials, either with IFL or with 5-FU/leucovorin regimens. The 4.5% ACM was therefore compared to the previously reported death rate for the IFL regimen, which was 0.9%. However, the previously reported death rate was the treatment-related death rate, the percentage of deaths judged by the investigators to have been caused by treatment, not all deaths within 60 days of starting therapy. Of further concern to the safety monitoring committee, however, the experimental arms (FOLFOX-4 and IROX) each showed 60-day ACMs of 1.8% (compared to 4.5% for the IFL control arm). This information was difficult to put into context, however, because the efficacy of the two experimental arms had not yet been established.

TABLE 40.14

Comparison of First-line Use of Irinotecan versus Oxaliplatin in Conjunction with the Same Simplified Biweekly Infusional 5-Fluorouracil/Leucovorin Schedule

	FOLFIRI (*n* = 109 Patients Treated)	FOLFOX-6 (*n* = 111 Patients Treated)	*P* Value
Major objective response rate (partial plus complete responses)	56%	54%	0.68
Time to tumor progression (on first-line regimen)	8.5 mo	8.1 mo	0.65
Time to tumor progression (after first- and second-line regimen)	14.4 mo	11.5 mo	0.65
Overall survival (from initial randomization)	20.4 mo	21.5 mo	0.9
Two-yr overall survival	41%	45%	
Grade 3–4 neutropenia	25%	44%	
Neutropenic fever	6%	1%	
Grade 3–4 diarrhea	14%	11%	
Neuropathy (grade 3)	0%	34%	
Alopecia (grade 2)	24%	9%	

FOLFIRI, irinotecan, 5-fluorouracil, leucovorin; FOLFOX, folinic acid, 5- fluorouracil, oxaliplatin.
With permission from Tournigand C, Andre T, Achille E, et al. FOLFIRI followed by FOLFOX6 or the reverse sequence in advanced colorectal cancer: a randomized GERCOR study. *J Clin Oncol* 2004;22:229–237.

In fact, the 60-day ACM on the original phase 3 trial of IFL (subsequently calculated after N9741 was halted) was 6.7%, and the 60-day ACM for the Mayo Clinic control arm of that trial was found to be 7.3%. Although the 7.3% 60-day ACM appeared subjectively to be unusually high, no historical baseline data on 60-day ACM in 5-FU–based regimens were readily available. To help interpret these data, an analysis was undertaken to determine the 60-day ACM in multiple large-scale randomized trials that had used 5-FU/leucovorin schedules over the prior decade. This analysis confirmed that 60-day ACM regularly was encountered at a rate of 5% to 8% in the treatment of metastatic CRC.[453] Thus the 60-day ACM for the IFL regimen was actually *lower* on N9741 than in previous trials and was *lower* than what had been seen consistently with 5-FU/leucovorin regimens alone. In the final analysis of N9741, the 60-day ACM seen in the IFL, FOLFOX-4, and IROX arms were 4.5%, 2.6%, and 2.7%, respectively, and these differences were not statistically significant.

The efficacy results of N9741, however, were statistically significant and showed superior outcome for the patients randomized to FOLFOX-4, as compared to those randomized to either IFL or IROX, in terms of response rate, time to tumor progression, and overall survival (Table 40.15).[454] Toxicity for FOLFOX-4 was also superior for virtually all parameters, except of course neurotoxicity. The results of the IROX arm did not statistically significantly differ from those of the IFL arm in terms of toxicity, response, or time to tumor progression; however, survival was borderline statistically significantly better in the IROX arm than the IFL arm ($P = 0.04$).

Taken together, where do these trials leave us in terms of first-line use of oxaliplatin- and irinotecan-based regimens? Data from trial N9741 indicate that FOLFOX-4 is superior to IFL in both response rate and time to tumor progression. Overall survival was superior in the FOLFOX-4 arm versus IFL as well; however, interpretation of the survival results of N9741 is somewhat complicated due to imbalances between arms in availability of effective second-line therapy. Second-line CPT-11 was available to all patients who had received FOLFOX-4. Oxaliplatin, however, was not commercially available in the United States during the course of N9741. To what degree this imbalance in second-line therapy may have influenced the survival result is unknown. Also, as IFL contains bolus 5-FU, while FOLFOX-4 contains infusional leucovorin/5-FU2, it is difficult to isolate the irinotecan versus oxaliplatin component from the 5-FU bolus versus 5-FU infusion component.

Two other trials indicate that the FOLFOX and FOLFIRI regimens have similar safety and efficacy, with differing toxicity profiles.[449,451] Thus, FOLFOX has comparable efficacy to FOLFIRI, whereas FOLFOX has a superior response rate, time to tumor progression, and possibly some degree of survival benefit over IFL. Toxicity with irinotecan-based regimens shows a higher degree of alopecia. Diarrhea and neutropenia are increased on the bolus 5-FU schedule but are similar between FOLFOX and FOLFIRI. Oxaliplatin-based regimens, however, have neurotoxicity, absent from the irinotecan-based regimens, which can be problematic in some patients. It would therefore seem reasonable at this time to favor the use of a high-dose intermittent infusional 5-FU/leucovorin schedule plus either oxaliplatin (i.e., FOLFOX) or CPT-11 (i.e., FOLFIRI). Data do not support continued routine use of the bolus IFL schedule, nor are there randomized data to support the routine use of a bolus 5-FU/leucovorin schedule with oxaliplatin in the metastatic setting. Routine use of IROX is also not supported by the currently available body of data.

Whether to use an irinotecan-based or oxaliplatin-based combination in first-line treatment of good performance status patients can be considered a matter of patient preference, and discussion of the differing toxicity profiles is appropriate to help individuals decide. It is hoped that in the near-term future molecular prognostic indicators and pharmacogenomics will provide useful guidance for the individualization of therapies, but such approaches remain investigational at this time.

The only oxaliplatin schedule registered for use in the United States is FOLFOX-4; however, the modified FOLFOX-6 would appear at this time to be a very reasonable schedule for routine clinical use when the decision is made to use an oxaliplatin/5-FU combination.

The recognition of neurotoxicity as a major limitation of the FOLFOX regimens led to the investigation of optimization of oxaliplatin (the OPTIMOX study).[455] In this trial, patients were randomly assigned to receive either standard FOLFOX-4 until progression or 12 weeks of FOLFOX-7, followed by planned cessation of oxaliplatin and continuation of the LV5FU2. As designed, the study called for a reintroduction of oxaliplatin after 6 months of LV5FU2, although this actually occurred in the minority of

TABLE 40.15

Results of Intergroup Trial N9741: Irinitecan Plus Bolus 5-Fluorouracil/Leucovorin, Oxaliplatin Plus Infusional 5-Fluorouracil/Leucovorin, and Irinotecan Plus Oxaliplatin in First-line Treatment of Patients with Metastatic Colorectal Cancer

	IFL (n = 264)	FOLFOX-4 (n = 267)	IROX (n = 264)	*P* Value (IFL vs FOLFOX)
Major objective response rate (partial plus complete responses)	31%	45%	35%	0.03
Time to tumor progression	6.9 mo	8.7 mo	6.5 mo	0.001
Overall survival	15.0 mo	19.5 mo	17.4 mo	0.0001
Received second-line therapy with active drug not included in first-line regimen	24% (oxaliplatin)	60% (irinotecan)	50% (fluorouracil)	Not given
Grade 3–4 neutropenia	40%	50%	36%	0.35
Neutropenic fever	15%	4%	11%	0.001
Grade 3–4 diarrhea	28%	12%	24%	0.001
Grade 3–4 nausea	16%	6%	19%	0.001
Grade 3 neuropathy	3%	18%	7%	0.001
60-d all-cause mortality	4.5%	2.6%	2.7%	Not significant

IFL, 5-fluorouracil, leucovorin; FOLFOX4, oxaliplatin plus infusional 5-fluorouracil/leucovorin; IROX, irinotecan plus oxaliplatin.
Data from Goldberg RM, Sargent DJ, Morton RF, et al. A randomized controlled trial of fluorouracil plus leucovorin, irinotecan, and oxaliplatin combinations in patients with previously untreated metastatic colorectal cancer. *J Clin Oncol* 2004;22:23–30.

patients and outcomes were superior in those patients in whom reintroduction of oxaliplatin occurred.[456] The primary end point was duration of disease control, the time from initiation of treatment until either progression through all agents (including failure after reintroduction of oxaliplatin if this was done), or death. Duration of disease control as well as progression-free survival and overall survival were not statistically significantly different between the two arms. As anticipated, toxicity, including neurotoxicity, was substantially reduced in the OPTIMOX arm. This OPTIMOX strategy of planned interruption of oxaliplatin can be considered a standard care option in metastatic disease. It is important to discuss plans for such planned interruptions with patients at the beginning of therapy so that they will not be surprised or alarmed at the removal of one of the drugs.

Although a regimen of oxaliplatin, weekly bolus 5-FU, and low-dose weekly leucovorin (bFOL) appeared promising in an initial phase 2 trial, two sequential randomized phase 2 trials, known as TREE-1 and TREE-2, suggest modestly inferior activity for the bFOL schedule compared with FOLFOX or Cape/Ox.[457,458] Thus, in the metastatic setting, oxaliplatin with bolus 5-FU schedules is therefore not recommended for routine use.

Use of planned sequential administration of FOLFOX and FOLFIRI has also been proposed, both in terms of pretreatment for potentially resectable patients with liver metastases and in terms of adjuvant treatment of earlier stage disease. Several groups are also exploring the use of "triple therapy" with oxaliplatin, CPT-11, and 5-FU/leucovorin (FOLFOXIRI). A randomized phase 3 trial conducted in Italy reported that FOLFOXIRI is tolerable and offers a modest survival benefit over FOLFIRI. These trials utilized higher doses of 5-FU than are likely to be tolerable by American patients, and use of this combination has not gained widespread acceptance.[459]

As discussed previously, a trial comparing FOLFIRI to capecitabine/irinotecan suggested inferior outcome for the capecitabine-containing regimen.[434] However, a large randomized trial has now compared Cape/Ox versus FOLFOX. The study also had a two-by-two randomization to with or without bevacizumab (discussed subsequently). This trial demonstrated the Cape/Ox regimen to be noninferior to FOLFOX, and each had acceptable toxicity, indicating that Cape/Ox is an acceptable alternative to FOLFOX.[460] It should be noted that the Cape/Ox regimen requires a motivated, reliable patient who will be able to take multiple pills of oral medication on a complex schedule, even in the setting of potentially emetogenic oxaliplatin.

Duration of Therapy

Controversy continues to exist regarding the optimal duration of chemotherapy for palliation of metastatic disease. Traditional practice for many years had been to continue chemotherapy until either unacceptable toxicity, clinical deterioration, or disease progression. When efficacy of treatment was more limited, with the duration of therapy typically limited to a small number of months, the issue of treatment breaks did not seem relevant. Now, with patients typically living multiple years with metastatic CRC and with some treatments maintaining control for more extended periods of time, the need for patients to have breaks (often referred to as "treatment holidays" or "chemotherapy-free intervals" [CFI]) is greater, and there is considerable interest in using these approaches. Both physically and psychologically, many patients appear to both need and derive benefit from these treatment interruptions.

The concept of noncontinuous chemotherapy has been investigated for some time now. Maughan et al.[461] conducted a randomized trial of continuous versus interrupted treatment in 354 patients who were responding or who had stable disease after receiving 12 weeks of either 5-FU– or raltitrexed-based chemotherapy. Patients were randomized to either continue chemotherapy until progression or to stop chemotherapy after the first 12 weeks, followed by a planned restarting on the same chemotherapy at the time of progression. At randomization, 41% of patients had achieved a major objective response and 59% had stable disease. There was no evidence of a difference in overall survival, the primary end point, between the two groups, with an HR of 0.87 ($P = 0.23$, favoring the intermittent arm).

The idea of planned early cessation of all chemotherapy was investigated in the OPTIMOX-2 trial.[462] A total of 202 patients with previously untreated metastatic CRC were treated. All patients received six cycles of modified FOLFOX-7 followed by either continued LV5FU2 until progression or a complete cessation of all chemotherapy (a CFI). Patients on both arms were planned to receive retreatment with FOLFOX following tumor progression. The results of this study did not support the use of this planned, early interruption in therapy, as the median duration of disease control, progression-free survival, and overall survival were all inferior in the arm with the early planned CFI. This should not be misconstrued as evidence that CFIs are contraindicated, but rather that early planned CFIs for all patients is not an appropriate strategy. The authors suggest that this study indicates there are no pretreatment parameters that can identify a priori those patients who can successfully benefit from a CFI. Thus, specific decisions regarding use and timing of CFIs cannot be made in advance of starting treatment; rather, clinical judgment must be exercised in deciding on treatment interruptions for CFIs in responding patients after a favorable response. In a retrospective review of 822 patients in the two OPTIMOX studies, after excluding those patients who had early progression within the first 3 months of treatment as well as those who underwent complete gross resection of metastatic disease within 3 months of stopping chemotherapy, Perez-Staub et al.[465] noted that there was no indication of a detriment in survival when comparing those patients who took a CFI versus those who did not. In fact, in this retrospective, nonrandomized analysis, the median survival was 37.5 months in patients who had a CFI versus 21.2 months in matched patients who did not have a CFI. Of note, median overall survival of patients who stopped chemotherapy earlier than 3 months was 24 months, whereas it was 42 months when a CFI was taken between 3 and 9 months into therapy, and 44 months when a CFI was taken later than 9 months into chemotherapy. These studies were accomplished prior to the use of bevacizumab. Some clinicians have advocated the continuation of bevacizumab during CFIs. Such an approach is not supported by data, and given the absence of activity of single-agent bevacizumab in CRC, use of single-agent bevacizumab during otherwise CFIs is not recommended at this time.

Other investigators specifically addressed the question of whether rechallenge with 5-FU after a planned treatment interruption could produce a response. A pooled analysis was conducted on 613 patients involved in three randomized trials of first-line 5-FU–based therapy.[464] All patients had a planned maximum treatment period of 6 months. Patients with responding or stable disease at the end of that period were observed off treatment with a plan for retreatment at the time of disease progression. Median time to rechallenge was 11.7 months. Seventeen percent of patients had an objective response to rechallenging. Median survival for the group was 14.8 months. These nonrandomized data indicate that patients have a meaningful response rate at time of reinstitution of chemotherapy.

A similar approach was explored in patients who received second-line irinotecan therapy.[465] A total of 333 patients entered into a trial to receive 24 weeks of irinotecan. Patients who remained in the study at the end of that time were to be randomized to either continue treatment or to stop therapy. Of the 333 patients, most came off the study due to progression or toxicity before reaching the 24-week mark. Fifty-five patients with responding or stable disease agreed to randomization. Although the numbers available for comparison were small, there were no differences between the arms in progression-free survival or overall survival, nor were there differences in quality-of-life scores.

TABLE 40.16

Sequential versus Combination Chemotherapy with Capecitabine, Irinotecan, and Oxaliplatin in Advanced Colorectal Cancer: Efficacy End Points

	Overall Survival	One-Yr Survival	Progression-Free Survival (First Line)	Response Rate
Sequential Cape, then Iri, then Cape/Ox ($n = 401$)	16.3 mo	64%	5.8 m	20%
Combination Cape/Iri, then Cape/Ox ($n = 402$)	17.4 mo	67%	7.8 m	41%
P value	0.33	0.38	0.0002	0.0001

Cape, capecitabine; Iri, irinotecan; Ox, oxaliplatin.
With permission from Koopman M, Antonini NF, Douma J, et al. Sequential versus combination chemotherapy with capecitabine, irinotecan, and oxaliplatin in advanced colorectal cancer (CAIRO): a phase III randomised controlled trial. *Lancet* 2007;370:135–142.

Overall, there appears to be no compelling evidence that continuation of chemotherapy indefinitely is necessary for optimal control of metastatic disease. The option of discontinuation of therapy after a reasonable period of time appears to be an appropriate consideration in standard practice.

Combination versus Single-agent Cytotoxic Chemotherapy

Given that combination regimens are invariably associated with more toxicity than single agents, the question of the need for universal upfront use of these combinations was investigated. The CAIRO (CApecitabine, IRinotecan, Oxaliplatin) trial randomized 820 patients to sequential versus concurrent therapies (Table 40.16).[466] In the sequential arm, first-line therapy was single-agent capecitabine. Upon failure, single-agent irinotecan was used, and then third-line therapy was Cape/Ox (since single-agent oxaliplatin is essentially inactive in 5-FU–refractory CRC). The combination arm used Cape/Ox as first-line therapy and capecitabine and irinotecan as second-line therapy. The primary end point, median overall survival, was not statistically significantly different between the two arms (17.4 months for combination vs 16.3 months for the sequential arm; $P = 0.33$). Dose-limiting toxicity (grade 3 or 4) was not significantly different between the two groups; in fact, grade 3 hand-foot syndrome was somewhat more common in the sequential arm (13% vs 7%; $P = 0.004$).

Similar findings were reported in the FOCUS study.[467] In this trial, a total of 2,135 patients were randomized to one of three arms. Arm A was sequential therapy, with initial treatment given with 5-FU on the leucovorin/5-FU2 (biweekly infusional) schedule until progression, at which point second-line therapy was given with single-agent irinotecan. Arm B also gave biweekly LV5FU2 until failure, and then LV5FU2 was continued with the addition of either oxaliplatin or irinotecan (this was a second randomization within this arm). Thus, second-line therapy was a change from biweekly LV5FU2 to either FOLFOX or FOLFIRI in this arm. In arm C, patients began with combination chemotherapy, and within this arm were randomized to either FOLFOX or FOLFIRI. The primary end point, median overall survival, for arm A was 13.9 months. For arm B, survival was 15.0 months for irinotecan and 15.2 months for oxaliplatin. Arm C had a median overall survival of 16.7 months for the FOLFIRI patients and 15.4 months for those treated with FOLFOX. Only the difference between arm A and the irinotecan arm of arm C reached statistical significance ($P = 0.01$). Arm B (initial LV5FU2 followed by FOLFOX or FOLIRI) was noninferior to arm C (initial FOLFOX or FOLIRI) (HR = 1.06; 90% CI = 0.97 to 1.17).

Taken together, the CAIRO and FOCUS trials provide a strong argument that not all patients with unresectable metastatic disease require exposure to the toxicity of combination therapy, and that initial use of fluorinated pyrimidine alone in patients with previously untreated metastatic CRC is a treatment alternative that needs to be carefully considered.

Bevacizumab

Bevacizumab is a humanized monoclonal antibody that binds to VEGF, thereby substantially reducing the amount of circulating ligand and thus preventing receptor activation.[468,469] The first trial of bevacizumab in CRC was a modest-sized, three-arm, randomized phase 2 trial in which a total of 104 patients were randomly assigned to either one of two different doses levels of bevacizumab (5 mg/kg or 10 mg/kg) plus weekly 5-FU/leucovorin or to 5-FU/leucovorin alone.[470] The response rate, time to tumor progression, and overall survival were superior in the 5-FU/leucovorin with 5 mg/kg bevacizumab arm. Despite the small size and limited statistical power of this study, this result would serve as the basis for design of the pivotal phase 3 trial of bevacizumab in CRC.

The initial design of the phase 3 pivotal trial was a comparison between 5-FU/leucovorin plus placebo to 5-FU/leucovorin plus 5 mg/kg of bevacizumab. However, as the randomized phase 2 trial, discussed previously, was nearing completion, the randomized phase 3 trial was reported, which demonstrated a modest but statistically significant survival advantage for the IFL regimen (irinotecan plus weekly bolus 5-FU/leucovorin) compared with 5-FU/leucovorin alone.[318] As a result of this trial, the IFL regimen was then felt to be the appropriate control arm for subsequent phase 3 trials. There were no safety data at the time, however, on the combination of bevacizumab plus IFL. As a result of this, a three-arm trial was designed that contained[37] 5-FU/leucovorin/bevacizumab,[38] IFL/bevacizumab,[39] and IFL/placebo (the control arm).[471] The design included a preplanned analysis of safety on all arms when enrollment reached 100 patients per arm, with a further plan to close the 5-FU/leucovorin/bevacizumab arm at that time if the safety data indicated acceptable tolerability and safety of the IFL/bevacizumab arm.

In the final efficacy analysis, the IFL/bevacizumab cohort experienced superior outcome compared to the IFL/placebo group in response rate (45% vs 35%; $P < 0.003$), progression-free survival (10.6 months vs 6.2 months; $P < 0.00001$), and overall survival (20.3 months vs 15.6 months; $P = 0.00003$) (Table 40.17A). It should be noted that no crossover to second-line bevacizumab in the IFL/placebo control arm was allowed in this trial.

In order to better understand the effects of bevacizumab in conjunction with 5-FU/leucovorin, Kabbinavar et al.[472] combined the data from three separate modest-sized trials to create a more robust data set. In this combined analysis of 5-FU/leucovorin with or without bevacizumab, there was a statistically significant survival advantage for the patients who received bevacizumab. Given the favorable aspects of the biweekly infusional LV5FU2 schedule used in the FOCUS trial, the LV5FU2 schedule would seem most appropriate for combination with bevacizumab.[473] At the same time as the pivotal front-line study was accruing, the Eastern Cooperative Oncology Group (ECOG) also performed a trial (ECOG-3200) to evaluate the use of bevacizumab in the second-line setting.[474] This trial randomized patients who had failed irinotecan and 5-FU, but were naive to bevacizumab, to one of three arms: bevacizumab/FOLFOX, FOLFOX alone, or bevacizumab. The investigators

TABLE 40.17A

Efficacy Outcomes, First-line Treatment of Metastatic Colorectal Cancer: Irinotecan Plus Bolus 5-Fluorouracil/Leucovorin Plus Placebo versus 5-Fluorouracil/Leucovorin Plus Bevacizumab

Regimen	No. of Patients	Response Rate	Progression-free Survival	Overall Survival
IFL + placebo	411	34.8%	6.2 mo	15.6 mo
IFL + bevacizumab	402	44.8% $p = 0.004$	10.6 mo $p < 0.001$	20.3 mo $p < 0.001$

IFL, 5-fluorouracil, leucovorin.

chose to investigate a 10 mg/kg bevacizumab dose. Overall, a modest but statistically significant improvement in median overall survival was demonstrated for FOLFOX-4 with bevacizumab versus FOLFOX-4 alone (12.5 months vs 10.7 months; $P = 0.0024$), and grade 3 or 4 toxicities were not increased. The bevacizumab-alone arm had substantially inferior progression-free survival and an investigator-adjudicated response rate of 3%, suggesting that single-agent bevacizumab does not have meaningful activity in CRC and should not be used. It is important to note that this trial was performed *exclusively* in patients who had *not* received bevacizumab in the first-line setting. This trial provides no data on whether use of bevacizumab with a second-line regimen after progression on a first-line bevacizumab-containing regimen is efficacious.

Although it was performed in second-line patients, the ECOG-3200 trial was the first trial to provide safety data for the combination of bevacizumab plus FOLFOX. As a result of this, even before front-line data were available, bevacizumab plus FOLFOX had become widely accepted as a front-line option in the United States for metastatic CRC. More recently, the NO16966 trial directly addressed the question of front-line bevacizumab plus oxaliplatin-based therapy (Table 40.17B).[475] In this trial, 1,400 patients with previously untreated CRC were randomly assigned to either FOLFOX-4 or Cape/Ox and then to either placebo or bevacizumab, in a two-by-two randomization. Although the study did show a statistically significant progression-free survival advantage for the addition of bevacizumab (9.4 months vs 8.0 months for chemotherapy with bevacizumab vs chemotherapy with placebo, respectively; HR = 0.83; $P = 0.003$), this difference was more modest than the 4.4-month progression-free survival difference seen in the initial bevacizumab with IFL front-line trial. Overall survival improvement with bevacizumab approached, but did not reach, statistical significance (21.3 months vs 19.9 months; HR = 0.89; $P = 0.077$) Also, in the NO16966 trial, the addition of bevacizumab to front-line oxaliplatin-based chemotherapy did not confer any response benefit. It is noteworthy that the majority of patients on this trial discontinued treatment, presumably due to nonbevacizumab-related toxicity issues, before progression. This may have diminished the impact of bevacizumab on survival and progression-free survival but would not have impacted the response rate.

Toxicity

In terms of toxicity, grade 3 hypertension was higher in the bevacizumab/IFL arm than in the placebo/IFL arm (11% vs 2%) in the IFL/bevacizumab study.[471] Hypertension is now widely recognized as a common side effect of bevacizumab, and monitoring for and treatment of hypertension with antihypertensive medications is a routine part of bevacizumab management. Incidences of overall thromboembolic events and proteinuria were not statistically different between the two arms. However, two rare but extremely serious toxicities were encountered with increased frequency in the bevacizumab-containing arm: GI perforations and arterial thrombotic events.

The GI perforations were a group of events that included a perforated gastric ulcer, small bowel perforations, and free air under the diaphragm without identified sources. Although these were somewhat heterogeneous in nature, it was noted that six such events occurred on the bevacizumab-containing arm (one fatal) compared with none on the chemotherapy-alone arm. No clear risk factors for these perforations could be identified from this trial. Interestingly, GI perforations were not frequent occurrences in large cooperative group trials in patients with lung cancer or breast cancer; however, an unusually high GI perforation rate has recently halted accrual on a trial of bevacizumab in patients with ovarian cancer. These ovarian observations illustrate an important aspect about GI perforations in association with bevacizumab: there is not an association between the presence of an intact primary tumor in the colon and a GI perforation. Concerns have been expressed by some clinicians about the possibility of needing to remove an asymptomatic primary colorectal tumor in a patient with synchronous stage IV disease before using bevacizumab out of an unsubstantiated fear that the primary will put the patient at risk for perforation. At present, there are no data to support this assumption, and surgery for an asymptomatic primary tumor in a stage IV patient is not routinely indicated, regardless of whether there are plans to use a bevacizumab-containing chemotherapy regimen.[414]

The other rare but very serious identified increased risk with bevacizumab-containing treatment was that of arterial thrombotic events. Initially, no clear indication of this risk was detected in the

TABLE 40.17B

Efficacy Outcomes, First-line Treatment of Metastatic Colorectal Cancer: Capeox/Folfox Plus Placebo versus Capeox/Folfox Plus Bevacizumab

Regimen	No. of Patients	Response Rate	Progression-Free Survival	Overall Survival
CapeOx/FOLFOX + placebo	701	49%	8.0 mo	19.9 mo
CapeOx/FOLFOX + bevacizumab	699	47% $p = 0.31$	9.4 mo $p = 0.0023$	21.3 mo $p = 0.078$

CapeOx, capecitabine, oxaliplatin; FOLFOX, fluorouracil, leucovorin, oxaliplatin.
With permission from Saltz LB, Clarke S, Diaz-Rubio E, et al. Bevacizumab in combination with oxaliplatin-based chemotherapy as first-line therapy in metastatic colorectal cancer: a randomized phase III study. *J Clin Oncol* 2008;26:2013–2019.

pivotal phase 3 trial. However, in a combined analysis of several trials, an important observation was made. Here again, multiple events were combined into one metric. Thus, cerebral vascular accidents, myocardial infarctions, transient ischemic attacks, and angina were combined to create the metric of arterial thrombotic events. The observed incidence of these events was 2.5% in the nonbevacizumab-containing control arms versus 5.0% in the bevacizumab-containing experimental arms. It was noted that patients who had histories of cardiovascular or atherosclerotic disease appeared to be at greater risk for increased bevacizumab-related arterial thrombotic complications. In addition, a further analysis of these events suggested that the risk was essentially linear over time, indicating that the risk of a new arterial thrombotic event was the same in earlier versus later months of exposure.[476]

Another complication of bevacizumab that has been rarely described in the literature is fistula formation. Ganapathi et al.[477] reports a 4.1% incidence of this problem in a review of 222 patients with metastatic CRC. Two-thirds were perineal or anal with the remainder colovesicular, occuring an average of 3.9 months after initiation of treatment. Cessation of bevacizumab led to fistula healing in nearly all cases; however, three patients required fecal diversion. The authors suggest that this complication has been underreported thus far and stress the importance of early recognition.[477]

FOLFOXIRI plus Bevacizumab

Building on prior experience with FOLFOXIRI, Loupakis et al. randomized 508 patients with previously untreated metastatic colorectal cancer to receive either FOLFOXIRI plus bevacizumab or FOLFIRI plus bevacizumab. All patients had an excellent performance status (mostly ECOG 0). The median progression free survival was 12.1 months versus 9.7 months, respectively ($P = 0.003$). FOLFOXIRI was also superior in terms of response rate (65% vs 53%, $P = 0.006$). Overall survival was improved from 25.8 months to 31.0 months with AP value that just missed statistical significance ($P = 0.054$). Grade 3 or 4 toxicities including neurotoxicity, stomatitis, diarrhea, and neutropenia were significantly higher in the FOLFOXIRI group. It is again noteworthy that the 5-FU infusional dose of 3,200 mg/m² over 46 hours was higher in the FOLFOXIRI than in the FOLFIRI group, and with this study having been done only in Europe, the question of whether or not such a fluorouracil dose would be tolerable in American patients remains unknown. In general, the FOLFOXIRI plus bevacizumab regimen, possibly with fluorouracil dose modifications for American patients, is a reasonable consideration for first-line treatment of excellent performance status patients, especially those in need of a substantial rapid response.[478]

Bevacizumab Beyond Progression

Prior studies had demonstrated the activity of bevacizumab when added to first-line chemotherapy, and the ECOG 3200 trial had shown that bevacizumab added to second-line chemotherapy in bevacizumab-naïve patients. Until recently, however, data were lacking regarding the question of continuation of bevacizumab with second-line therapy after progression of disease through a first-line, bevacizumab-containing regimen. This question has now been addressed by the TML trial.[479] In this trial, 820 patients who had progressed through a first-line, bevacizumab-containing regimen were assigned to a noncross-resistant chemotherapy regimen (irinotecan plus fluoropyrimidine if previously treated with oxaliplatin or oxaliplatin-fluoropyrimidine if previously treated with irinotecan) and then randomized to receive bevacizumab with this second-line chemotherapy or not. The arm receiving bevacizumab showed a modest but statistically significant survival benefit of 1.4 months (overall survival of 11.2 months vs 9.8 months; HR = 0.81; 95% CI = 0.69 to 0.94; $P = 0.0062$). Grade 3 to 5 bleeding or hemorrhage (8 [2%] vs 1 [1%]), GI perforation (7 [2%] vs 3 [1%]), and venous thromboembolisms (19 [5%] vs 12 [3%]) were more common in the bevacizumab plus chemotherapy group than in the chemotherapy-alone group.

Aflibercept

Aflibercept is a fusion molecule containing the binding domains of VEGF receptors 1 and 2 bound to the human immunoglobulin (Ig) G Fc fragment, forming a VEGF trap molecule. Aflibercept binds all human VEGF A isoforms, VEGF B, and placental growth factor with greater affinity than the native receptors for these ligands. Aflibercept has been evaluated in the large scale phase 3 VELOUR trial, in which 1,226 patients who had progressed on a first-line oxaliplatin-containing regimen were randomized to receive second-line FOLFIRI plus aflibercept 4 mg/kg versus FOLFIRI plus placebo. All treatments were given every 2 weeks. Thirty percent of patients had received prior bevacizumab with their first-line treatment regimen, while the remainder were naïve to anti-VEGF therapy. The group receiving aflibercept achieved a modest but statistically significant overall survival benefit of 1.4 months (13.50 months vs 12.06 months; HR = 0.817; 95% CI = 0.713 to 0.937; $P = 0.0032$).[480] PFS was also statistically significantly improved from 4.67 months to 6.9 months with the addition of aflibercept ($P < 0.0001$). Reponse rate was improved from 11.1% to 19.8%.

The findings of the VELOUR and TML trials show striking similarities to one another. Each utilized an anti-VEGF strategy in second-line treatment in conjunction with active chemotherapy, and each shows a 1.4-month survival benefit. Given these findings, use of either with second-line FOLFIRI (if second-line FOLFIRI is deemed appropriate) would seem reasonable. Aflibercept has not demonstrated benefit in conjunction with oxaliplatin-based regimens at the time of this writing, and so use of aflibercept with oxaliplatin-based chemotherapy is not recommended. Furthermore, a change from FOLFIRI-aflibercept to FOLFIRI-bevacizumab (or vice versa) is not supported by current data, nor is use of either aflibercept or bevacizumab as a single agent. As such, aflibercept presents an option for second-line therapy in conjunction with FOLFIRI but does not create a new line of therapy in the continuum of treatment.

Ramucirumab

Ramucirumab is a monoclonal antibody that targets VEGF receptor 2. In a clinical trial of patients receiving second-line FOLFIRI, the addition of ramucirumab versus placebo resulted in a median survival benefit of 1.6 months with a hazard ratio of 0.84. Toxicities were similar to those of other vaginal inhibitors.

VEGF Inhibition Beyond Progression

Given the above data, a reasonable interpretation would be that in a patient treated with a bevacizumab-containing first-line regimen, similar modest benefit could be expected from use of either continued bevacizumab, or aflibercept, or ramucirumab, with second-line treatment. The side effect profiles of these three agents appear to be similar as well. Consensus guidelines offer the opinion that given the similar efficacy and safety, use of ramucirumab in this setting would be difficult to justify over the other, substantially less expensive alternatives. It should be noted that no data suggest that any one of these agents would be likely to have salvage activity after failure of another; rather, any of these agents may potentiate a potentially active second-line regimen.

Cetuximab and Panitumumab

The EGFR, also called HER-1, is a transmembrane glycoprotein receptor. When the external binding domain of the EGFR binds specific ligands, such as epidermal growth factor or TGF-α, receptor dimerization occurs (either homodimerization with another

TABLE 40.18

Efficacy Outcomes: Cetuximab Plus Irinotecan versus Cetuximab Alone in Irinotecan-refractory Colorectal Cancer

	No. of Patients	Response Rate (95% CI)	Disease Control (95% CI)	Median TTP (mo)	Median OS (mo) (95% CI)
Cetuximab	111	11% (6% to 18%)	32% (24% to 42%)	1.5	6.9 (5.6–9.1)
Cetuximab + irinotecan	218	23% (18% to 29)[a]	56% (49% to 62%)[b]	4.1[c]	8.6 (7.6–9.6)

[a] $p = 0.0074$.
[b] $p = 0.0001$.
[c] $p < 0.0001$.
CI, confidence interval; TTP, time to progression; OS, overall survival.
Data from Cunningham D, Humblet Y, Siena S, et al. Cetuximab monotherapy and cetuximab plus irinotecan in irinotecan-refractory metastatic colorectal cancer. *N Engl J Med* 2004;351:337–345.

EGFR or heterodimerization with another member of the EGFR family). This in turn stimulates phosphorylation of the tyrosine kinases on the intracellular domain of the receptor, which initiates a signaling cascade, which ultimately regulates cell proliferation, migration, adhesion, differentiation, and survival.[481–483] Cetuximab (c-mab) is a chimeric IgG_1 monoclonal antibody that recognizes and binds to the extracellular domain of the EGFR. Panitumumab (p-mab) is a fully human IgG_2 monoclonal antibody that also targets the EGFR. Binding of either c-mab or p-mab to this receptor does not cause receptor activation but rather results in a steric interference with the ligand binding site.[484]

Preclinical models of cetuximab, or its murine precursor, demonstrated more substantial activity when given in combination with cytotoxic chemotherapy. Based on these observations, and on a single anecdotal report of a major response to cetuximab plus irinotecan in a young woman with irinotecan-refractory CRC, a multicenter phase 2 trial was initiated. This trial, reported in abstract form only, was conducted in patients who were determined by their treating investigator to have progressed on irinotecan.[485] Patients were treated with cetuximab at a dose of 400 mg/m^2 loading dose week 1 over 2 hours, followed by weekly 250 mg/m^2 over 1 hour. Irinotecan was given on the same dose and schedule as had previously failed. Irinotecan dose reductions made previously, prior to study entry, were maintained upon initiation of the study treatment.

One hundred twenty patients with irinotecan-refractory CRC were identified and enrolled. In addition, in a parallel portion of the trial, 28 patients with clinically and radiographically stable disease after receiving a minimum of 3 months of irinotecan therapy were also enrolled and treated by the addition of cetuximab to their ongoing irinotecan therapy. The response outcome of this "stable disease cohort" was not reported; only those patients who were felt to be irinotecan refractory were included in the initial report. As reported by an independent response assessment committee, 22.5% of irinotecan-refractory patients achieved a major objective response. The irinotecan-related toxicity was relatively mild in this population, at least in part because many patients had already had irinotecan dose modifications made prior to starting on this trial. Of the side effects specifically attributable to cetuximab, 3% of patients developed an allergic, anaphylactoid reaction requiring discontinuation of cetuximab therapy, and 75% of patients experienced a skin rash (12% grade 3), a rash now recognized to be characteristic of all EGFR inhibitors. This rash superficially resembles acne, leading to its initial description as an acneiform rash. However, microscopically this is not acne, and topical acne medications are ineffective in its management. An interesting observation from this trial, which has since been corroborated in multiple trials, is that the presence and severity of the rash appeared to be associated with response in this study.

The results seen in the phase 2 cetuximab plus irinotecan combination trial raised the question, both from a scientific and from a regulatory perspective, of the activity of single-agent cetuximab in irinotecan-refractory CRC. A small phase 2 trial was therefore quickly designed and accrued. In this trial, 5 of 57 patients (9%) achieved a partial response confirmed by an independent radiologic review.[486]

Based on the preliminary results of the initial phase 2 cetuximab plus irinotecan study, described previously, a subsequent larger trial, ultimately reported by Cunningham et al.,[316] was designed to provide confirmatory evidence of the activity of cetuximab in CRC (Table 40.18). This large, randomized phase 2 trial in patients with irinotecan-refractory CRC, which has become known as the BOND trial, compared cetuximab plus irinotecan to cetuximab monotherapy. A total of 329 patients were randomized in a two-to-one schema. The response rates of 22.9% for cetuximab plus irinotecan and 10.8% for cetuximab alone were virtually identical to the response rates that had been reported previously in the two US phase 2 trials, confirming the activity of this agent in CRC. Time to tumor progression in the Cunningham et al.[316] study was 4.1 months for the combination versus 1.5 months for single-agent cetuximab. Survival in the two arms was not significantly different; however, the study was neither designed nor powered to address the issue of a survival advantage for cetuximab, and cetuximab was given to all patients on both arms of the study.

A National Cancer Institute Canada phase 3 trial compared cetuximab plus best supportive care to best supportive care alone in 572 patients who had exhausted standard treatment options.[487] The median overall survival was improved by 1.5 months (from 4.6 months to 6.1 months) in the cetuximab group compared to supportive care alone. Partial responses occurred in 23 patients (8.0%) in the cetuximab group versus none in the supportive care group ($P < 0.001$).

Similar results were reported with panitumumab. As seen with single-agent cetuximab, phase 2 evaluations of panitumumab in patients with CRC indicate approximately a 10% response rate, with over 90% of patients experiencing some degree of acneiform-like rash.[488,489] The fully human nature of this antibody appears to reduce the likelihood of anaphylactoid infusion reactions, with only 1 of the 148 patients treated after experiencing a dose-limiting allergic reaction. A randomized trial of panitumumab versus best supportive care in the salvage setting demonstrated a modest (8 weeks vs 7.3 weeks) but highly statistically significant improvement in median progression-free survival for single-agent panitumumab over best supportive care (HR = 0.54; $P < 0.000000001$). Response rate to p-mab was 10%. There was no difference in overall survival; however, there was extensive postprogression crossover, which obscures this end point.[490]

RAS and Other Determinants of Anti-epidermal Growth Factor Receptor Resistance

Perhaps the most important development in the use of anti-EGFR agents over the past decade has been the recognition that these

agents only have the potential to be beneficial to patients whose tumors have nonmutated, or wild-type, *KRAS* and NRAS genes. KRAS and NRAS are signal transduction proteins that are a critical intermediate in transmission of growth and survival signals from the EGFR to the nucleus. Initially it was noted that mutations in exon 2 of the gene that encodes for the KRAS protein led to constitutive activation of this signaling pathway, which rendered blocking of the EGFR-binding site on the surface useless. Several small retrospective series identified KRAS mutations as being incompatible with responses to cetuximab.[491,492] Subsequently, Amado et al.[493] demonstrated that the activity of panitumumab in the registration study referenced previously was limited to those patients with wild-type KRAS. In this trial, 92% of patients had tissue available for KRAS genotyping, and 43% of tumors were found to harbor a mutation in codons 12 or 13 in exon 2 of KRAS. The objective response rate to single-agent panitumumab was 17% in KRAS wild-type tumors and 0% in those tumors that had a KRAS mutation. The progression-free survival in the patients with KRAS wild-type tumor who received panitumumab was 12.3 weeks versus 7.3 weeks for best supportive care. In patients with KRAS-mutated tumors, there was no difference in progression-free survival with panitumumab versus best supportive care. Again, overall survival in this trial could not be interpreted due to extensive postprogression crossover.

Similarly, analysis of the National Cancer Institute Canada study, discussed previously, demonstrated that activity of cetuximab as a single agent in chemotherapy-refractory disease was limited to the KRAS wild-type patients only (Table 40.19).[494] Approximately 70% of patients in this trial had tissue available for KRAS genotyping, and 42% were found to have mutated KRAS. Those with mutated KRAS showed no evidence of clinical benefit from cetuximab, while patients whose tumors had wild-type K-ras showed a 4.7-month improvement in median overall survival with cetuximab versus best supportive care (9.5 months vs 4.8 months; HR for death = 0.55; 95% CI = 0.41 to 0.74; $P < 0.001$). In the control group who received best supportive care, KRAS mutation status had no impact on median overall survival ($P = 0.97$). Although some data have suggested that tumors with KRAS exon 2, codon 13 mutations might still garner some benefit from first-line treatment with cetuximab or panitumumab,[495] subsequent data have refuted this, and any KRAS exon 2 mutation remains a firm contraindication to treatment with an EGFR inhibitor. More recently, a retrospective analysis suggested that in addition to exon 2 KRAS mutations, tumors with mutated BRAF, NRAS, or PIK3CA were significantly associated with a low response rate to cetuximab or panitumumab.[496] The question of the impact of other KRAS mutations outside of exon 2 has been further addressed, as has the impact of NRAS mutations. Data strongly suggest that the presence of any KRAS or NRAS mutation confers resistance to cetuximab and panitumumab and that these agents should not be used in patients whose tumors harbor any of these mutations.[199] The role of BRAF genotyping remains less definitive. While an unpreplanned retrospective subset analysis of a combination of two first-line trials suggests some possible contribution of cetuximab in patients with BRAF-mutated tumors, most published datasets of patients treated in the non–first-line setting show essentially no activity for cetuximab or panitumumab in patients with BRAF-mutated tumors.

Genotyping for all RAS mutations (KRAS, and NRAS, exons 2, 3, and 4) should now be regarded as standard practice in all patients with stage IV disease, and cetuximab and panitumumab should only be considered in patients with nonmutated RAS.[497] Importantly, using a matched set of resected metastases and primary tumors from 84 patients, Vakiani et al.[498] demonstrated near-complete concordance between primary and metastasis for RAS, BRAF, and PIK3CA. Therefore, genotyping of the primary tumor is sufficient and patients do not need to be subjected to a biopsy of a metastatic site for the purposes of tissue genotyping.

It is prudent to obtain RAS genotyping at the time that stage IV disease is diagnosed, not necessarily because the information is needed for first-line therapy, as only a minority of patients will be appropriate for first-line anti-EGFR therapy, but because whether a patient can consider the use of cetuximab or panitumumab in the course of multiple lines of therapy is easier to both determine and to deal with early on when there are multiple options, rather than waiting until all other options are exhausted. At present, there is no role for determining RAS status in stage I, II, or III disease, as there is no basis for use of EGFR agents in other than stage IV.

Preliminary data suggest that KRAS status might play a role in oxaliplatin sensitivity as well. In vitro analysis published by Lin et al.[501] demonstrated that this mutation caused downregulation of ERCC1 (excision repair cross-complementation group 1), which led to enhanced oxaliplatin sensitivity. Overexpression of ERCC1 has been shown previously to be associated with resistance to platinum-based therapy in a number of cancers. Improved understanding of this pathway may prove critical toward defining new targets for RAS-mutant CRC treatment.[499]

More recent data suggest that the small (approximately 2%) of metastatic colorectal cancers that are RAS wild type but HER-2 amplified are highly unlikely to respond to anti-EGFR monoclonal antibodies. However, preliminary data suggest that combination HER-2 targeted therapy, such as trastuzumab plus lapatinib, may be active in this molecularly identified subset of patients.[500]

Cetuximab or Panitumumab in First-line Therapy

A 1,200-patient study of FOLFIRI with or without cetuximab, known at the CRYSTAL trial, has been reported.[13] The primary end point of the trial, progression-free survival, was statistically significantly improved with the addition of cetuximab, albeit by only 0.9 month, or 27 days. When the study was analyzed in terms of KRAS genotype, those patients with mutated KRAS showed no benefit, while those with wild-type KRAS showed a progression-free survival improvement of 1.2 months, or 37 days, but also a median overall survival improvement from 20.0 months to 23.5 months (HR = 0.796; $P = 0.0093$).[501] Response rates were also statistically significantly higher (57.3% vs 39.7%) in the cetuximab arm for the KRAS wild-type tumors. Skin rash and diarrhea were increased in the cetuximab arm. As has been noted in virtually all

TABLE 40.19

Cetuximab versus Best Supportive Care in Metastatic Chemotherapy-refractory Colorectal Cancer

	Overall Survival		Progression-free Survival		Response Rate	
	KRAS Wt	KRAS Mut	KRAS Wt	KRAS Mut	KRAS Wt	KRAS Mut
Cetux	9.5 mo	4.5 mo	3.7 mo	1.8 mo	12.8%	1.2%
BSC	4.8 mo	4.6 mo	1.9 mo	1.8 mo	0%	0%
P value	0.001	0.89	0.001	0.96	NR	NR

Wt, wild type; Mut, mutant; Cetux, cetuximab; BSC, best supportive care; NR, not reported.
Data from Karapetis CS, Khambata-Ford S, Jonker DJ, et al. K-ras mutations and benefit from cetuximab in advanced colorectal cancer. *N Engl J Med* 2008;359:1757–1765.

trials of anti-EGFR agents, there was a strong correlation between severity of skin rash and clinical benefit, with progression-free survival advantage being limited to those patients with grade 2 or 3 skin rash. BRAF was confirmed as a poor prognostic factor but was not predictive of response to cetuximab in this trial. The numbers of BRAF-mutated tumors limit the power to detect a predictive role of BRAF in this trial, however.

FOLFOX with or without cetuximab had been initially investigated in a randomized phase 2 trial in which the primary end point was response rate.[502] A total of 337 patients were treated. The overall response rate was improved by the addition of cetuximab from 36% to 46%, a result that did not achieve statistical significance ($P = 0.64$). For the KRAS wild-type patients, however, the result was more robust, with an improvement from 37% to 61% ($P = 0.011$). Progression-free survival was very modestly, albeit statistically significantly, improved in the KRAS wild-type patients by a median of 15 days; however, progression-free survival was statistically significantly *worse* in the cetuximab arm in those patients whose tumors had mutated KRAS.

However, several larger phase 3 trials have cast considerable doubt on the role of cetuximab in conjunction with oxaliplatin-based therapy. In the Medical Research Council COIN trial, 1,603 patients with previously untreated metastatic CRC were treated with oxaliplatin plus a fluoropyrimidine (either the FOLFOX or Cape/Ox regimen) and randomized to receive cetuximab in addition or not. In the patients with KRAS wild-type tumors (367 with cetuximab plus oxaliplatin-containing chemotherapy and 362 with oxaliplatin-containing chemotherapy alone), there was no improvement in overall survival (17.9 months vs 17 months; HR = 1.04; $P = 0$ .67) or in progression-free survival (8.6 months vs 8.6 months in the two arms). Overall response rate increased from 57% to 64% with the addition of cetuximab ($P = 0.049$). The data from this trial do not support use of cetuximab with front-line oxaliplatin-based chemotherapy. Outcomes were worse with the addition of cetuximab to capecitabine plus oxaliplatin, and the use of that combination is specifically not recommended.[503] In addition, the Nordic VII trial investigated the addition of cetuximab to the Nordic FLOX regimen of oxaliplatin, bolus 5-FU, and leucovorin in a 571-patient phase 3 trial.[504] Patients were randomized to one of three arms: Nordic FLOX given continuously, cetuximab plus Nordic FLOX given continuously, or cetuximab plus Nordic FLOX given intermittently. There were no statistically significant differences in the median progression-free survival between the three arms (7.9 months, 8.3 months, and 7.3 months, respectively). Overall survival also showed no difference between the three arms (20.4 months, 19.7 months, and 20.3 months, respectively). Even in patients with KRAS wild-type tumors, cetuximab did not provide demonstrable benefit.

Taken in the aggregate, these data would not appear to lend substantial support to the use of cetuximab with oxaliplatin-based chemotherapy. Curiously, however, the results with panitumumab are somewhat different. In a phase 3 trial of FOLFOX with or without panitumumab (6 mg/kg every 14 days), 1,183 patients were randomized, of whom 60% had wild-type KRAS.[505] Progression-free survival was modestly but statistically significantly improved with panitumumab in the KRAS wild-type patients (9.6 months vs 8 months; $P = 0.02$), and response rate was increased from 48% to 55%. However, as was seen with cetuximab, in the patients with KRAS-mutated tumors, the addition of panitumumab resulted in a statistically significant *worsening* of median progression-free survival from 8.8 months in the control arm to 7.3 months in the panitumumab-containing arm. In an updated analysis focusing on KRAS wild-type patients only, as statistically significant survival benefit was seen with the addition of panitumumab in this cohort.[199] The addition of panitumumab resulted in increased skin rash, diarrhea, and hypomagnesemia.

Also reported in abstract form is a phase 3 trial of second-line FOLFIRI with or without panitumumab.[506] Despite this being a second-line study, patients had an excellent performance status (ECOG 0 or 1 in 94% of patients). KRAS mutations were present in 45% of patients. For the patients with KRAS wild-type tumors, the progression-free survival was statistically significantly improved in the panitumumab-containing arm (5.9 months vs 3.9 months; $P = 0.004$), and response rate was improved as well (35% vs 10%). Overall survival differences in favor of the panitumumab arm (14.5 months vs 12.5 months) did not reach statistical significance ($P = 0.1$). There were no differences in efficacy outcomes with the addition of panitumumab in the patients with KRAS-mutated tumors. Again, skin rash, diarrhea, and hypomagnesemia were increased in the panitumumab arm.

More recently the long-awaited initial results of the NCI cooperative group 80405 study have been reported in abstract form. In this trial, patients with previously untreated metastatic colorectal cancer with wild-type KRAS were enrolled. Patients were assigned to receive either FOLFOX or FOLFIRI per physician preference and then randomized to receive either cetuximab or bevacizumab. There was no difference in overall survival, the primary study end point, between the cetuximab and bevacizumab arms (29.0 months vs 29.9 months, $P = 0.34$).[507]

Toxicities of Anti-epidermal Growth Factor Receptor Monoclonal Antibodies

The primary toxicity of cetuximab and panitumumab is an acne-like rash, which is seen to some degree in from 75% to 100% of patients treated. This rash is not acne, and it is accompanied by skin dryness and paronychial cracking. Other than moisturizers, which are recommended, no topical agents have been shown to be of benefit in the treatment of this rash. Drying agents and retinoids, such as are used in the treatment of acne, are contraindicated. Anecdotal reports of benefit for topical steroids or antibiotics do not have supportive randomized data, and as the natural history of the rash is to wax and wane, interpretation of these anecdotal reports is problematic. There are data suggesting that prophylactic use of oral antibiotics may somewhat mitigate the severity of the rash.[508] Importantly, it has been well established that there is a clear correlation between severity of skin rash and favorable outcome with EGFR agents.[509] The mechanism of this correlation has not yet been determined; however, it is clear that benefit from these agents is virtually confined to those patients who experience a grade 2 or 3 skin rash. A severe rash does not guarantee a response or clinical benefit; however, absence of a rash after the first month of therapy is virtually incompatible with clinical benefit from these agents. This is an important point to consider, especially in consideration of front-line use; only those patients with a very substantial rash stand a chance of benefit.

Hypersensitivity reactions, which are anaphylactoid in nature and are completely separate and distinct from the skin rash toxicity discussed previously, occur in approximately 3% of patients with cetuximab and <1% of patients with panitumumab. Almost all of these reactions are first-dose events. Dramatic regional differences in the frequency of these reactions have been noted, with serious hypersensitivity reactions to cetuximab noted in up to 20% of patients in North Carolina and Tennessee, while the serious hypersensitivity reaction rate in the northeastern United States is <1%.[510] Subsequently, it has been demonstrated that there is a high prevalence of cetuximab-specific IgE in Tennessee, suggesting cross-reactivity with an environmental allergen.[511] Panitumumab does not appear to exhibit this marked regional variation in incidence of hypersensitivity reaction and would be the clearly preferred agent over cetuximab in these areas of high incidence of cetuximab hypersensitivity.

It should be noted that the incidence of skin rash appears to be quite similar between cetuximab and panitumumab, as does the degree of clinical activity. Thus, outside of the areas that see high frequency of cetuximab hypersensitivity reactions, there appears to be little reason to favor one agent over the other. Case reports and anecdotal evidence suggest that patients who experience hypersensitivity to

cetuximab do typically tolerate panitumumab. There is no basis, however, for using one of these agents after clinical failure on the other.

Another more recently recognized toxicity of anti-EGFR therapy is hypomagnesemia.[512] This result is due to hypermagnesemia, presumably promoted by EGFR antagonism in the loop of Henle. Regular monitoring of serum magnesium levels, and intravenous magnesium supplementation when indicated, should be practiced routinely with anti-EGFR therapies. Oral magnesium is unlikely to provide adequate supplementation, as diarrhea from this is often dose limiting.

Cetuximab and Panitumumab in Epidermal Growth Factor Receptor–Negative Patients

From the outset of clinical development, the assumption was made that quantitative EGFR expression would be predictive of the activity, or lack thereof, of an anti-EGFR antibody, and that an absence of demonstrable EGFR expression would therefore preclude clinical activity of cetuximab or panitumumab. For this reason, all early trials with these agents required that the patient's tumor shows EGFR positivity by IHC as a criterion for study eligibility. This assumption, that EGFR expression would be predictive, has never been supported by clinical or preclinical data and has been refuted by all clinical data that have addressed the issue. All of the reported cetuximab trials to date, the earlier ones that excluded EGFR-negative patients altogether, have demonstrated absolutely no correlation between the intensity of the EGFR expression and clinical response.[316,485] Additionally, the results of a small cohort of nine patients who were EGFR negative and treated with cetuximab were reported in abstract form.[513] Two major objective responses were reported by the investigators, one of which was confirmed as a major response by third-party review and one of which was not.

On the basis of the lack of correlation between EGFR staining intensity and response, as well as the small data set outlined previously, a decision was made at Memorial Sloan Kettering Cancer Center in New York that patients with EGFR-negative CRC would not be excluded from standard off-protocol treatment with cetuximab simply on the basis of EGFR status. Subsequently, a retrospective review was conducted using the computerized pharmacy records to identify all patients who had received nonresearch cetuximab-based therapy at Memorial Sloan Kettering Cancer Center in the first 3 months of cetuximab's commercial availability. This review identified 16 patients with irinotecan-refractory, EGFR-negative CRC who had been treated. Fourteen of these patients had received cetuximab in combination with irinotecan and two had received cetuximab alone. Of the 16, four patients experienced major objective (response rate = 25%; 95% CI = 4% to 46%), demonstrating that the hypothesis that a negative EGFR stain would preclude the possibility of response to cetuximab is false.[514] A similar lack of correlation of EGFR staining and activity with panitumumab has also more recently been reported.[515]

Because current EGFR IHC techniques have no predictive value, these techniques have no role in current management of CRC. The exclusion of a patient from cetuximab-based or panitumumab-based therapy solely on the basis of EGFR IHC is not appropriate. Likewise, no patient, with CRC or otherwise, should be given an anti-EGFR treatment solely on the basis of a high EGFR IHC expression.

First-line EGFR versus VEGF

Two large randomized clinical trials have addressed the question of use of first-line anti-VEGF versus anti-EGFR monoclonal antibodies in RAS wild-type patients. The larger of these is the CALGB/SWOG C80405 trial.[516] In this trial, 1,137 patients with K-ras wild-type metastatic colorectal cancer were permitted to select either FOLFOX or FOLFIRI and then were randomized to receive this with either cetuximab or bevacizumab. Pre-specified primary end point was overall survival. No significant difference was seen, with a median overall survival of 29.0 months in the bevacizumab group versus 30.0 months in the cetuximab group. The median progression-free survival was 10.6 months and 10.5 months, respectively. Response rates also were not significantly different with 55.2% of bevacizumab patients and 59.6% of cetuximab patients responding ($P = 0.13$). No differences were seen in any subsets evaluated, including patients treated with FOLFIRI or patients treated with FOLFOX as their chemotherapy backbone. In a subsequent exploratory analysis looking at an all RAS wild-type population, the results remained unchanged. Notably, however, activity of cetuximab was seen only in those patients with left-sided tumors. Thus for patients with either RAS-mutated tumors or right-sided tumors in whom a monoclonal antibody is to be used, an anti-VEGF strategy with bevacizumab would be warranted. For patients with RAS wild-type left-sided tumors, data would support use of either an anti-EGFR agent (cetuximab or panitumumab) or bevacizumab as the accompanying antibody. Even in this population, an argument can be made in favor of bevacizumab on cost and toxicity issues.

Another clinical trial that addressed the question of EGFR versus the EGFR strategy is the FIRE-3 trial.[517] In this trial, 592 patients with K-ras wild-type metastatic colorectal cancer were randomized to receive FOLFIRI plus cetuximab versus FOLFIRI plus bevacizumab. The prespecified primary end point of this trial was overall response rate and there was no significant difference between the two arms (62% vs 58%, $P = 0.18$). Overall survival was also not different at 10.0 versus 10.3 months, $P = 0.55$). Interestingly, there was a significant difference in overall survival with the cetuximab group having a median survival of 28.7 months versus 25.0 months in the bevacizumab group ($P = 0.017$). This survival difference was even more impressive in a post hoc all RAS wild-type subset analysis. At first glance the results of FIRE-3 would appear to contradict those of CALGB/SWOG 80405. However, it should be noted that the prespecified primary end point of FIRE-3, overall response rate, showed no difference in the arms. With the specified primary end point being negative, any other conclusions drawn from the trial must necessarily be guarded as hypothesis generating, rather than definitive. Furthermore, there appears to be a substantial number of patients in the bevacizumab arm that never received an anti-EGFR monoclonal antibody for their RAS wild-type colorectal cancer. This relative lack of second-line usage may have substantially contributed to the survival difference seen in the trial, especially because the curves begin to separate at approximately the 2- year mark, substantially beyond the point of progression.

Bevacizumab Plus Anti-epidermal Growth Factor Receptor Agents

Given the reported activity of both bevacizumab and cetuximab, investigators logically became interested in the idea of concurrent use of these agents. Limited preclinical data and the mechanistic understandings of potential interaction between anti-EGFR and anti-VEGF pathways supported the concept. As will be discussed subsequently, this concept serves as yet another example of perfectly logical, well-thought out assumptions supported by preliminary clinical evidence that turned out to be incorrect when subjected to the appropriately rigorous test of an adequately powered clinical trial.

The first study to attempt to administer bevacizumab and cetuximab concurrently was a small randomized phase 2 study of bevacizumab added to cetuximab alone or to cetuximab plus irinotecan in patients with irinotecan-refractory CRC.[518] This was a feasibility trial to assess the safety of concurrent administration of these agents and to look for preliminary evidence of efficacy. The study concluded that coadministration of these two monoclonal antibodies together was feasible and that the preliminary data were

TABLE 40.20A

Capecitabine, Oxaliplatin, and Bevacizumab with or without Cetuximab in Metastatic Colorectal Cancer

Overall	Median Progression-free Survival	Median Response	Objective Survival Rates
COB ($n = 332$)	20.3 mo	10.7 mo	50%
COB plus cetux ($n = 317$)	19.4 mo	9.4 mo	52.7%
P value	0.16	0.01	0.49

COB, capecitabine, oxaliplatin, bevacizumab; cetux, cetuximab.
From Tol J, Koopman M, Cats A, et al. Chemotherapy, bevacizumab, and cetuximab in metastatic colorectal cancer. *N Engl J Med* 2009;360:563–572.

encouraging. It should be noted, however, that this was a small feasibility trial with 41 and 40 patients, respectively, reported in each arm. Furthermore, this study was conducted in patients who were naive to both cetuximab and bevacizumab. As most patients now receive bevacizumab with their first-line regimen, the results in the bevacizumab-naive population might not necessarily have a bearing on current practice today. A small follow-up trial in patients with prior progression on a front-line bevacizumab-containing regimen showed far less activity, with 3 of 33 patients (9%) achieving a partial response and a median time to tumor progression of 3.9 months.[519] These small trials were designed to serve as the safety pilots for large-scale front-line studies that combined bevacizumab plus cetuximab with front-line chemotherapy. Two such studies have now been reported, with alarming results, which highlights the dangers of jumping to conclusions prior to the availability of mature, definitive data.

The CAIRO-2 study randomized 755 patients with previously untreated metastatic CRC to Cape/Ox-bevacizumab with or without concurrent cetuximab (Table 40.20A).[520] Not only was there not a benefit to the addition of cetuximab, but the group receiving cetuximab actually had a worse median progression-free survival of 9.4 months, compared to 10.7 months in the Cape/Ox-bevacizumab–alone arm ($P = 0.01$). Response rates were identical (44%) in the two arms. Furthermore, quality-of-life scores were lower in the cetuximab-containing arm. Overall survival was not statistically significantly different between the two groups. Even for the wild-type KRAS patients, there was no benefit in progression-free survival with the addition of cetuximab. As might now be anticipated (Table 40.20B), within the cetuximab-containing arm, patients whose tumors had mutated KRAS had statistically significantly decreased progression-free survival compared to those with wild-type KRAS tumors (8.1 months vs 10.5 months; $P = 0.04$). However, for the patients with KRAS mutations, those who received cetuximab also had a worse outcome than those on the noncetuximab-containing control arm (progression-free survival 8.1 months vs 12.5 months, $P = 0.003$; overall survival 17.2 months vs 24.9 months, $P = 0.03$).

TABLE 40.20B

Impact of KRAS Mutation Status on Addition of Cetuximab to Capecitabine, Oxaliplatin, and Bevacizumab

	Overall Survival COB + COB Cetux	Progression-Free Survival COB + COB Cetux
KRAS wild type	22.4 mo 21.8 mo ($P = 0.64$)	10.6 mo 10.5 mo ($P = 0.30$)
KRAS mutated	24.9 mo 17.2 mo ($P = 0.03$)	12.5 mo 8.1 mo ($P = 0.003$)

COB, capecitabine, oxaliplatin-bevacizumab; cetux, cetuximab.
From Tol J, Koopman M, Cats A, et al. Chemotherapy, bevacizumab, and cetuximab in metastatic colorectal cancer. *N Engl J Med* 2009;360:563–572.

Another study that investigated the use of combined bevacizumab plus anti-EGFR monoclonal antibody was the Panitumumab in Advanced Colon Cancer Evaluation (PACCE) trial.[521] This trial used FOLFOX-bevacizumab (823 patients) or FOLFIRI-bevacizumab (230 patients) and randomized them with or without concurrent panitumumab. Again, the result of adding the anti-EGFR to chemotherapy plus bevacizumab was not only not beneficial, but was actually detrimental. The median progression-free survival for the overall study was 10.0 months versus 11.4 months for the panitumumab-containing versus chemotherapy-bevacizumab–alone arm. The median overall survival was decreased by 5.1 months, from 24.5 months in the control arm to 19.4 months in the panitumumab arm. Toxicity, including not only skin rash but also diarrhea, infections, and pulmonary embolisms, was more frequent in the panitumumab-containing arm, and worse outcomes were seen in the panitumumab-containing arm regardless of KRAS mutation status.

Clearly, the concurrent use of anti-EGFR monoclonal antibodies, bevacizumab, and cytotoxic chemotherapy, despite supportive encouraging preliminary data, is not an acceptable treatment strategy. The reasons for this unanticipated negative interaction remain unknown at this time.

BRAF V600E Mutated Colorectal Cancer

The BRAF V600E mutation, which occurs in approximately 10% of patients with metastatic colorectal cancer, confers a particularly poor prognosis with median overall survival of 4 to 6 months after failure of initial therapy. BRAF inhibition alone have limited clinical efficacy due to rapid feedback activation through the epidermal growth factor receptor. Prior studies have shown increased antitumor activity when BRAF inhibition is combined with anti-EGFR antibodies or with a mitogen-activated protein kinase. The BEACON CRC trial was a global multicenter, randomized phase 3 study that aimed to evaluate whether triplet therapy (the combination of all three agents) or doublet therapy (anti-EGFR and BRAF inhibitor) would yield improved outcomes over standard therapy (cetuximab/irinotecan or cetuximab & FOLFIRI) in patients with the BRAF V600E mutation who had failed one or two previous regimens. Investigators found that both experimental arms resulted in a significant and clinically relevant improvement in overall survival and objective response rate compared to standard therapy. Median overall survival was 9 months for the triplet group, 8.4 months in the doublet group, and 5.4 months in the control arm. While the trial was not powered to compare the two experimental arms directly, analysis of overall survival showed a hazard ratio for death of 0.79 that favored the triplet regimen. The rate of adverse events was slightly higher in the control group but similar between the doublet and triplet allowing for maintenance of high dose intensity for the majority of patients.[522] Based on these initial results, both the doublet and triplet therapies were added to consensus guidelines as options for patients with BRAF V600E mutation-positive CRC. However, additional data presented at ASCO 2020, which also demonstrated oncologic equivalence of the two experimental regimens, reported higher rates of serious adverse events in the triplet arm prompting consensus guidelines to remove this option, concluding that only the doublet therapy should be considered.[523]

Immunotherapy in Mismatch Repair Deficient Colorectal Cancer

- While immunotherapy strategies have been largely unsuccessful in mismatch repair proficient (Microsatellite stable) colorectal cancer, the Programed Death Receptor 1 (PD-1)

agents have shown substantial activity in mismatch-repair deficient (dMMR) colorectal cancer patients. In a report by Le et al.,[524] the PD-1 inhibitor pembrolizumab demonstrated a 33% response rate in 124 patients with metastatic colorectal cancer that was refractory to one or two lines of prior chemotherapy. Similar results have been shown for nivolumab, another PD-1 inhibitor; in the Checkmate 142 trial by Overmal et al.,[525] 23 of 74 patients (31.1%) with dMMR colorectal cancer achieved a partial response. Studies have also been done combining nivolumab with the cytotoxic lymphocyte A-4 (CTLA-4) inhibitor ipilimumab.[526] A response rate of 55% was note with this combination immunotherapy regimen. Toxicity with the combination was higher than with single agents. At present, a randomized trial of nivolumab alone versus nivolumab plus ipilimumab in previously treated metastatic colorectal cancer is ongoing at the time of this writing.

- In a first-line study in metastatic dMMR colorectal cancer, Andre et al. reported that pembrolizumab showed superior progression-free survival (16.5 months vs 8.2 months) response rate (43.8% vs 33.1%).[527] More recently, the combination of nivolumab plus ipilimumab has been reported in first-line treatment of dMMR metastatic colorectal cancer.[528] A response rate of 69% was seen, with 13% achieving a complete response. With a median of 24 months of follow-up, neither median progression-free survival nor overall survival have been reached. Again, the relative benefits of nivolumab versus the combination of nivolumab plus ipilimumab have not been compared in a randomized fashion, and toxicity is high with the combination. Clinical decisions regarding single agent versus combination therefore need to be individualized.

Oral Epidermal Growth Factor Receptor, Vascular Endothelial Growth Factor, and Cyclooxygenase-2 Inhibitors

The limited experiences with the oral EGFR tyrosine kinase inhibitors gefitinib (ZD1839) and erlotinib (OSI-774) in CRC have been essentially negative, and at present there is no role for these agents in this disease.[529,530] This is consistent with the findings that the activating mutation seen in lung cancer required for anti-EGFR tyrosine kinase activity does not appear to occur in CRC.

Oral VEGF tyrosine kinase inhibitors, with the exception of regorafenib noted previously, have been similarly disappointing. Sunitinib showed essentially no activity as a single agent in chemotherapy-refractory disease, and front-line trials of chemotherapy with or without sunitinib, as well as chemotherapy with or without sorafenib, have now been closed early by their respective data monitoring committees for futility.[531] Two large, randomized trials of FOLFOX with or without the investigational VEGF tyrosine kinase inhibitor PTK-787 have also been reported as negative trials.

Regorafenib

Regorafenib is an orally administered small molecule multitargeted tyrosine kinase inhibitor. It is closely related to its parent compound, sorafenib, and differs only by the addition of a fluorine atom. After phase 1 trials identified preliminary evidence of activity in patients with refractory CRC, a large phase 3 trial of regorafenib versus placebo was undertaken.[532] A total of 760 patients, all with ECOG grade 0 or 1 performance status, who had progressed through all standard therapies, were randomized 2:1 to received regorafenib 160 mg orally daily versus placebo. The regorafenib group achieved a modest but statistically significant overall survival benefit of 1.4 months (6.4 months vs 5.0 months; HR = 0.77; 95% CI = 0.64 to 0.94; $P = 0.0052$). Response was essentially nonevident, with a response rate of 1% in the regorafenib arm. Grade 3 hand-foot syndrome (17%) and grade 3 fatigue (10%) were the most common toxicities encountered on the regorafenib arm. Regorafenib monotherapy can be considered as a standard care option for good performance status patients who have progressed through standard therapies. Studies assessing the use of regorafenib in earlier lines of therapy and in combination with cytotoxic agents are in progress at the time of this writing.

TAS-102

Tas-102, another orally administered agent, is a combination of trifluridine, a pyrimidine-based nucleic acid analog, and tipiracil hydrochloride, an inhibitor of thymidine phosphorylase that serves to potentiate the trifluridine. In a randomized trial of TAS-102 versus placebo, conducted in 800 colorectal cancer patients with performance status of ECOG 0–1 who had previously been treated with all available chemotherapy agents, TAS-102 resulted in a median 1.7 month improvement in survival (Hazard ratio 0.66: $P < 0.001$). The most common toxicity was neutropenia, with 38% grade ≥ 3, and 4% febrile neutropenia.[533]

Cyclooxygenase-2 Inhibitors

Cyclooxygenase-2 (COX-2) catalyzes the synthesis of prostaglandins in the inflammatory response process. COX-2 has been frequently shown to be upregulated in malignant and premalignant tissues. COX-2 expression has been correlated with increased invasiveness, resistance to apoptosis, and increased angiogenesis.[534] The science behind COX-2 inhibition appeared so compelling that many clinicians had chosen to add drugs such as celecoxib or rofecoxib (now withdrawn from the market for safety reasons) in the absence of efficacy data with the assumption that "it couldn't hurt." Evidence that use of either NSAIDs or selective COX-2 inhibitors has a beneficial role in the treatment of CRC is lacking. The large randomized BICC-C trial showed no benefit whatsoever for the use of celecoxib in terms of either safety or efficacy.[434] In the absence of any emerging data to the contrary, routine use of COX-2 inhibitors with chemotherapy is not recommended.

Aspirin

The role of aspirin as an adjuvant agent is also of particular interest, especially after a recent case-control study suggested that initiation of this medication after diagnosis reduced overall CRC specific mortality (HR = 0.53; 95% CI = 0.33 to 0.86).[535] The ASCOLT study is the first randomized, placebo-controlled trial designed to investigate this question in patients with stage III-IV and high-risk stage II disease. End points will include DFS and overall survival with anticipated follow-up of 5 years.[536]

In the meantime, a large observational study of 4,481 patients lends further support to the potential therapeutic benefits of aspirin in CRC. The authors report a 23% decrease in disease-specific mortality for patients who took aspirin for any length of time after diagnosis compared to nonaspirin users as well as a 30% lower disease-specific mortality for those who took aspirin for 9 months after diagnosis. A survival benefit was also found for prediagnosis users, although this was less pronounced at 12%.541

Other reports indicate that aspirin may only be beneficial for patients with mutation of the PIK3CA gene that is present in 15% to 20% of colorectal cancers and affects tumor apoptosis. In a study of 964 patients, Liao et al. demonstrated superior survival data (HR 0.54 for OS, HR 0.18 for cancer-specific survival) only for those with the mutation.[114] Domingo et al. produced comparable results with regards to aspirin use in his trial but interestingly found no such effect of the mutation on patients taking rofecoxib, despite similar mechanisms of action.[538] This topic remains controversial though as, more recently, Reimers et al. found no association between the mutation and survival benefit of aspirin. Instead, these authors found that this benefit was dependent on HLA class I antigen expression in the original tumor. Clearly, a better understanding of the biological

mechanisms underlying the anticancer effects of this medication is necessary.[539]

Other Novel Agents

The number of agents that are undergoing early evaluation in CRC is too large to allow a complete discussion in this chapter. Many are variations on the currently available agents and are unlikely to substantially move the field if successful in gaining approval. At present, no new agent with a unique mechanism of action has been identified as having meaningful activity in CRC. Furthermore, all of these at this point are of research interest only and do not have a role in standard treatment of CRC at this time.

MOLECULAR PREDICTIVE MARKERS

With the availability now of a number of active agents, the ability to prospectively select a particular drug or drug combination that would have an increased likelihood of efficacy or a decreased likelihood of toxicity would be clinically useful. Such means of rational selection do not yet exist.

One avenue of investigation has been the elucidation of markers of resistance to 5-FU based on knowledge of its metabolic pathways. Studies have indicated that high levels of either TS, DPD,[540] or thymidine phosphorylase, as measured in a tumor specimen by reverse transcription–polymerase chain reaction, predict for failure to respond to an infusional 5-FU regimen.[219,541,542] These observations are intriguing but are insufficient to exclude the use of 5-FU in a particular patient, and they need to be validated in large-scale prospective trials before being applied to routine practice. There is, at this time, no role for the use of these markers in standard practice. Others have investigated genomic analysis as an indicator of response or toxicity.[223,543] Although these approaches appear promising, they are not yet validated and should not be considered as part of standard care.

A recently reported mechanism of resistance thought to be independent of other mutations or MSI involves the tumor suppressor gene TFAP2E (transcription factor AP-2 epsilon) and its downstream target DKK4 (dickkopf homolog 4 protein). In an analysis of 220 patients with CRC, Ebert et al.[544] found that hypermethylation of TFAP2E led to decreased protein expression and was significantly associated with nonresponse to 5-FU–based chemotherapy, whereas hypomethylation yielded a sixfold higher likelihood of response. Moreover, TFAP2E hypermethylation in vitro led to overexpression of DKK4, which was in turn associated with increased chemoresistance to 5-FU but not irinotecan and oxaliplatin. The authors suggest that future studies focus on specific targeting of DKK4 to overcome this pathway of chemoresistance.[544]

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41 Cancer of the Rectum

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INTRODUCTION

Information concerning epidemiology and systemic approaches to the management of both colon and rectal cancer was given in Chapter 40 Cancer of the Colon. This chapter will focus on issues unique to rectal cancer with an emphasis on radiation, combined modality therapy, and sphincter-preserving surgery.

ANATOMY

The anatomy of the rectum can be quite confusing, as there are differing definitions of the relevant landmarks. In the upper portion of the rectum, there are changes both in the musculature of the large bowel and in the relationship to the peritoneal covering that roughly coincide. In the lower portion of the rectum, the mucosal changes occur at roughly the same location as the anal sphincter.

The anatomy of the rectum is usually divided into three portions (Fig. 41.1). The lower rectum is the area approximately from 3 to 6 cm from the anal verge. The midrectum goes from 5 to 6 to 8 to 10 cm and the upper rectum extends approximately from 8 to 10 to 12 to 15 cm from the anal verge, although the retroperitoneal portion of the large bowel often reaches its upper limit approximately 12 cm from the anal verge. In some patients, especially elderly women, the peritonealized portion of the large bowel can be located much lower than these definitions. The determination of the location of the boundary between rectum and sigmoid colon is important in defining adjuvant therapy, with the rectum usually being operationally defined as that area of the large bowel that is at least partially retroperitoneal.

Externally, the upper extent of the rectum can be identified where the tenia spread to form a longitudinal coat of muscle. The upper third of the rectum is surrounded by peritoneum on its anterior and lateral surfaces but is retroperitoneal posteriorly without any serosal covering. At the rectovesical or rectouterine pouch, the rectum becomes completely extra-/retroperitoneal. The rectum follows the curve of the sacrum in its lower two-thirds. It enters the anal canal at the level of the levator ani. The anorectal ring is at the level of the puborectalis sling portion of the levator muscles.

The location of a rectal tumor is most commonly indicated by the distance between the anal verge, dentate (pectinate or mucocutaneous) line, or anorectal ring and the lower edge of the tumor. These points of reference are all different for different individuals. In addition, these measurements differ depending on the method of measurement. This can be important clinically, as the measurement from a flexible endoscopy can substantially overestimate the distance to the tumor from the anal verge or other landmark. The distance from the anal sphincter musculature is clinically of more importance than the distance from the anal verge, as it has implications for the ability to perform sphincter-sparing surgery. The lack of a peritoneal covering over most of the rectum is a major reason for the higher risk of local failure after primary surgical management of rectal cancer compared to colon cancer. The mesorectum is usually used as the structure to define the extent of a total mesorectal excision (TME), with most of the perirectal fatty tissue and perirectal lymph nodes (LNs) contained within its boundaries.

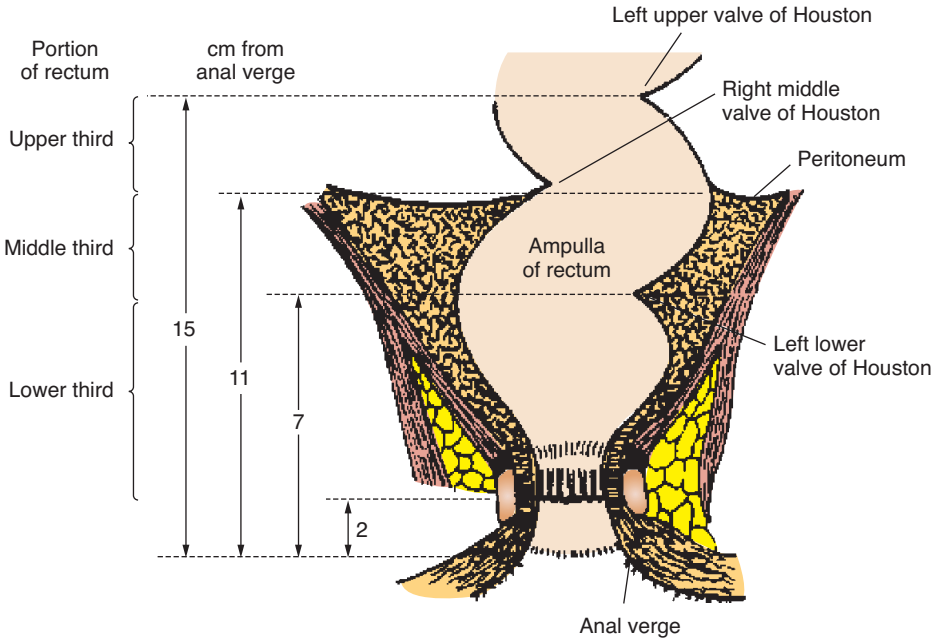


Figure 41.1 Division of the rectum into upper, middle, and lower thirds.